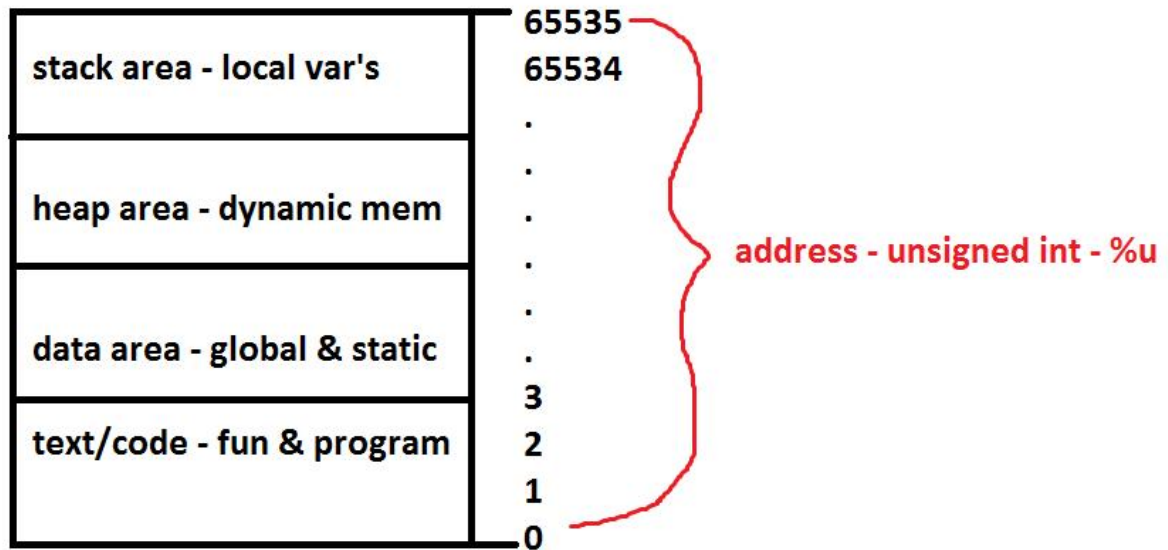


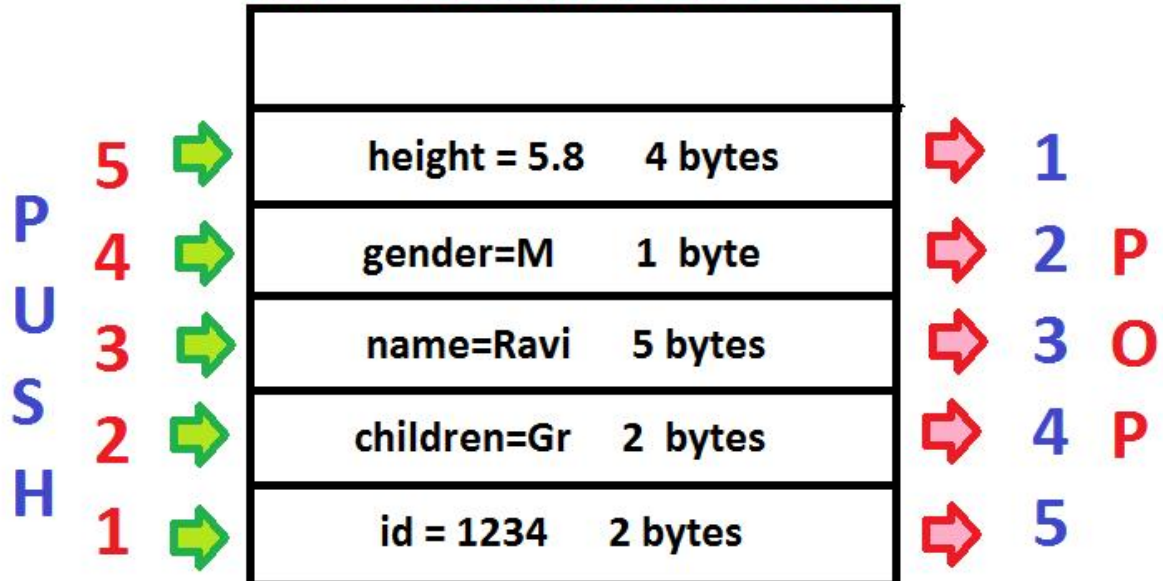
Memory allocation for variables:

RAM - stack - 64kb
65536 bytes



stack area

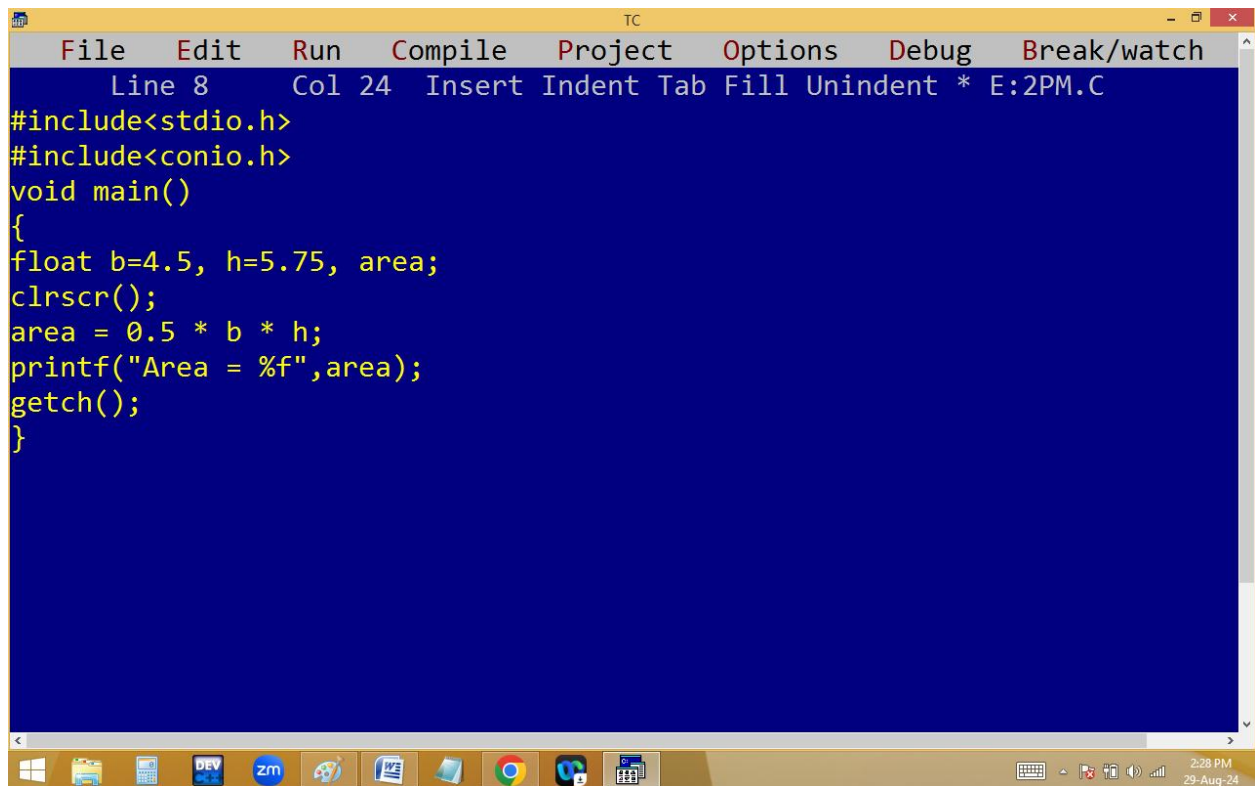
lifo - Last In First Out



filo - First In Last Out

Finding area of a triangle:

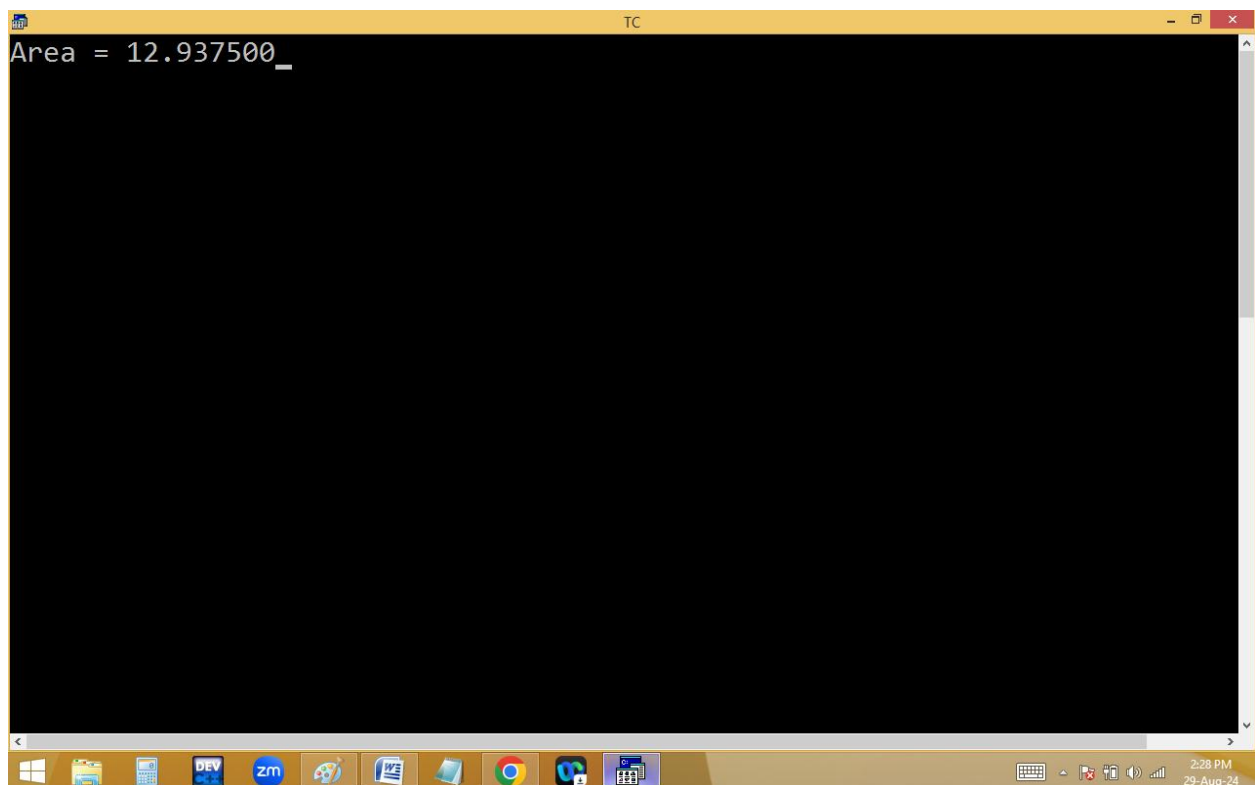
$$0.5 * b * h$$



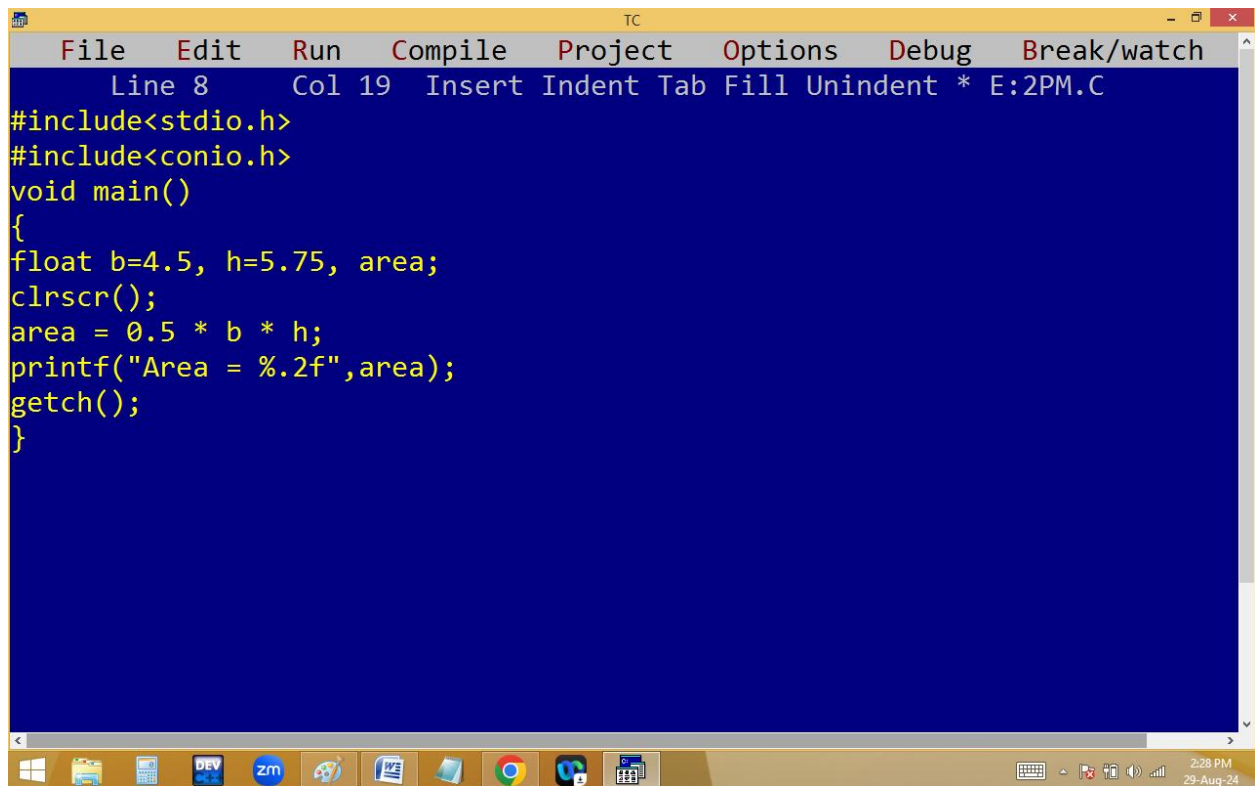
The screenshot shows the Turbo C++ (TC) IDE with a yellow title bar and menu bar. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates 'Line 8 Col 24 Insert Indent Tab Fill Unindent * E:2PM.C'. The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
float b=4.5, h=5.75, area;
clrscr();
area = 0.5 * b * h;
printf("Area = %f",area);
getch();
}
```

The Windows taskbar at the bottom shows various icons including Windows, File Explorer, Calculator, DEV C++, Zm, and others. The system clock in the bottom right corner displays '2:28 PM 29-Aug-24'.



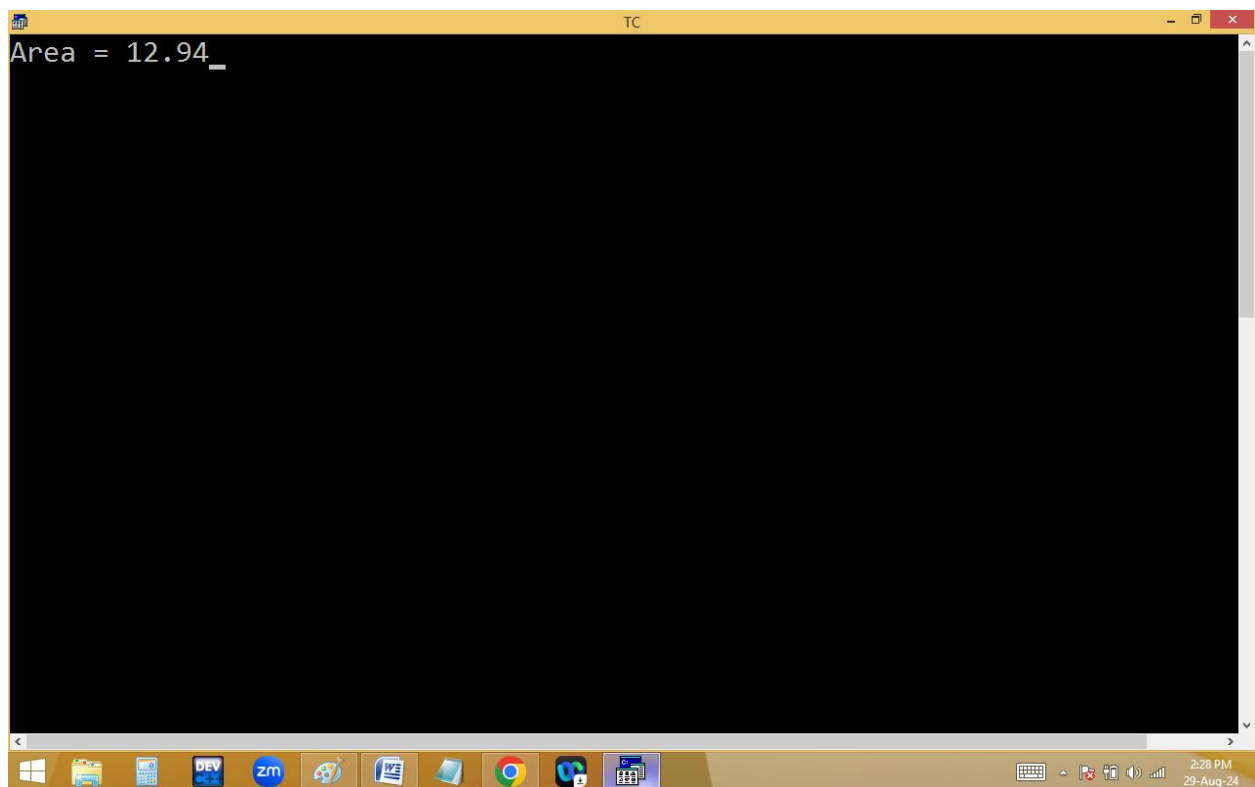
The screenshot shows the Turbo C++ (TC) IDE with a yellow title bar. The main editing area has a black background and displays the output of the program: 'Area = 12.937500_'. The Windows taskbar at the bottom is identical to the first screenshot, showing the same icons and system clock ('2:28 PM 29-Aug-24').



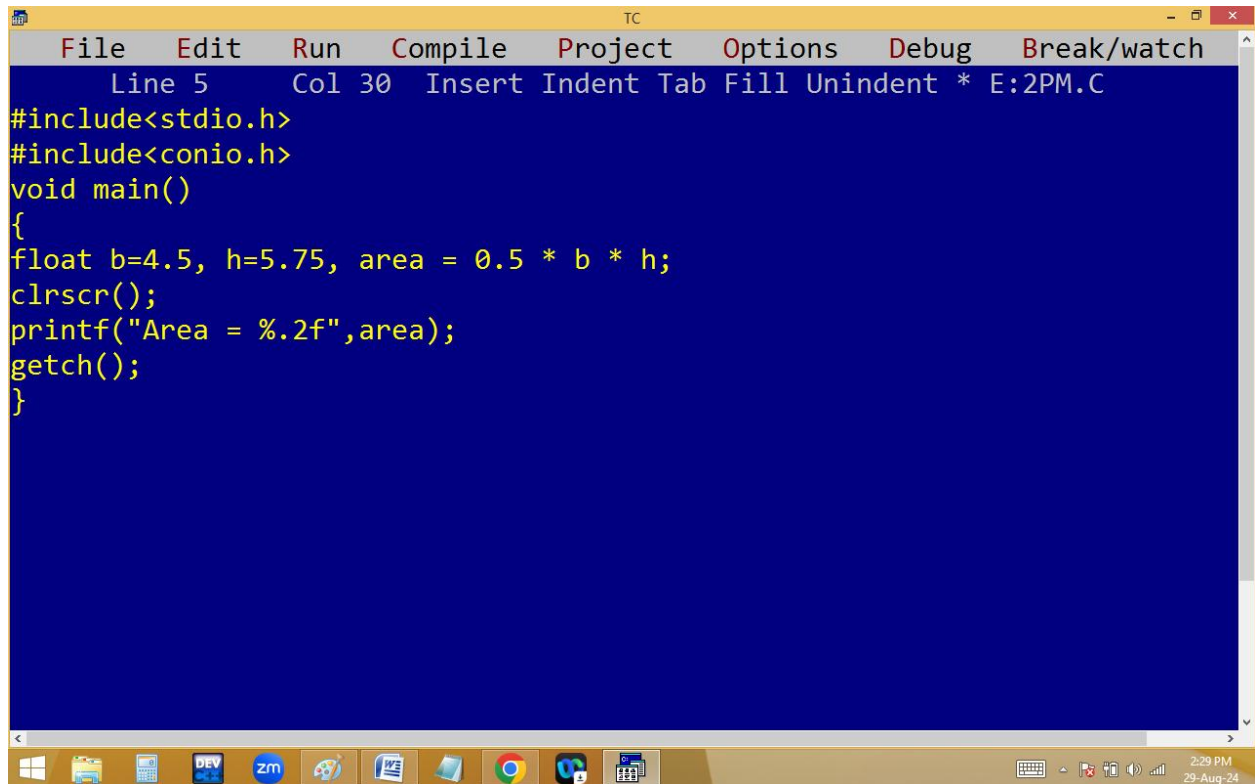
The screenshot shows the Turbo C++ (TC) IDE with a yellow title bar and menu bar. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates 'Line 8 Col 19 Insert Indent Tab Fill Unindent * E:2PM.C'. The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
float b=4.5, h=5.75, area;
clrscr();
area = 0.5 * b * h;
printf("Area = %.2f",area);
getch();
}
```

The Windows taskbar at the bottom shows various icons including the Start button, File Explorer, Calculator, DEV C++, Zoom, and several other applications. The system clock in the bottom right corner displays '2:28 PM' and '29-Aug-24'.



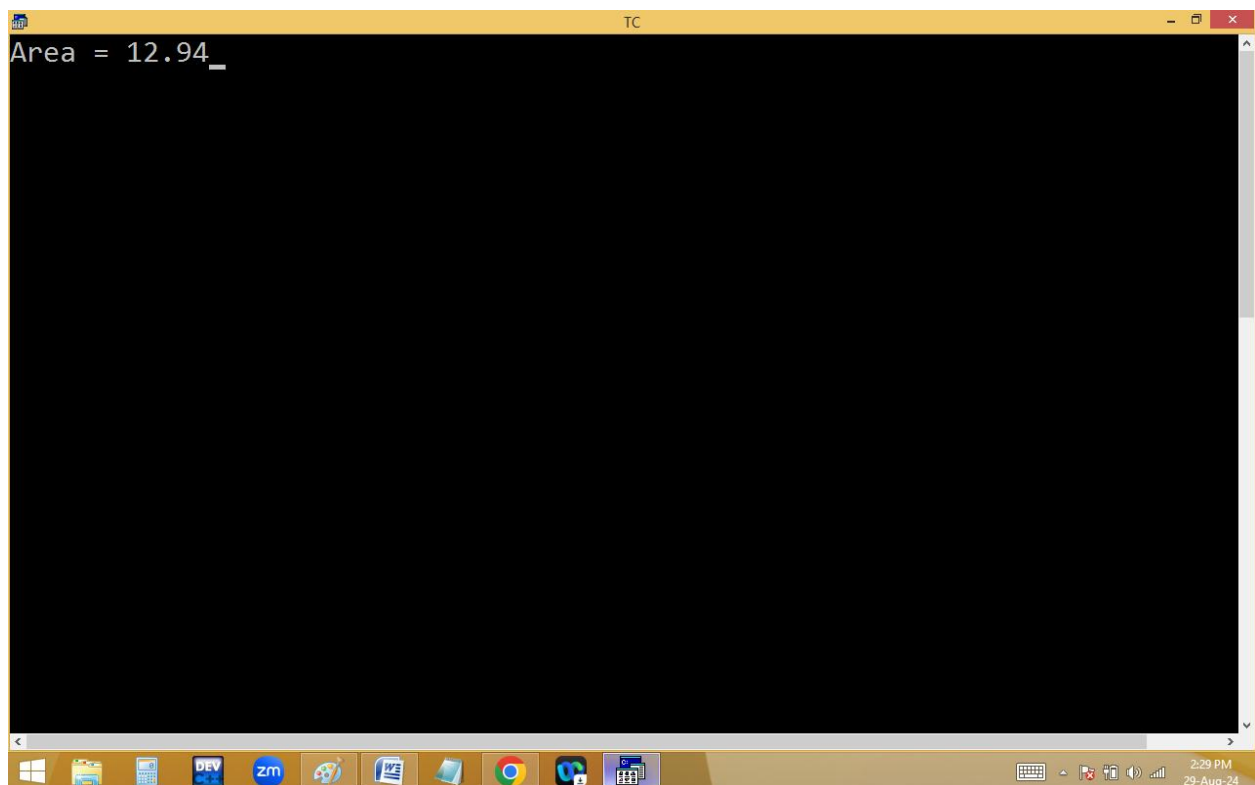
The screenshot shows the Turbo C++ (TC) IDE with a yellow title bar. The main editing area now has a black background and displays the output of the program: 'Area = 12.94_'. The Windows taskbar at the bottom is identical to the previous screenshot, showing the same set of application icons and system clock information ('2:28 PM' and '29-Aug-24').



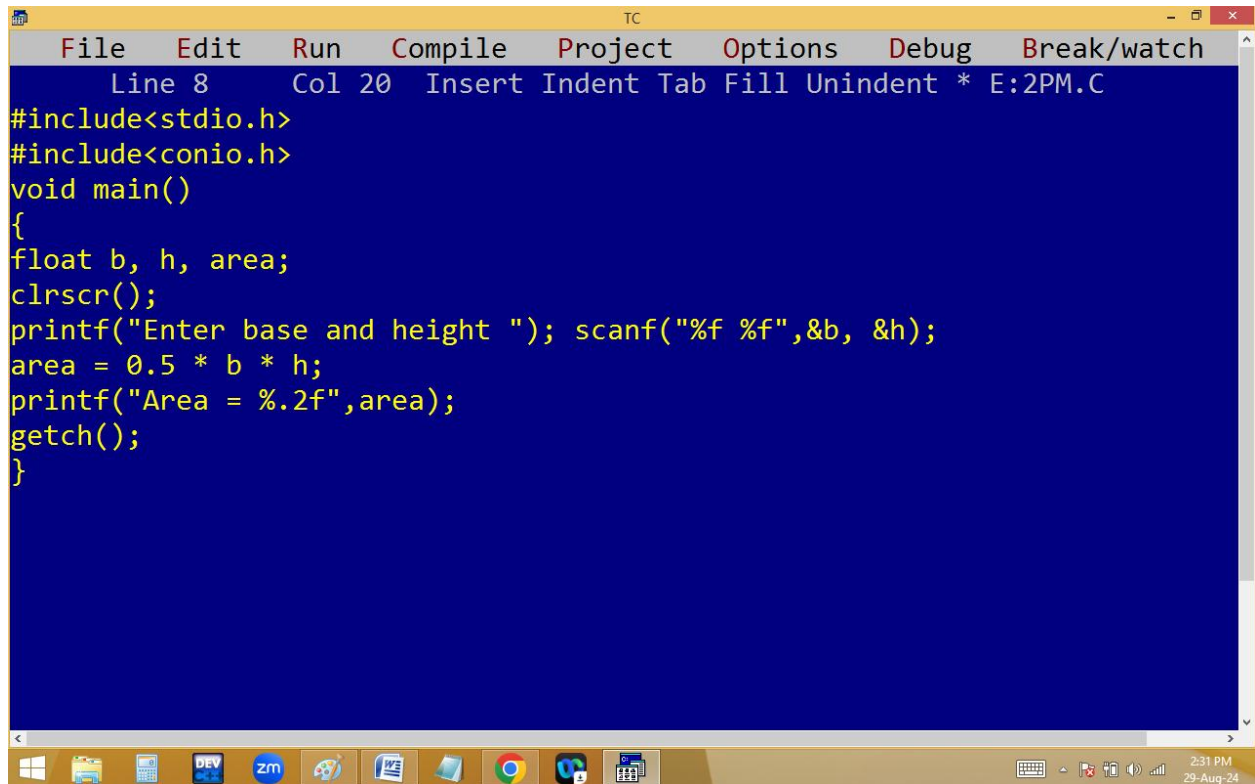
The screenshot shows the Turbo C++ (TC) IDE with a yellow title bar and menu bar. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates 'Line 5', 'Col 30', and 'Insert Indent Tab Fill Unindent * E:2PM.C'. The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
float b=4.5, h=5.75, area = 0.5 * b * h;
clrscr();
printf("Area = %.2f",area);
getch();
}
```

The Windows taskbar at the bottom shows various application icons, including Windows Explorer, DEV, zm, and others. The system clock in the bottom right corner displays '2:29 PM' and '29-Aug-24'.



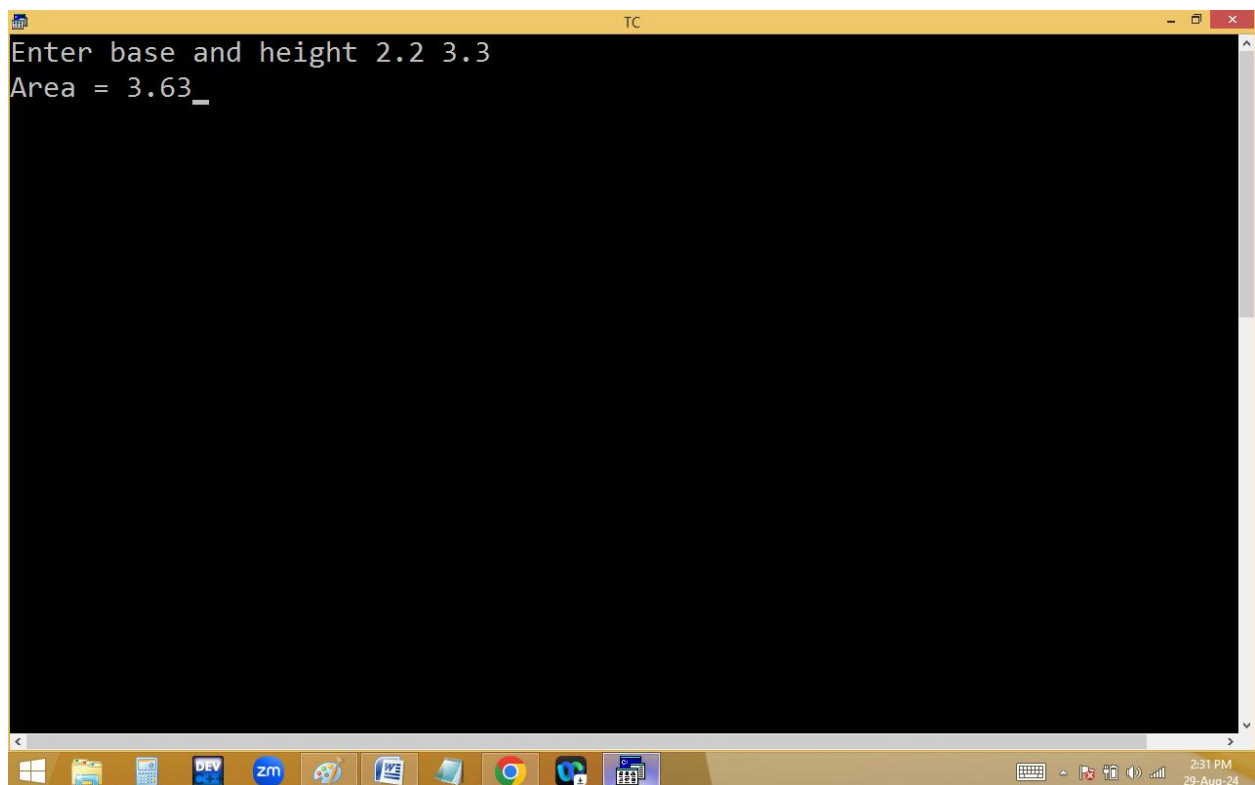
The screenshot shows the Turbo C++ (TC) IDE with a yellow title bar. The main editing area has a black background and displays the output of the program: 'Area = 12.94_'. The Windows taskbar at the bottom is identical to the first screenshot, showing the same application icons and system clock ('2:29 PM', '29-Aug-24').



The screenshot shows the Turbo C++ (TC) IDE with a yellow title bar and menu bar. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates 'Line 8 Col 20 Insert Indent Tab Fill Unindent * E:2PM.C'. The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
float b, h, area;
clrscr();
printf("Enter base and height "); scanf("%f %f",&b, &h);
area = 0.5 * b * h;
printf("Area = %.2f",area);
getch();
}
```

The Windows taskbar at the bottom shows various icons including the Start button, File Explorer, Calculator, DEV C++, Zm, a network icon, a printer icon, a folder icon, Google Chrome, a mail icon, and a calendar icon. The system tray on the right shows the time as 2:31 PM and the date as 29-Aug-24.



The screenshot shows the same Turbo C++ IDE window after the program has been executed. The output window, which has a black background, displays the following text:

```
Enter base and height 2.2 3.3
Area = 3.63_
```

The Windows taskbar and system tray are identical to the first screenshot, showing the same icons and system information (2:31 PM, 29-Aug-24).

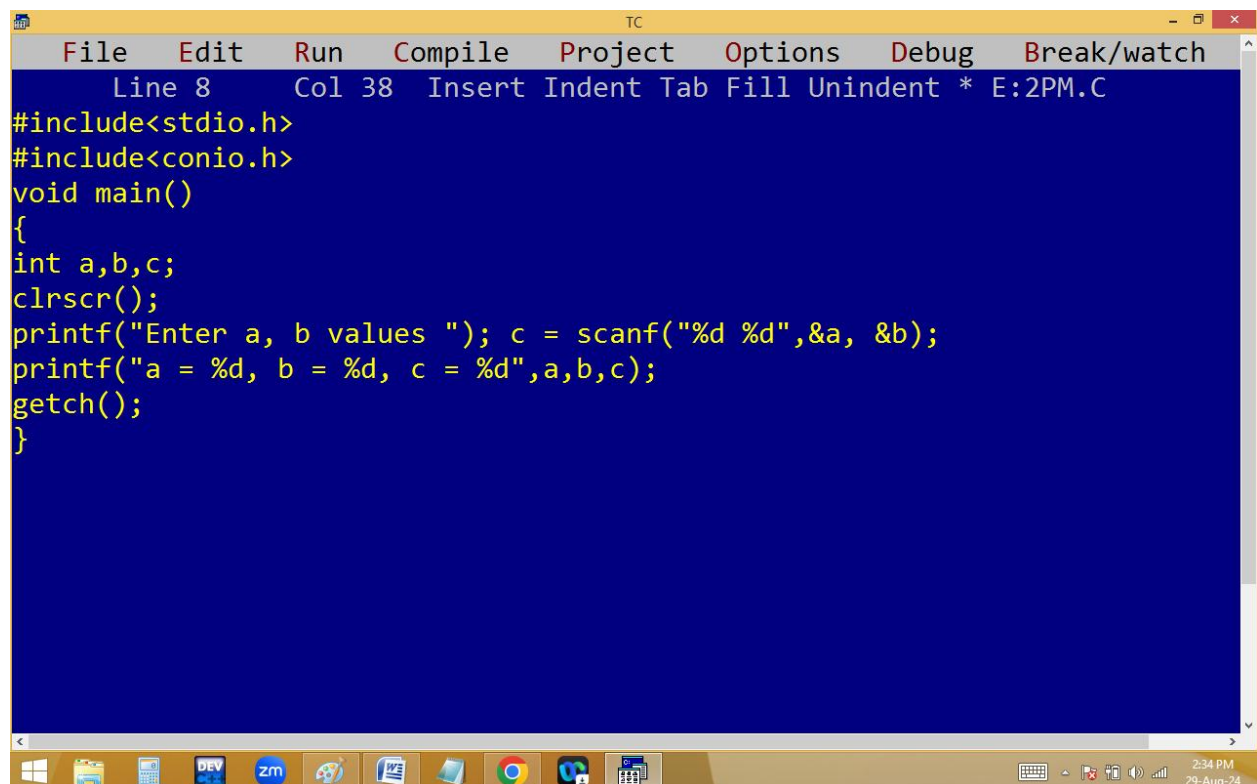
scanf(): It is the major input function in c.

it is available in stdio.h

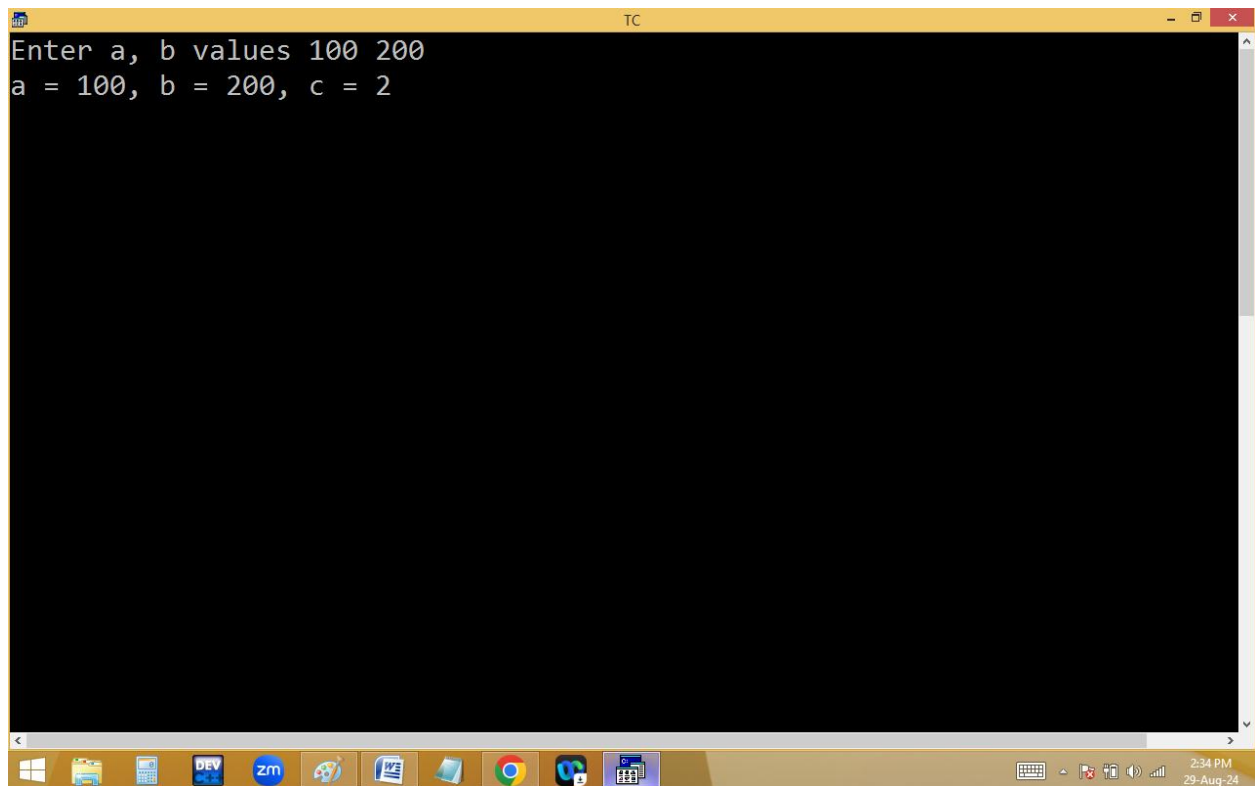
syntax:

int scanf("conversion characters", &variable, &var,...);

Finding no of conversion characters in scanf():

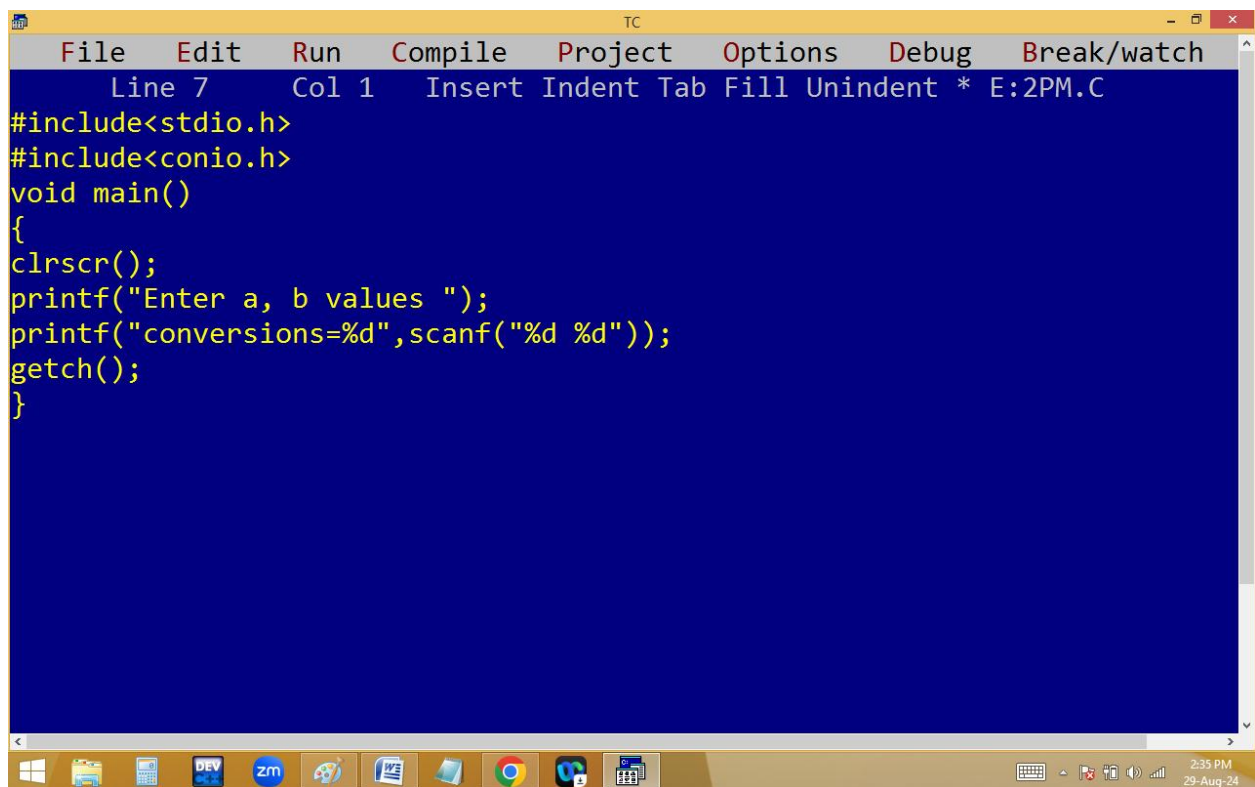


```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 8 Col 38 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a,b,c;
clrscr();
printf("Enter a, b values "); c = scanf("%d %d",&a, &b);
printf("a = %d, b = %d, c = %d",a,b,c);
getch();
}
```



A screenshot of a Turbo C++ (TC) console window. The window has a yellow title bar with the text "TC" and standard minimize, maximize, and close buttons. The main area is black with white text. The text displayed is "Enter a, b values 100 200" on the first line and "a = 100, b = 200, c = 2" on the second line. The Windows taskbar is visible at the bottom, showing icons for Windows, File Explorer, DEV, zm, and other applications. The system tray on the right shows the time as 2:34 PM and the date as 29-Aug-24.

```
Enter a, b values 100 200
a = 100, b = 200, c = 2
```



A screenshot of a Turbo C++ (TC) editor window. The window has a yellow title bar with the text "TC" and standard minimize, maximize, and close buttons. Below the title bar is a menu bar with the following items: File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. Below the menu bar is a status bar that reads "Line 7 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C". The main area has a blue background with yellow text. The code displayed is as follows:

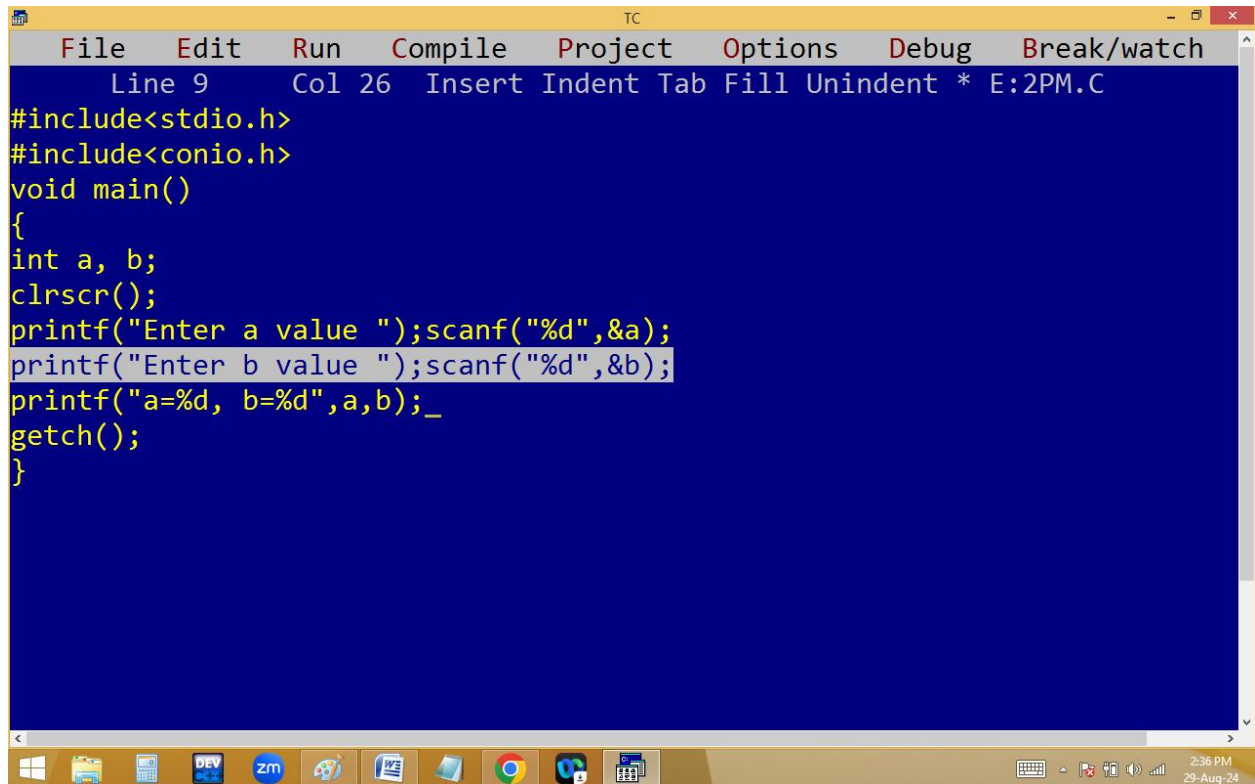
```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("Enter a, b values ");
printf("conversions=%d",scanf("%d %d"));
getch();
}
```

The Windows taskbar is visible at the bottom, showing icons for Windows, File Explorer, DEV, zm, and other applications. The system tray on the right shows the time as 2:35 PM and the date as 29-Aug-24.



```
Enter a, b values 3 9
conversions=2
```

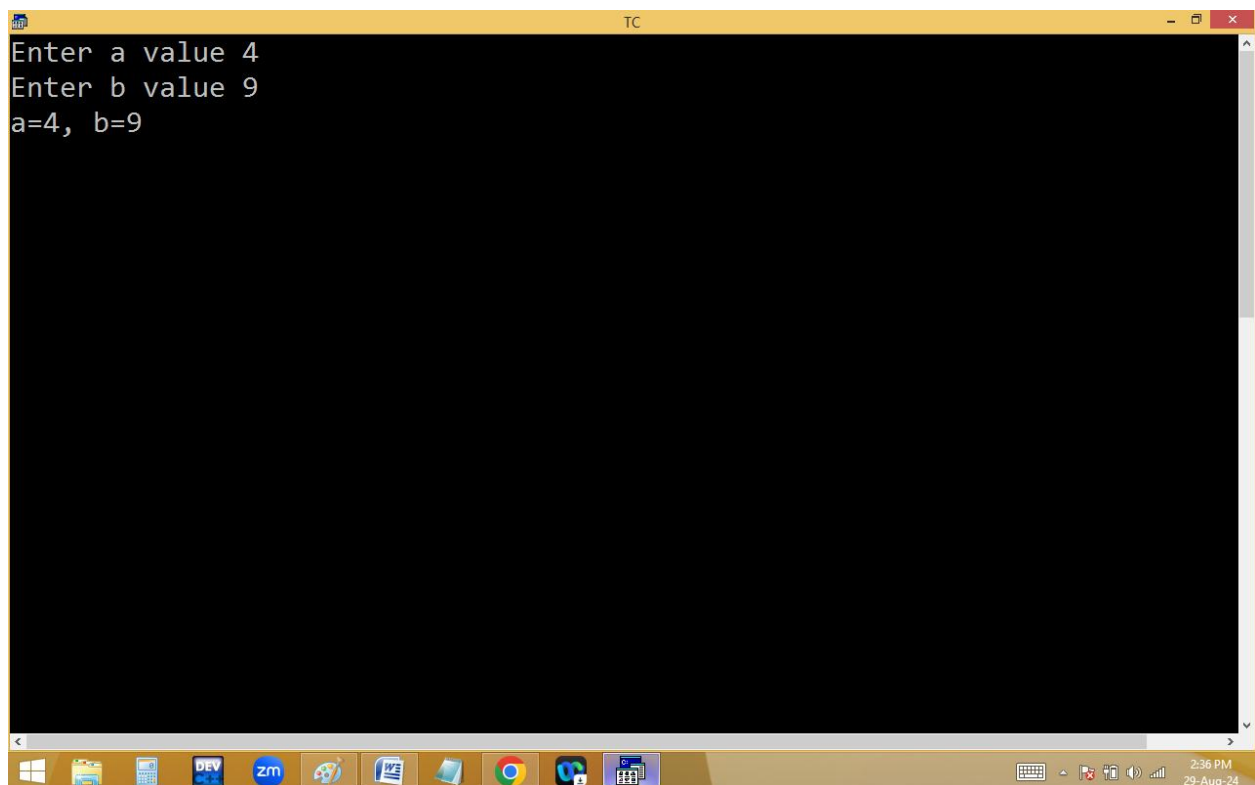
How to control inputs in scanf():



The screenshot shows the Turbo C++ (TC) IDE with a yellow title bar and menu bar. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates 'Line 9 Col 26 Insert Indent Tab Fill Unindent * E:2PM.C'. The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a, b;
clrscr();
printf("Enter a value ");scanf("%d",&a);
printf("Enter b value ");scanf("%d",&b);
printf("a=%d, b=%d",a,b);_
getch();
}
```

The taskbar at the bottom shows various application icons including Windows Explorer, DEV, zm, and others, along with system icons for keyboard, volume, and network, and a clock showing 2:36 PM on 29-Aug-24.

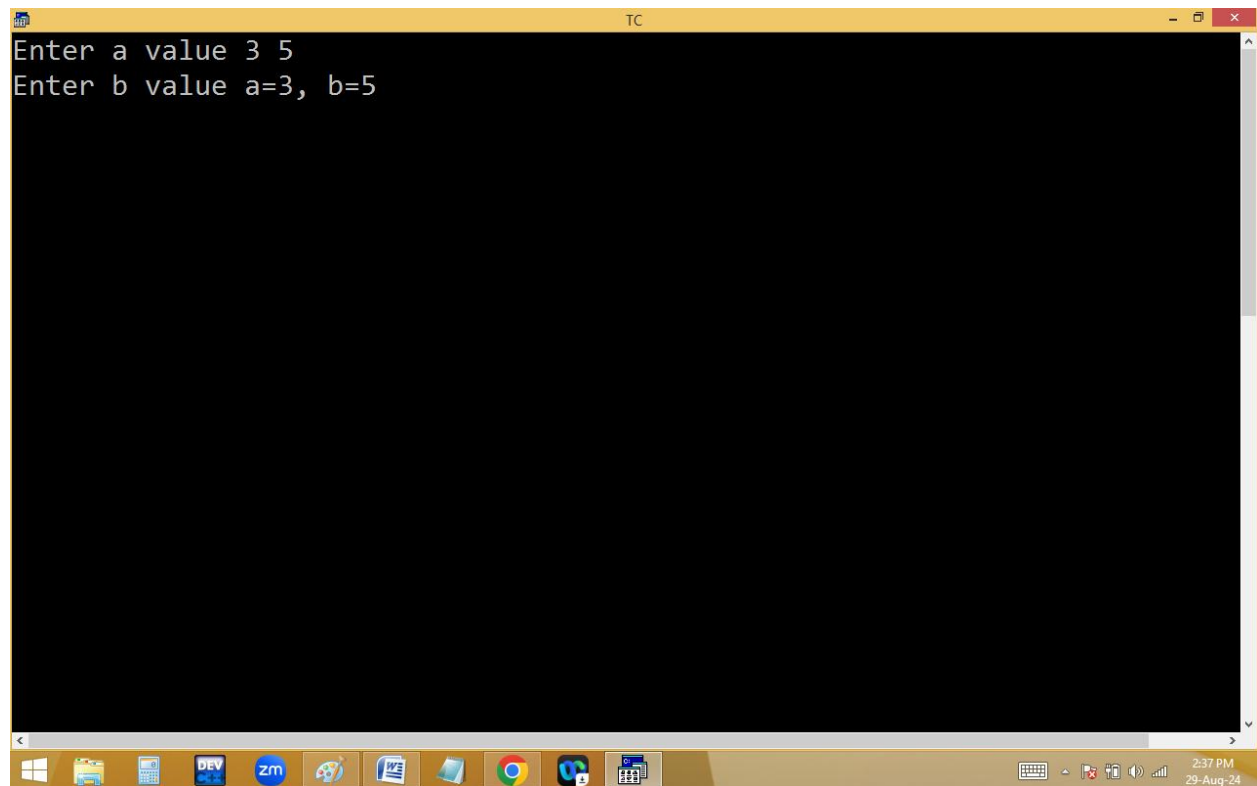


The screenshot shows the Turbo C++ (TC) IDE with a yellow title bar. The main editing area has a black background and displays the output of the program:

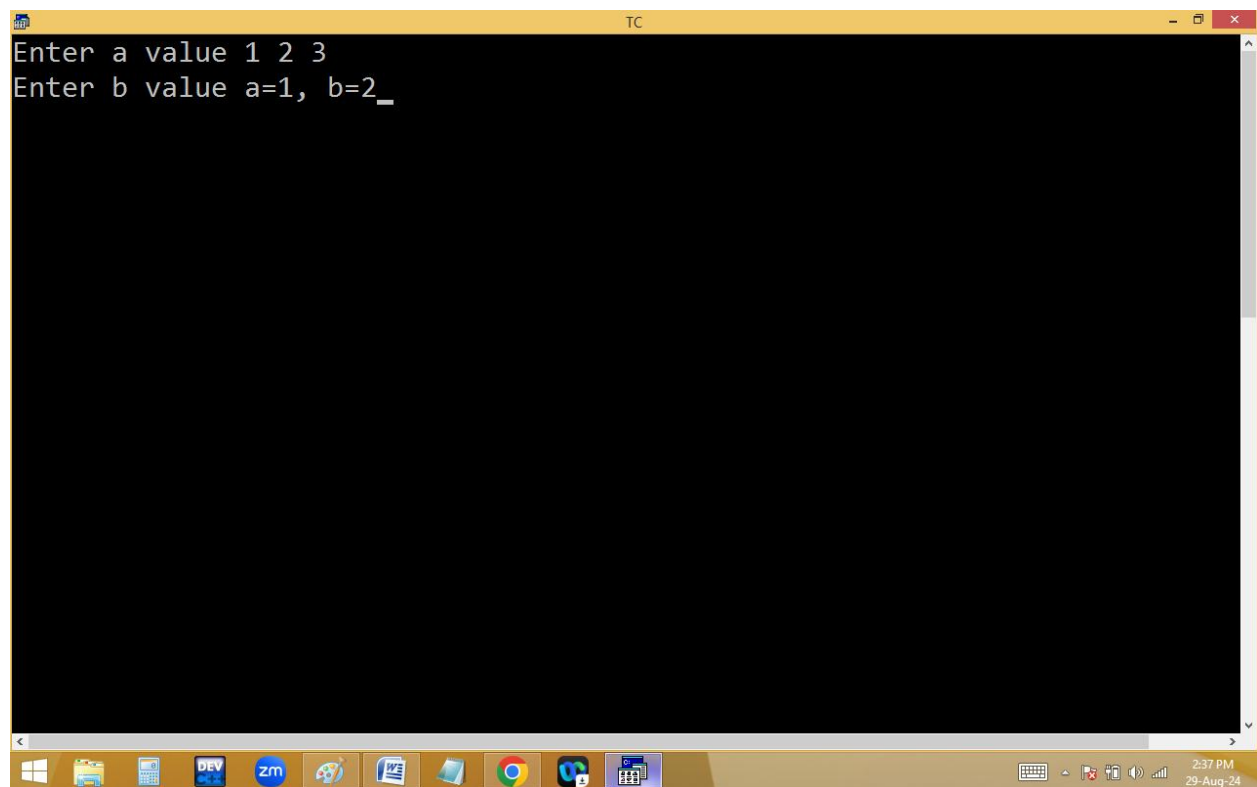
```
Enter a value 4
Enter b value 9
a=4, b=9
```

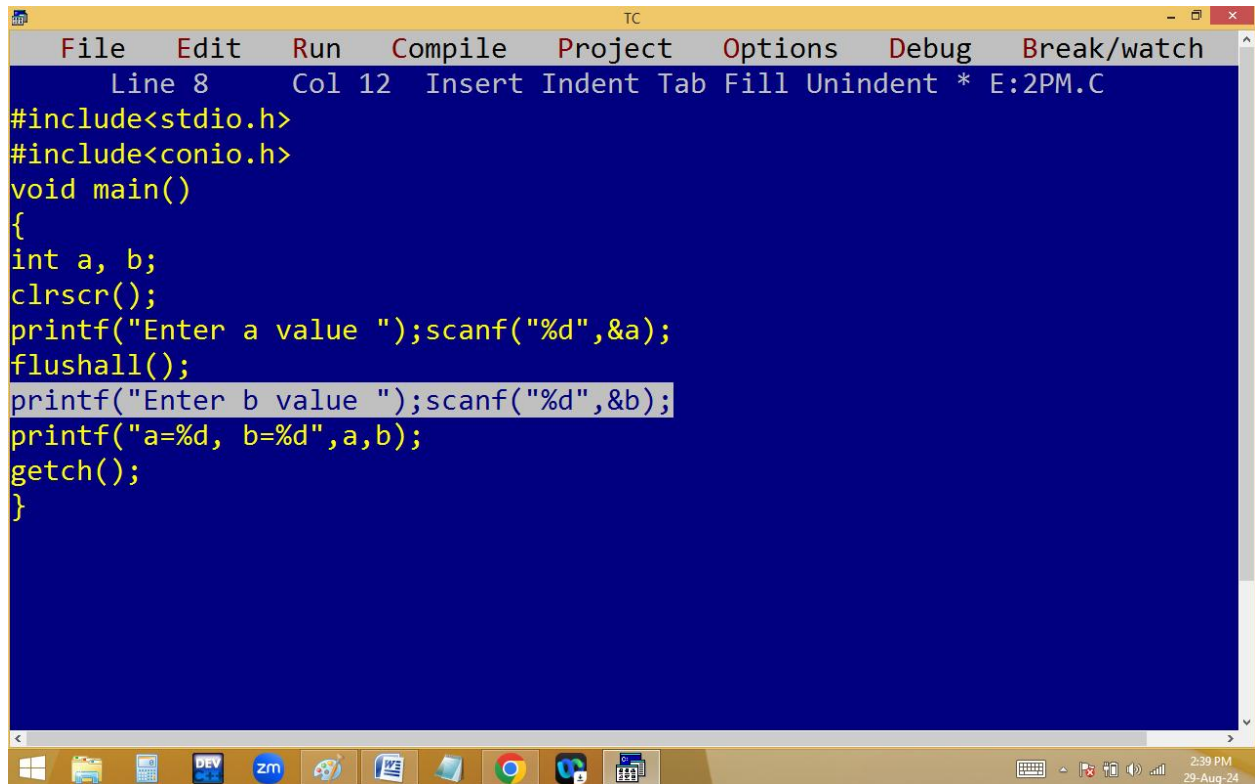
The taskbar at the bottom is identical to the first screenshot, showing the same application icons and system status information (2:36 PM on 29-Aug-24).

```
TC
Enter a value 3 5
Enter b value a=3, b=5
```



```
TC
Enter a value 1 2 3
Enter b value a=1, b=2_
```

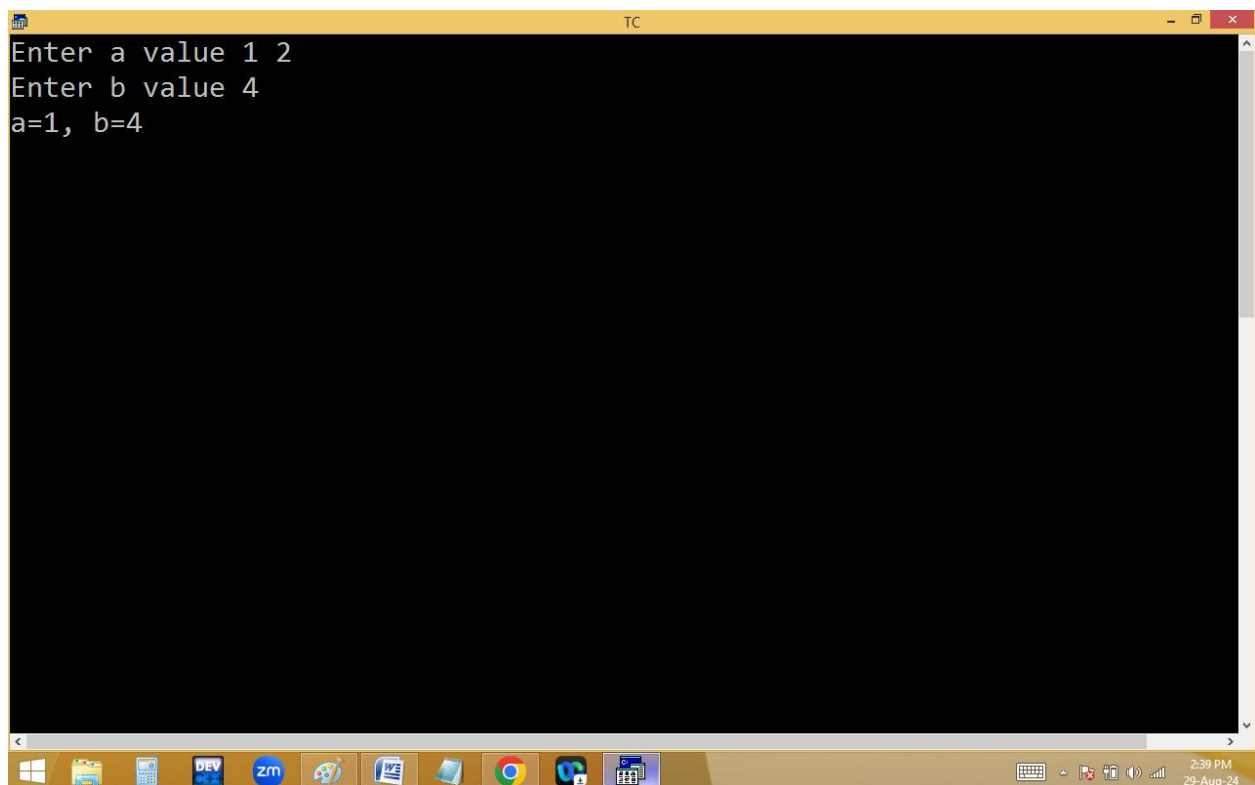




The screenshot shows the Turbo C++ (TC) IDE with a menu bar (File, Edit, Run, Compile, Project, Options, Debug, Break/watch) and a status bar (Line 8, Col 12, Insert, Indent, Tab, Fill, Unindent, * E:2PM.C). The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a, b;
    clrscr();
    printf("Enter a value ");scanf("%d",&a);
    flushall();
    printf("Enter b value ");scanf("%d",&b);
    printf("a=%d, b=%d",a,b);
    getch();
}
```

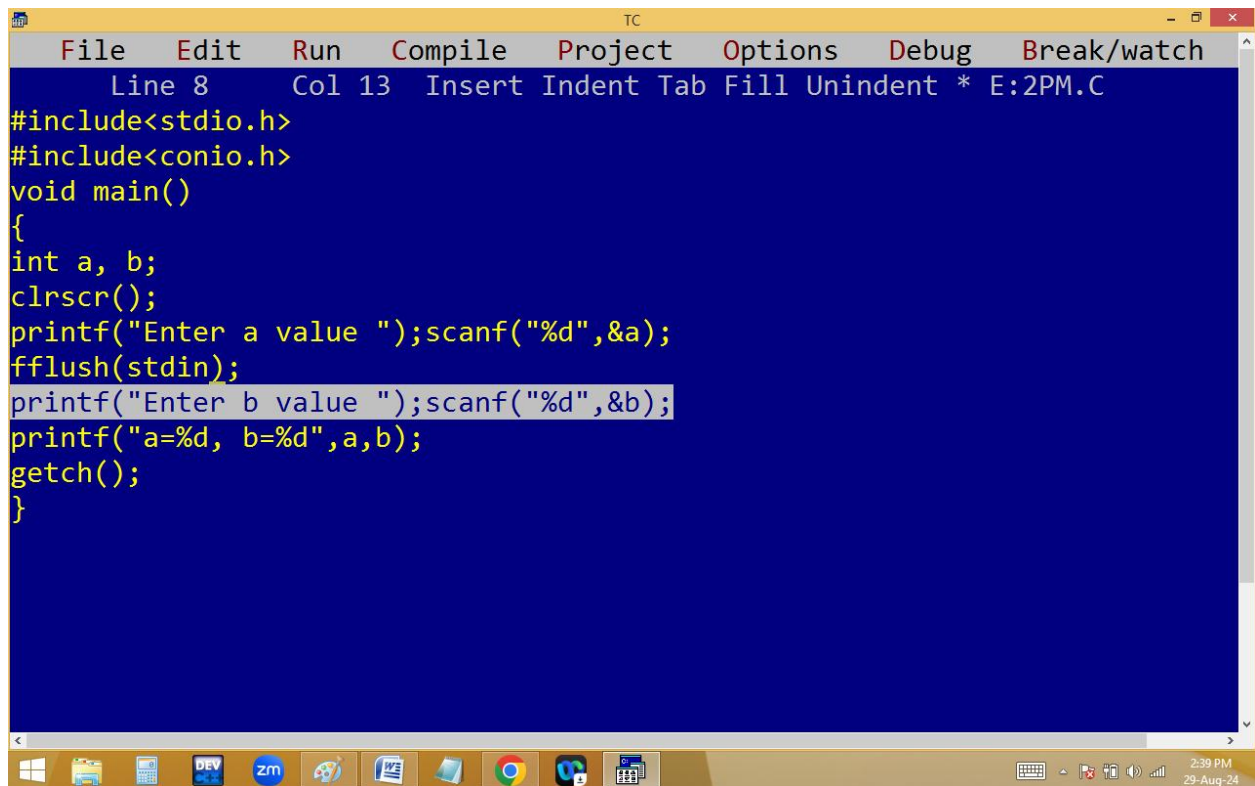
The taskbar at the bottom shows various application icons and the system clock indicating 2:39 PM on 29-Aug-24.



The screenshot shows the Turbo C++ (TC) IDE with the same menu bar and status bar. The output window displays the following text:

```
Enter a value 1 2
Enter b value 4
a=1, b=4
```

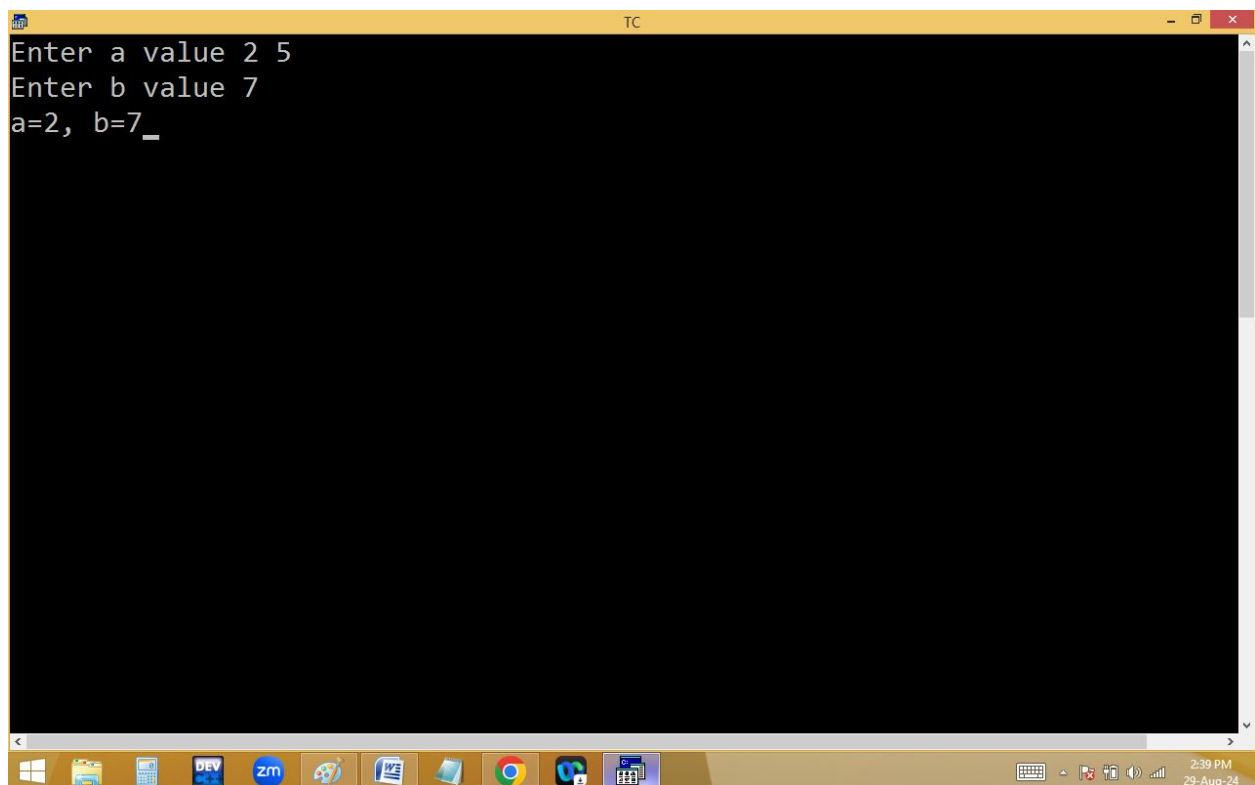
The taskbar at the bottom is identical to the first screenshot, showing the same application icons and system clock (2:39 PM on 29-Aug-24).



The screenshot shows the Turbo C++ (TC) IDE with a blue background. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates 'Line 8 Col 13 Insert Indent Tab Fill Unindent * E:2PM.C'. The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a, b;
clrscr();
printf("Enter a value ");scanf("%d",&a);
fflush(stdin);
printf("Enter b value ");scanf("%d",&b);
printf("a=%d, b=%d",a,b);
getch();
}
```

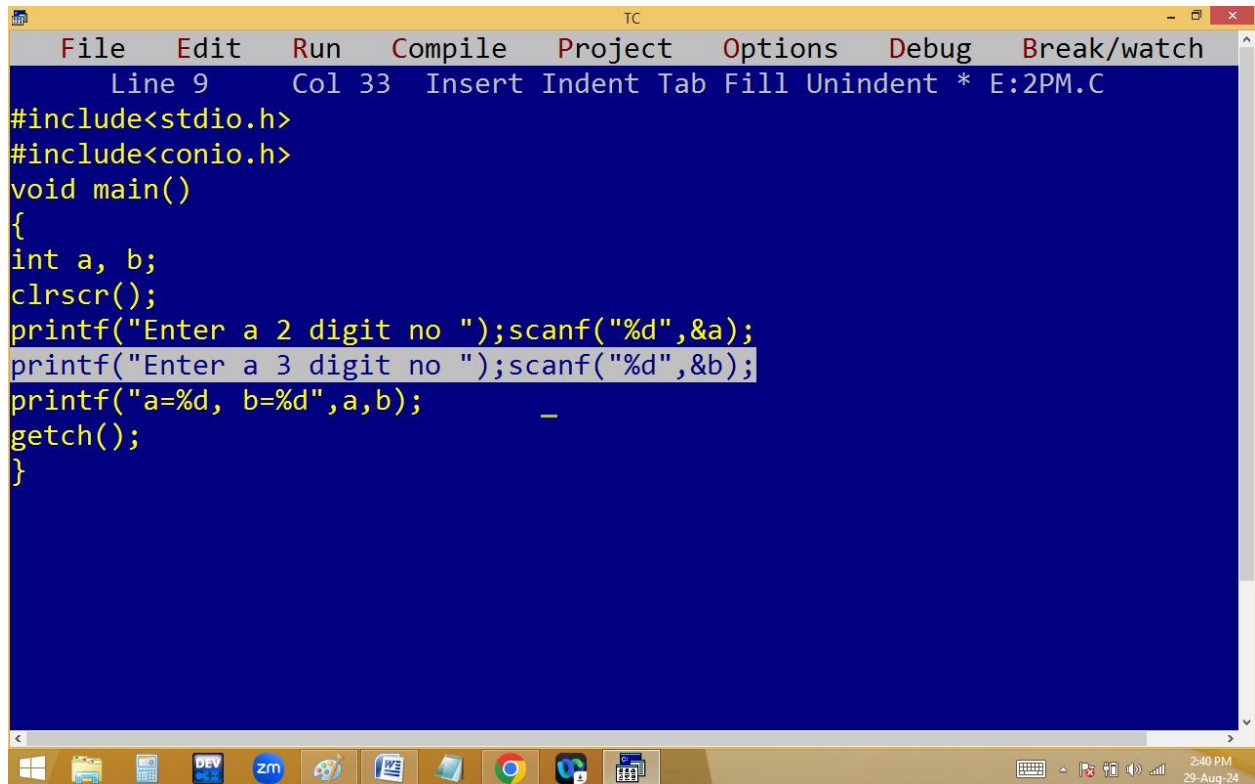
The taskbar at the bottom shows various icons including Windows, File Explorer, DEV, zm, and others. The system clock in the bottom right corner shows 2:39 PM on 29-Aug-24.



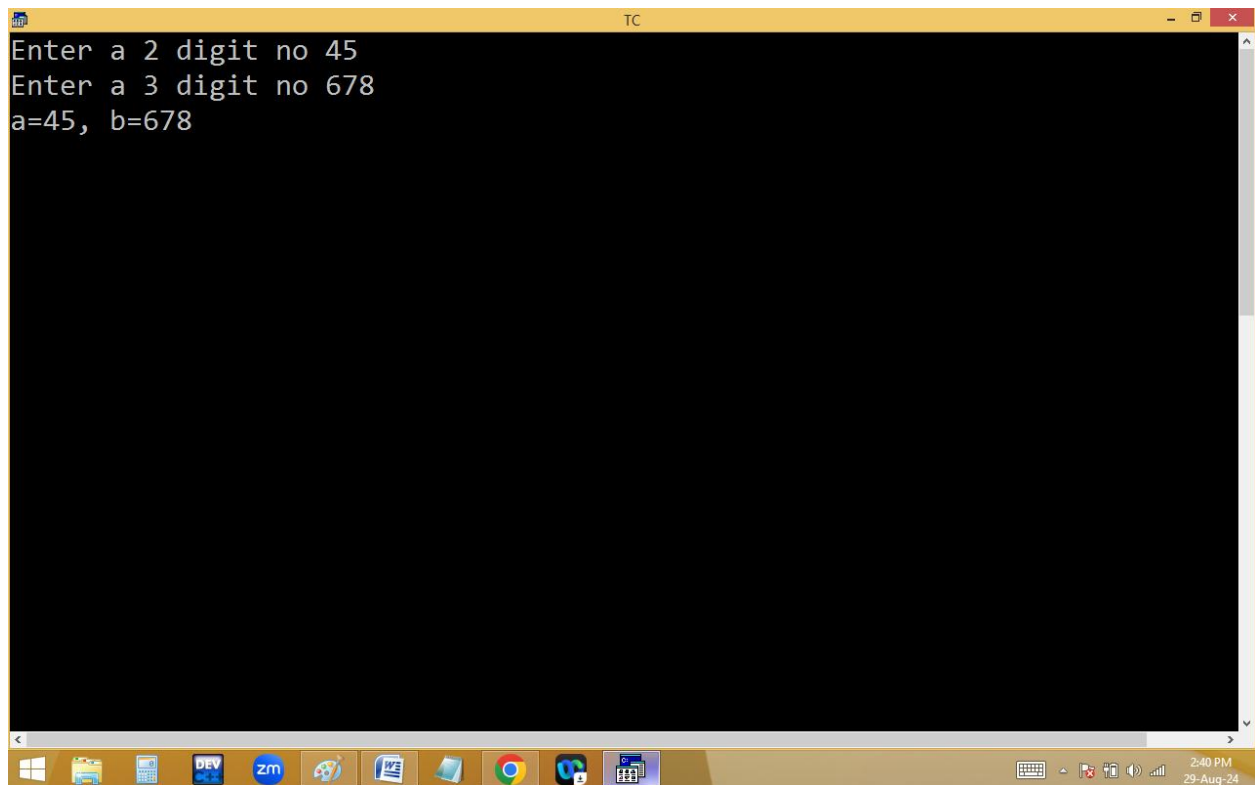
The screenshot shows the same Turbo C++ IDE window, but now displaying the program's output. The text in the window is:

```
Enter a value 2 5
Enter b value 7
a=2, b=7_
```

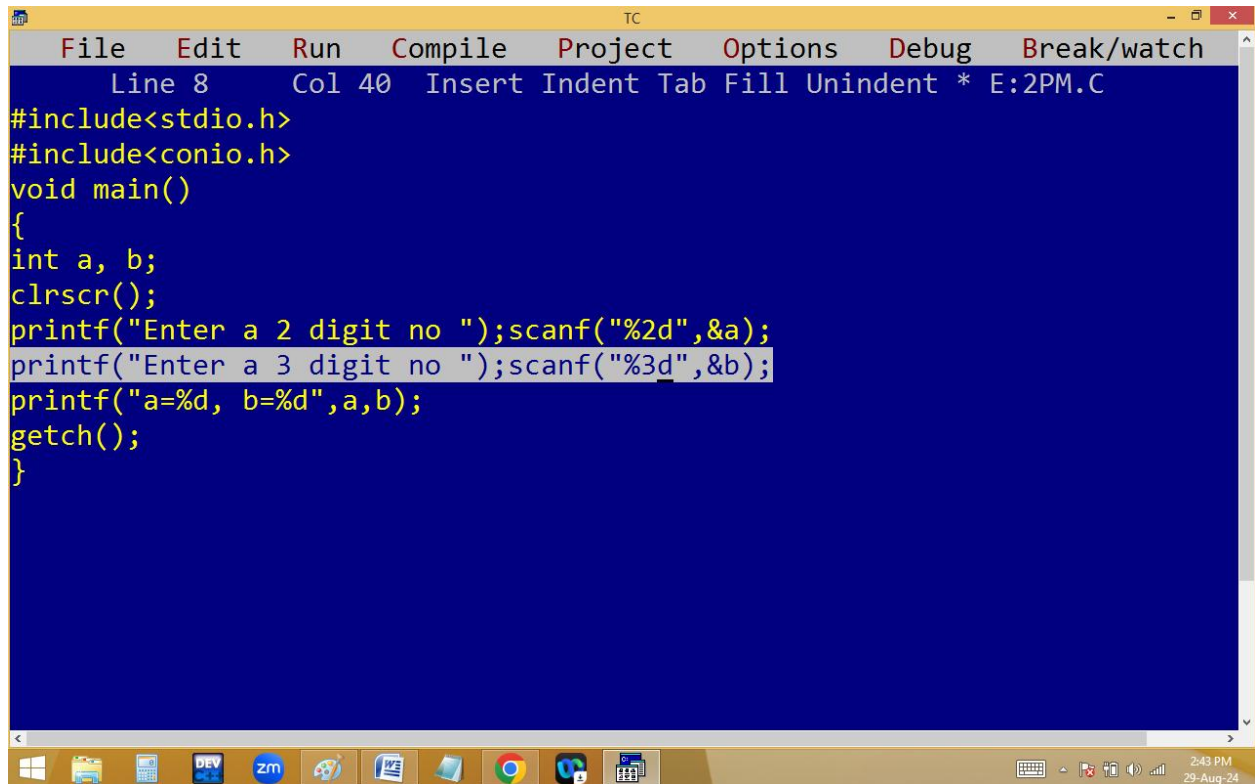
The taskbar and system clock are identical to the first screenshot, showing 2:39 PM on 29-Aug-24.



```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 9 Col 33 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a, b;
clrscr();
printf("Enter a 2 digit no ");scanf("%d",&a);
printf("Enter a 3 digit no ");scanf("%d",&b);
printf("a=%d, b=%d",a,b);
getch();
}
```



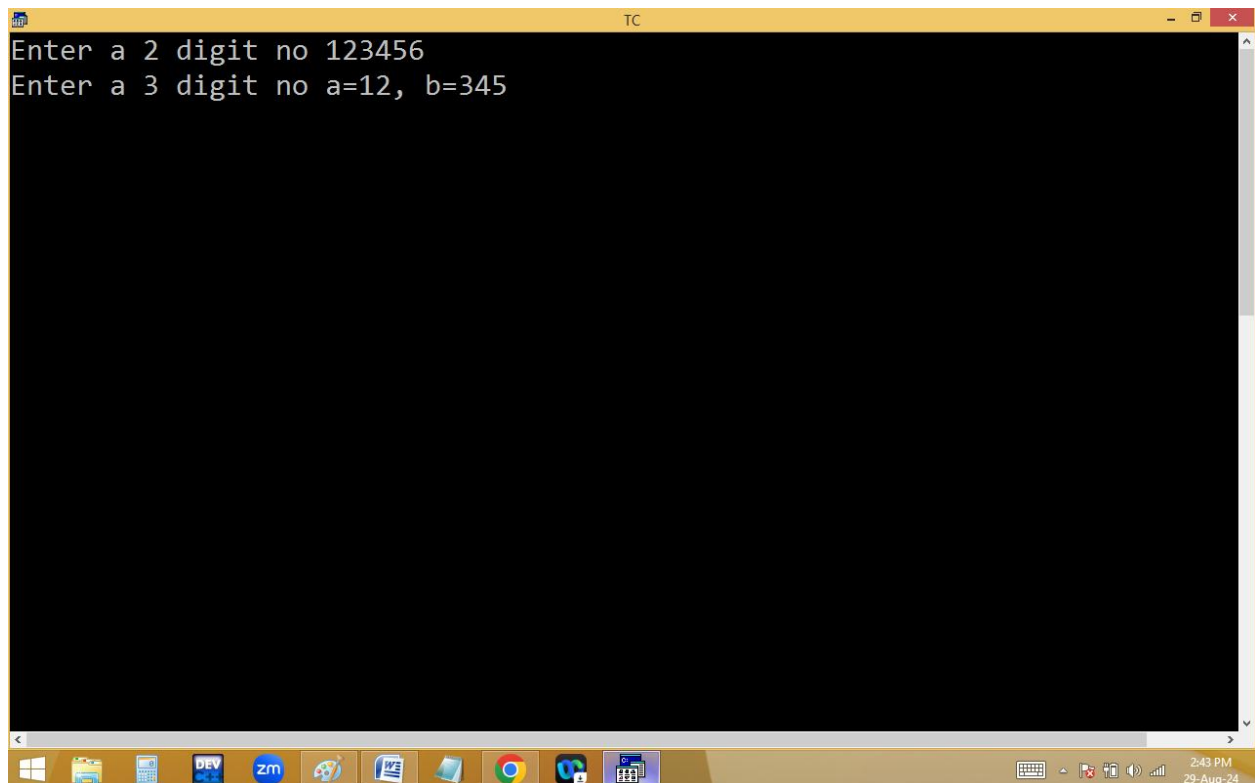
```
TC
Enter a 2 digit no 45
Enter a 3 digit no 678
a=45, b=678
```



The screenshot shows the Turbo C++ (TC) IDE with a yellow title bar and menu bar. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates 'Line 8 Col 40 Insert Indent Tab Fill Unindent * E:2PM.C'. The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a, b;
clrscr();
printf("Enter a 2 digit no ");scanf("%2d",&a);
printf("Enter a 3 digit no ");scanf("%3d",&b);
printf("a=%d, b=%d",a,b);
getch();
}
```

The taskbar at the bottom shows various application icons including Windows Explorer, DEV, zm, and others, along with system icons for keyboard, network, and volume. The system clock shows 2:43 PM on 29-Aug-24.

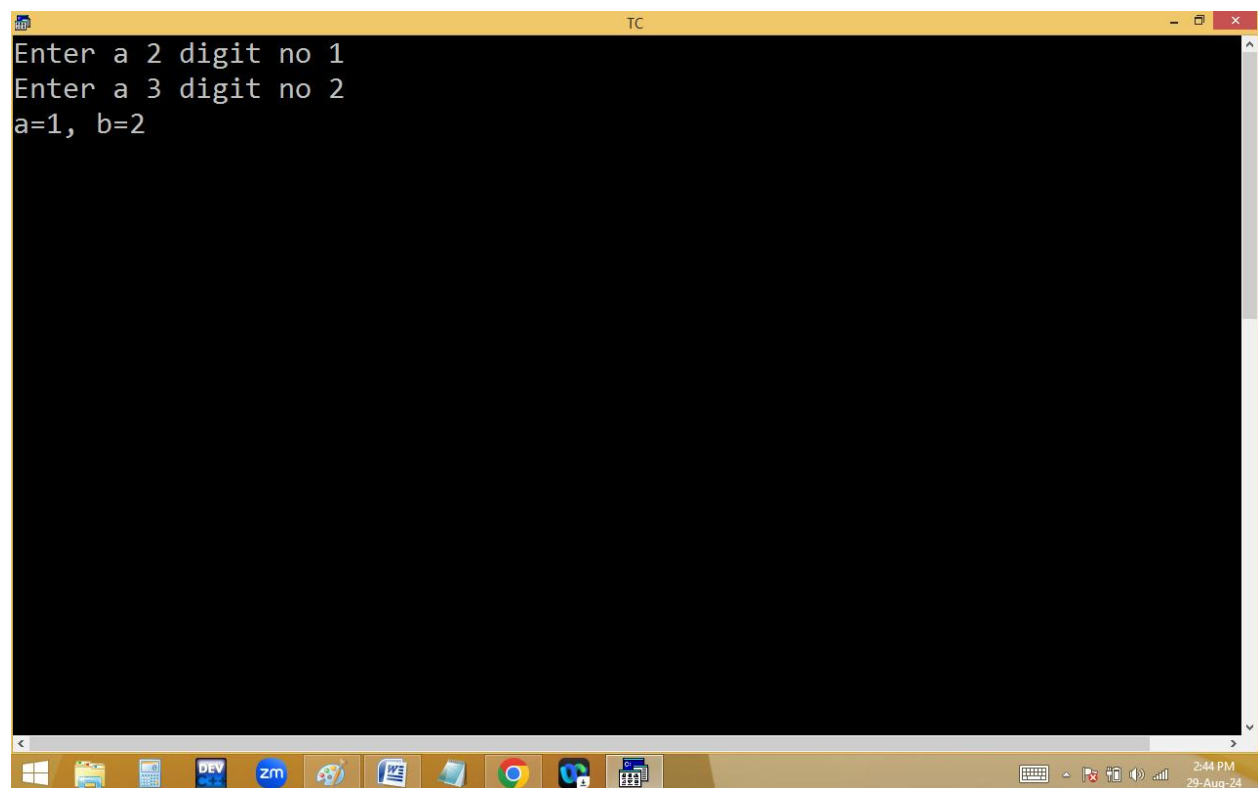


The screenshot shows the Turbo C++ (TC) IDE with a yellow title bar and menu bar. The main editing area has a black background and displays the output of the program execution:

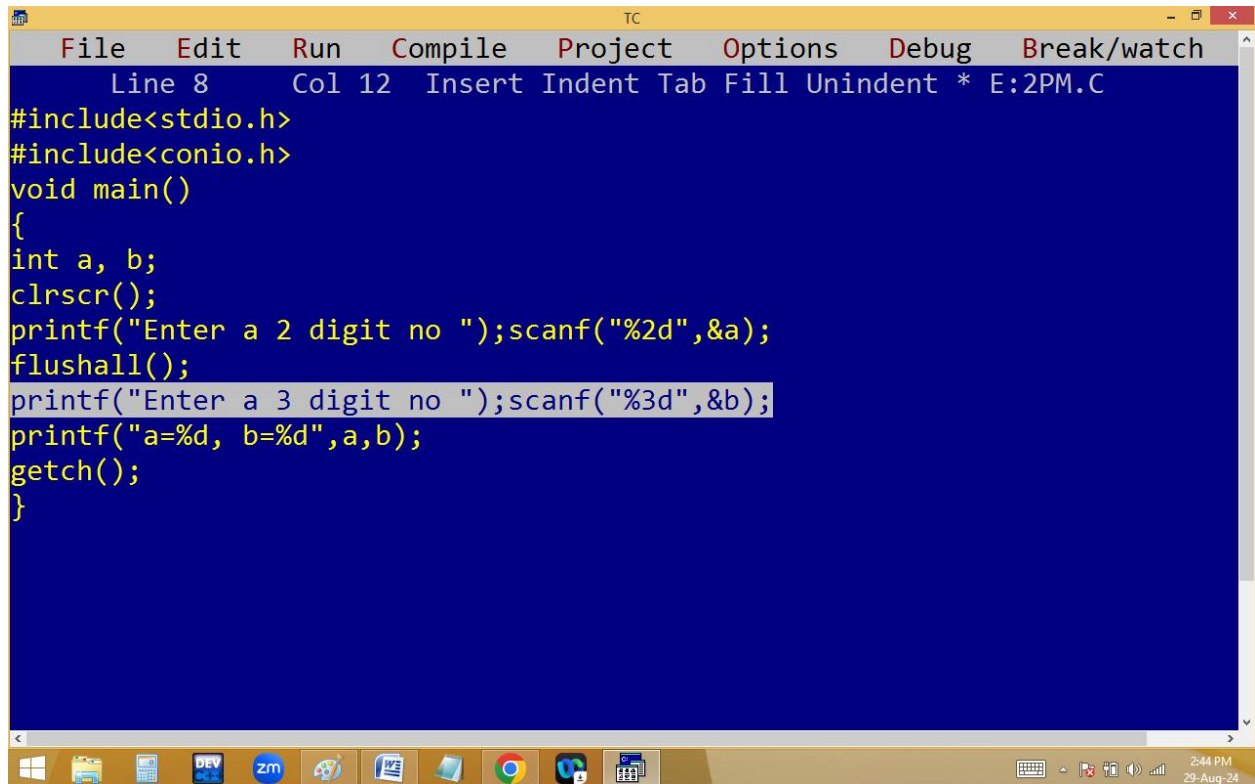
```
Enter a 2 digit no 123456
Enter a 3 digit no a=12, b=345
```

The taskbar at the bottom is identical to the first screenshot, showing the same application icons and system clock (2:43 PM on 29-Aug-24).

```
TC
Enter a 2 digit no 1
Enter a 3 digit no 2
a=1, b=2
```



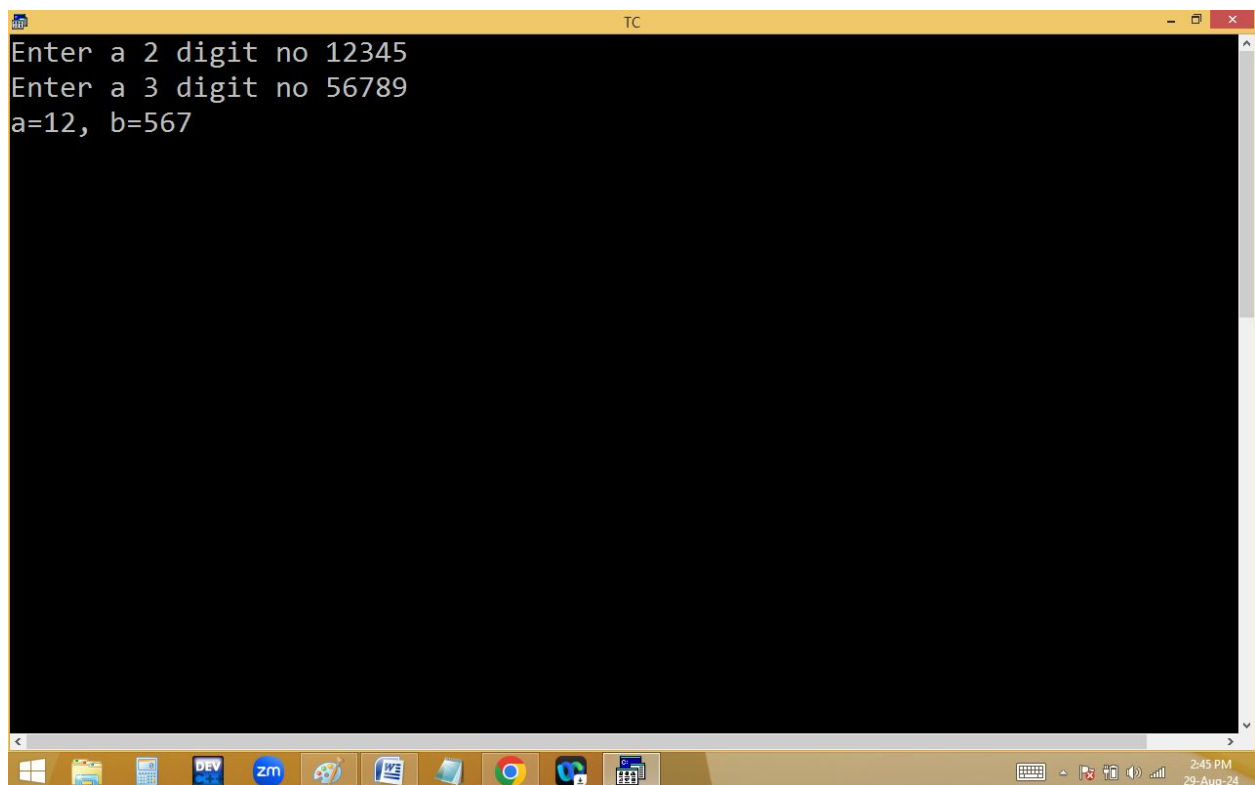
The image shows a Windows 10 desktop environment. A terminal window titled "TC" is open, displaying the following text: "Enter a 2 digit no 1", "Enter a 3 digit no 2", and "a=1, b=2". The taskbar at the bottom contains several icons, including the Start button, File Explorer, Calculator, DEV, Zoom (zm), a game controller, a document, a folder, Google Chrome, a blue icon, and a calendar. The system tray on the right shows the time as 2:44 PM and the date as 29-Aug-24.



The screenshot shows the Turbo C++ (TC) IDE with a yellow title bar and menu bar. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates 'Line 8 Col 12 Insert Indent Tab Fill Unindent * E:2PM.C'. The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a, b;
    clrscr();
    printf("Enter a 2 digit no ");scanf("%2d",&a);
    flushall();
    printf("Enter a 3 digit no ");scanf("%3d",&b);
    printf("a=%d, b=%d",a,b);
    getch();
}
```

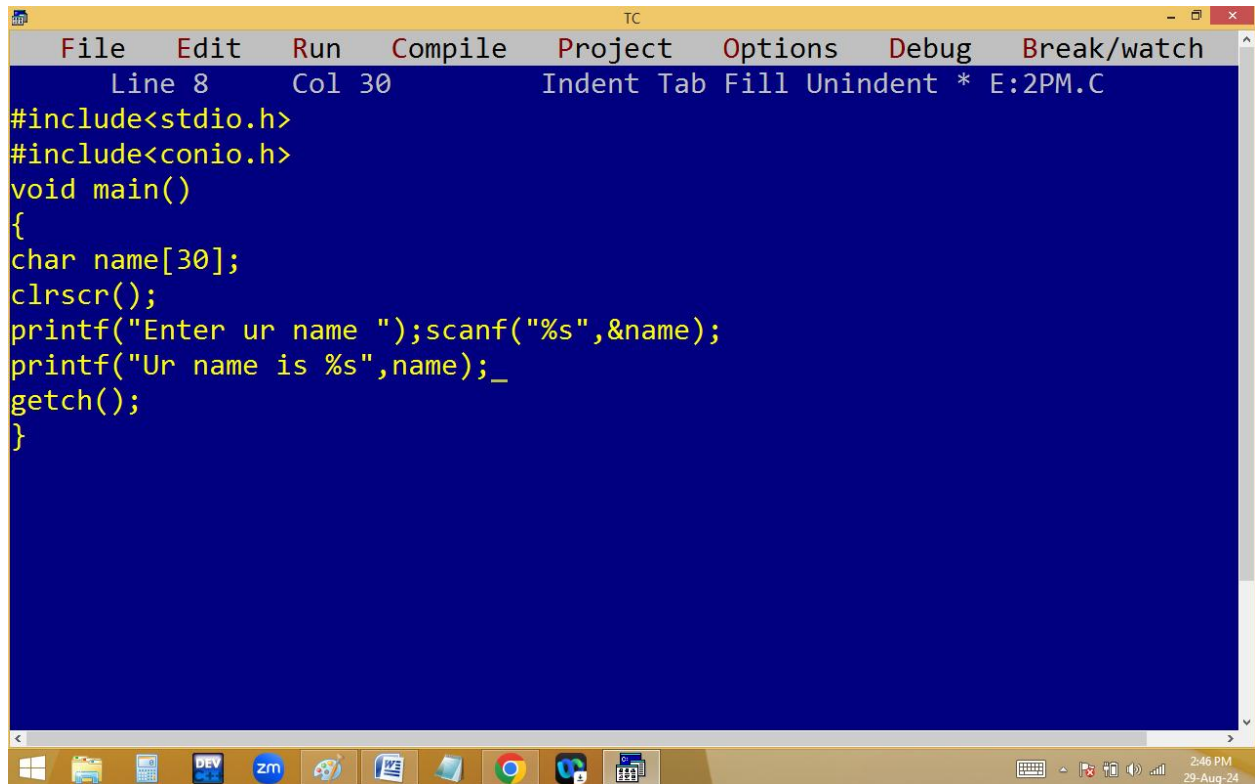
The taskbar at the bottom shows various application icons including Windows Explorer, DEV, zm, and others. The system clock in the bottom right corner shows 2:44 PM on 29-Aug-24.



The screenshot shows the same Turbo C++ IDE window, but now displaying the program's output. The text in the window is:

```
Enter a 2 digit no 12345
Enter a 3 digit no 56789
a=12, b=567
```

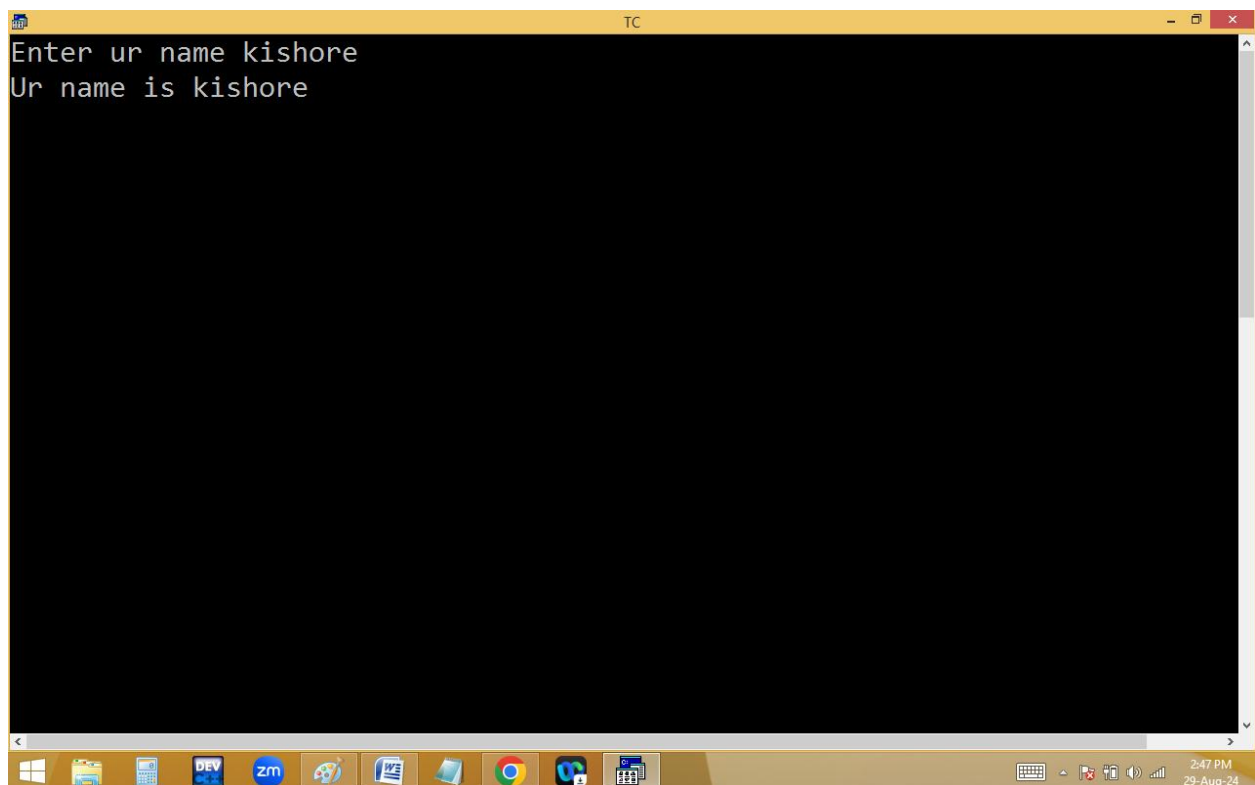
The taskbar and system clock are identical to the first screenshot, showing 2:45 PM on 29-Aug-24.



The screenshot shows the Turbo C++ (TC) IDE with a blue background. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates 'Line 8', 'Col 30', and 'Indent Tab Fill Unindent * E:2PM.C'. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
char name[30];
clrscr();
printf("Enter ur name ");scanf("%s",&name);
printf("Ur name is %s",name);_
getch();
}
```

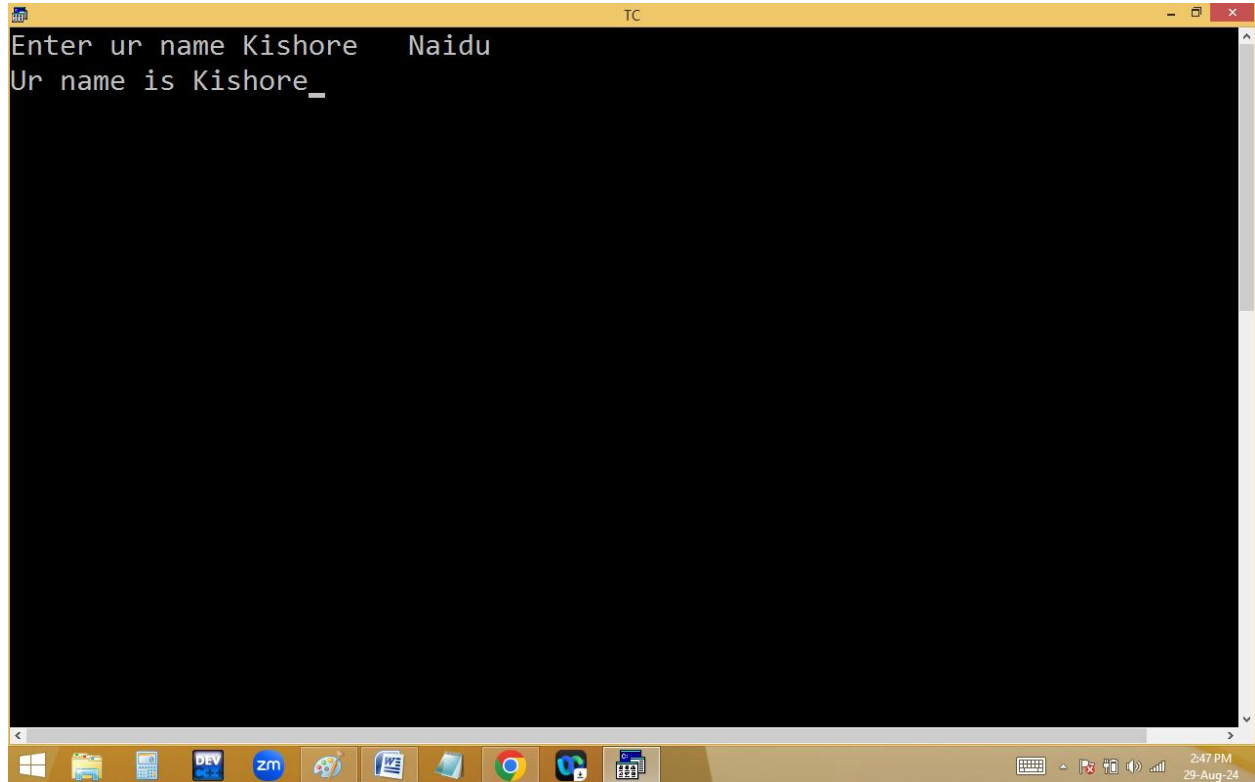
The Windows taskbar at the bottom shows various icons including Windows, File Explorer, DEV, zm, and others. The system clock in the bottom right corner displays '2:46 PM' and '29-Aug-24'.



The screenshot shows the same Turbo C++ (TC) IDE window after the program has been executed. The output window now displays the text:

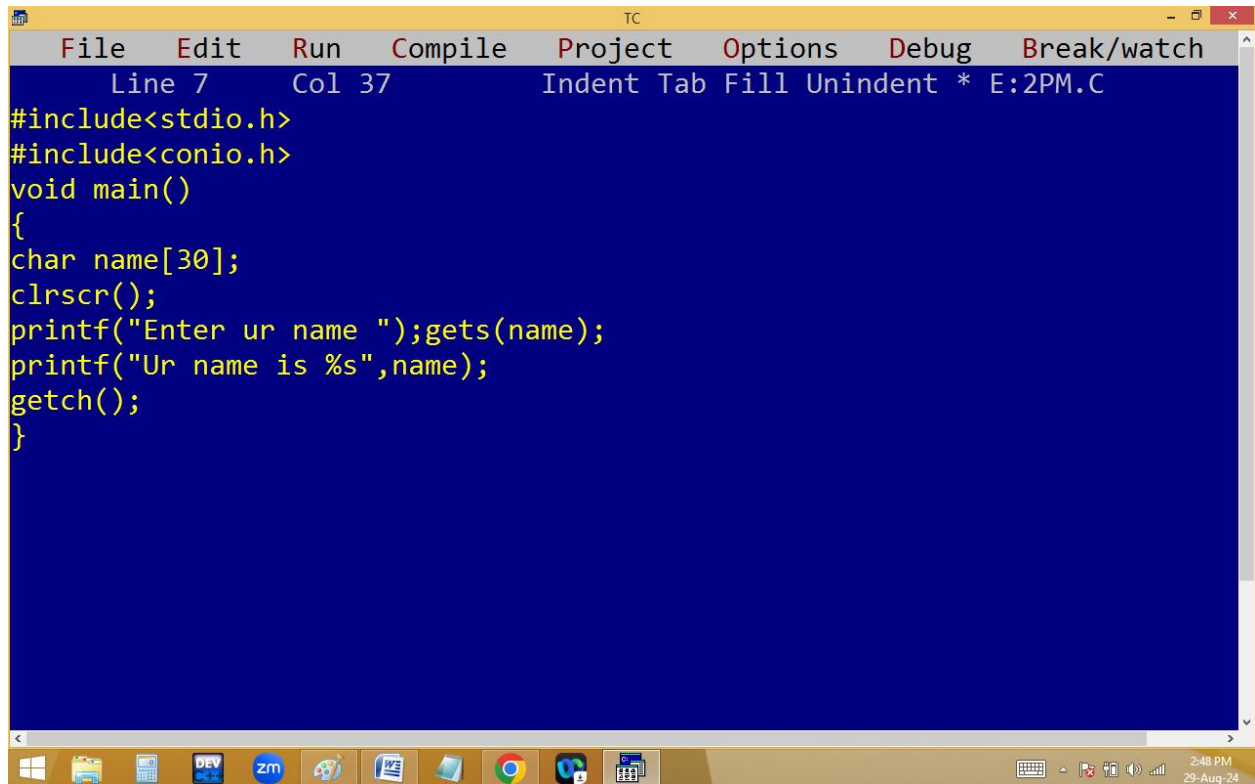
```
Enter ur name kishore
Ur name is kishore
```

The rest of the IDE interface, including the menu bar and taskbar, remains the same as in the previous screenshot. The system clock in the bottom right corner now displays '2:47 PM' and '29-Aug-24'.

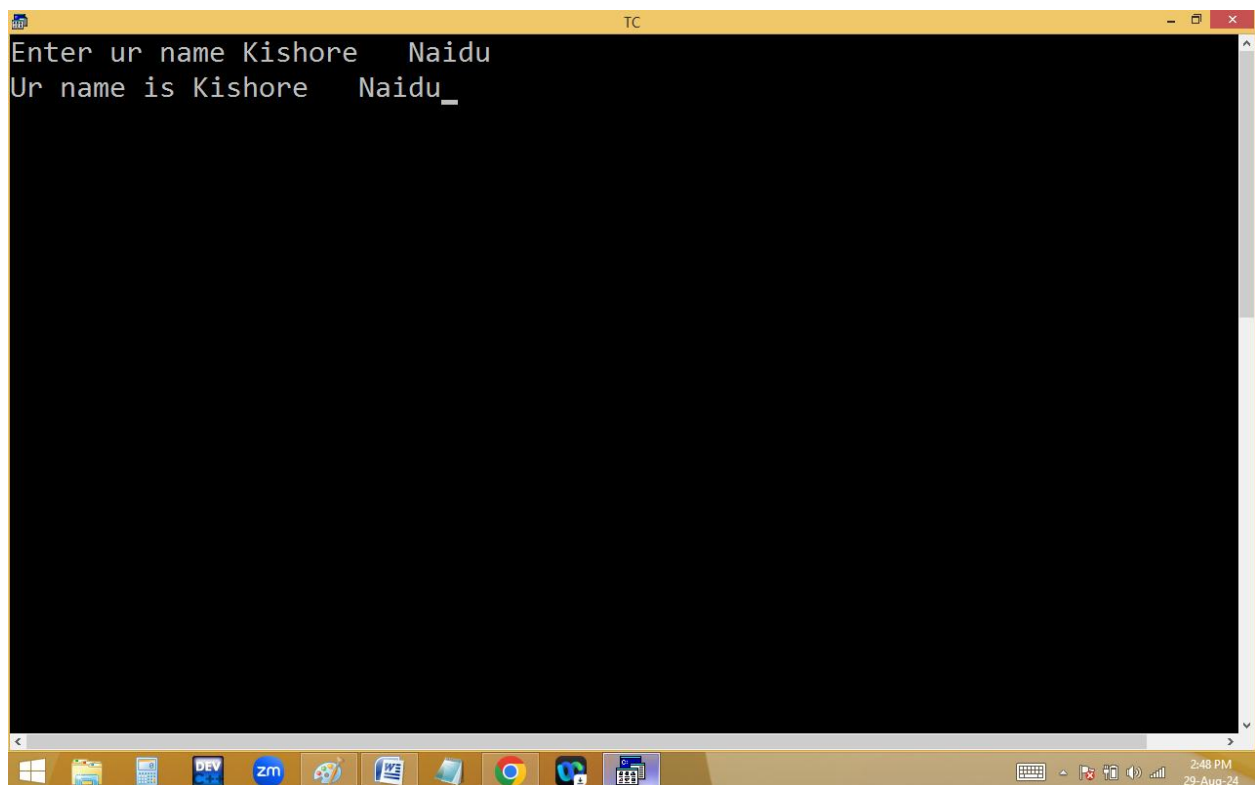


```
TC
Enter ur name Kishore  Naidu
Ur name is Kishore_
```

gets(): It reads the string with spaces until pressing of enter key.



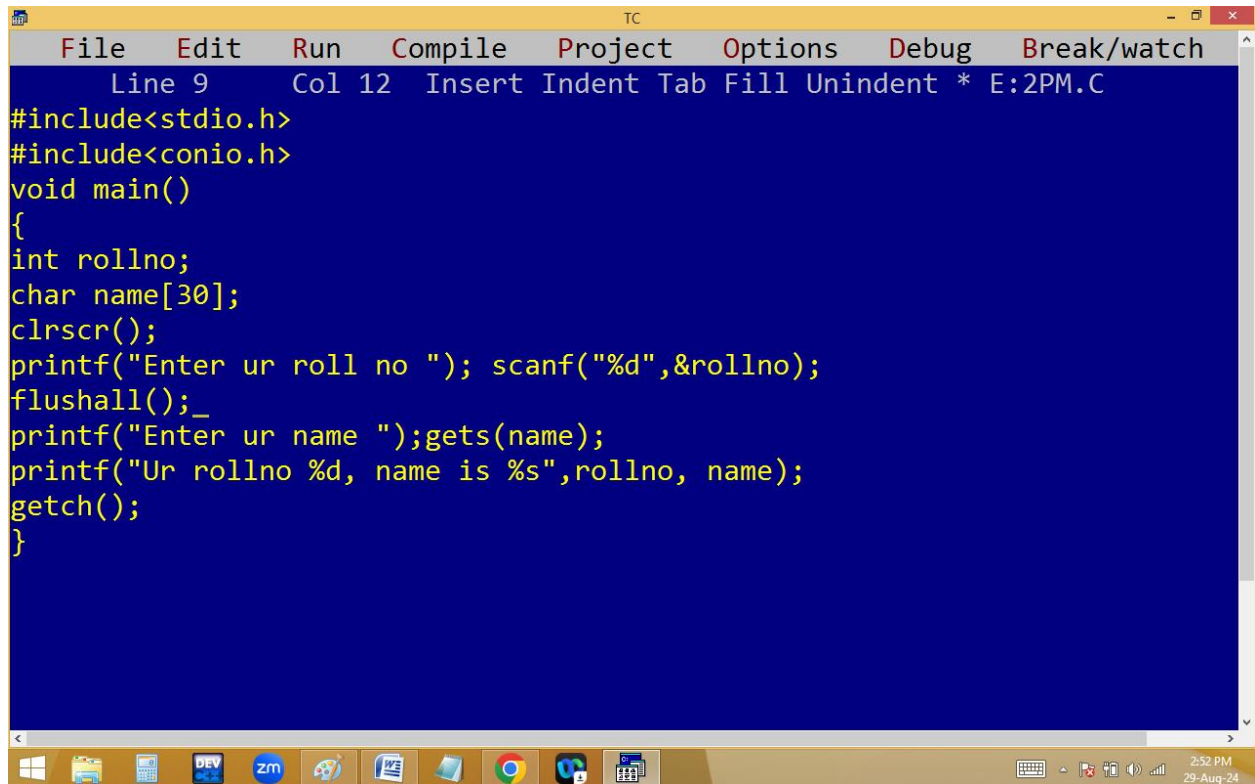
```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 7 Col 37 Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char name[30];
clrscr();
printf("Enter ur name ");gets(name);
printf("Ur name is %s",name);
getch();
}
```



```
TC
Enter ur name Kishore Naidu
Ur name is Kishore Naidu_
```

```
TC
Enter ur name Sachin    Ramesh    Tendulkar
Ur name is Sachin    Ramesh    Tendulkar
```

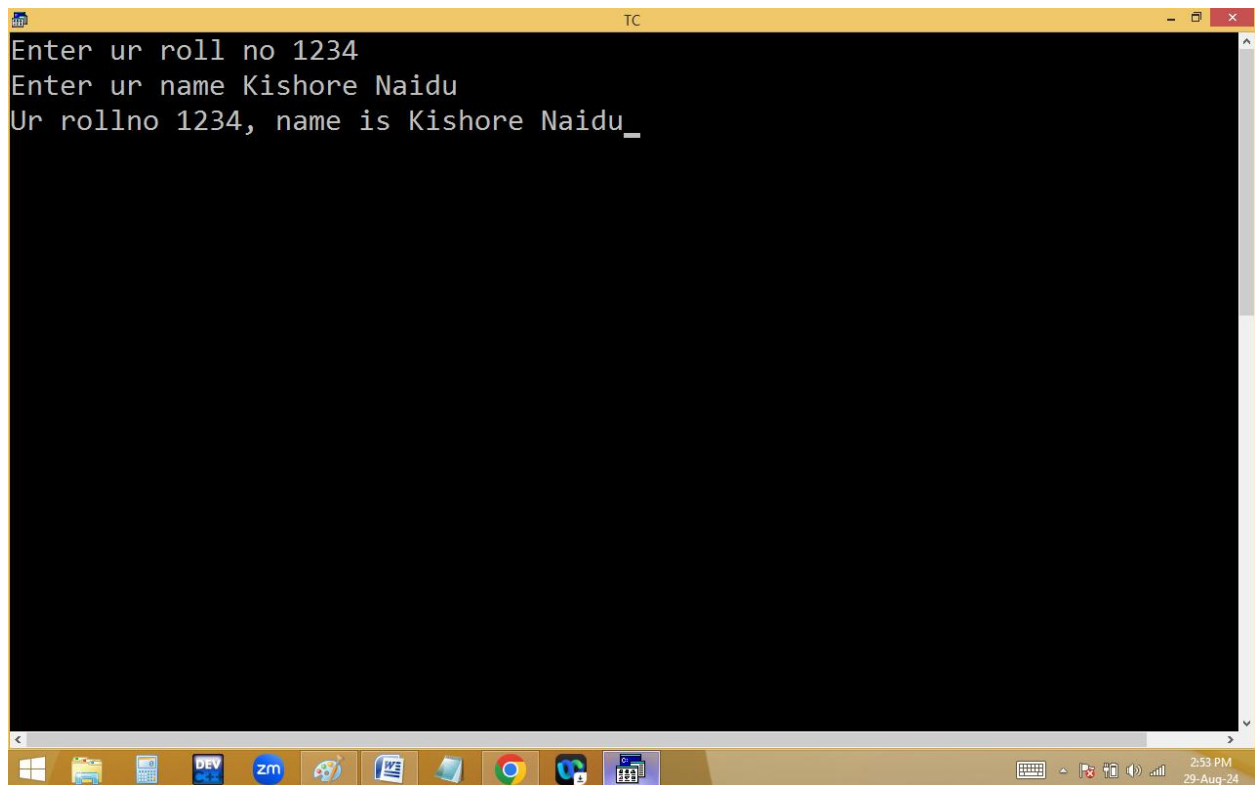
```
TC
Enter ur name
Ur name is
```



The screenshot shows the Turbo C++ (TC) IDE with a blue background. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates 'Line 9 Col 12 Insert Indent Tab Fill Unindent * E:2PM.C'. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int rollno;
char name[30];
clrscr();
printf("Enter ur roll no "); scanf("%d",&rollno);
flushall();_
printf("Enter ur name ");gets(name);
printf("Ur rollno %d, name is %s",rollno, name);
getch();
}
```

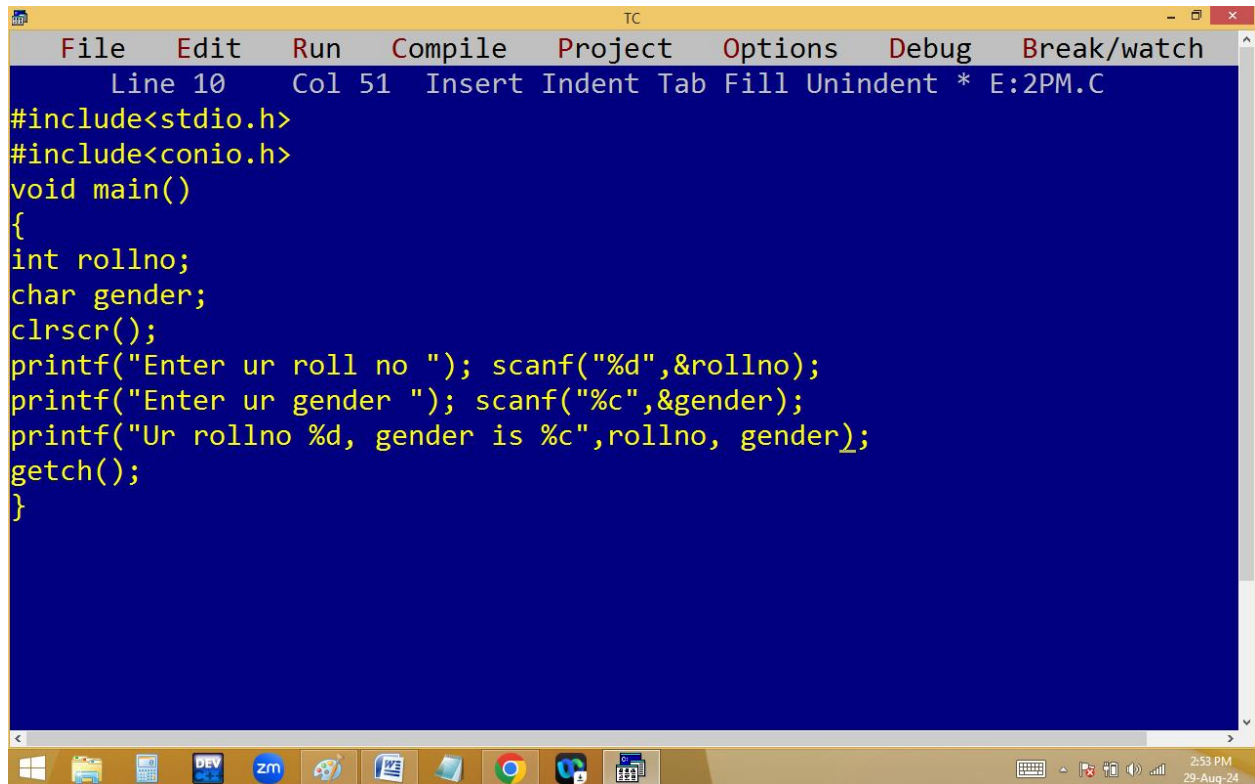
The Windows taskbar at the bottom shows the time as 2:52 PM on 29-Aug-24.



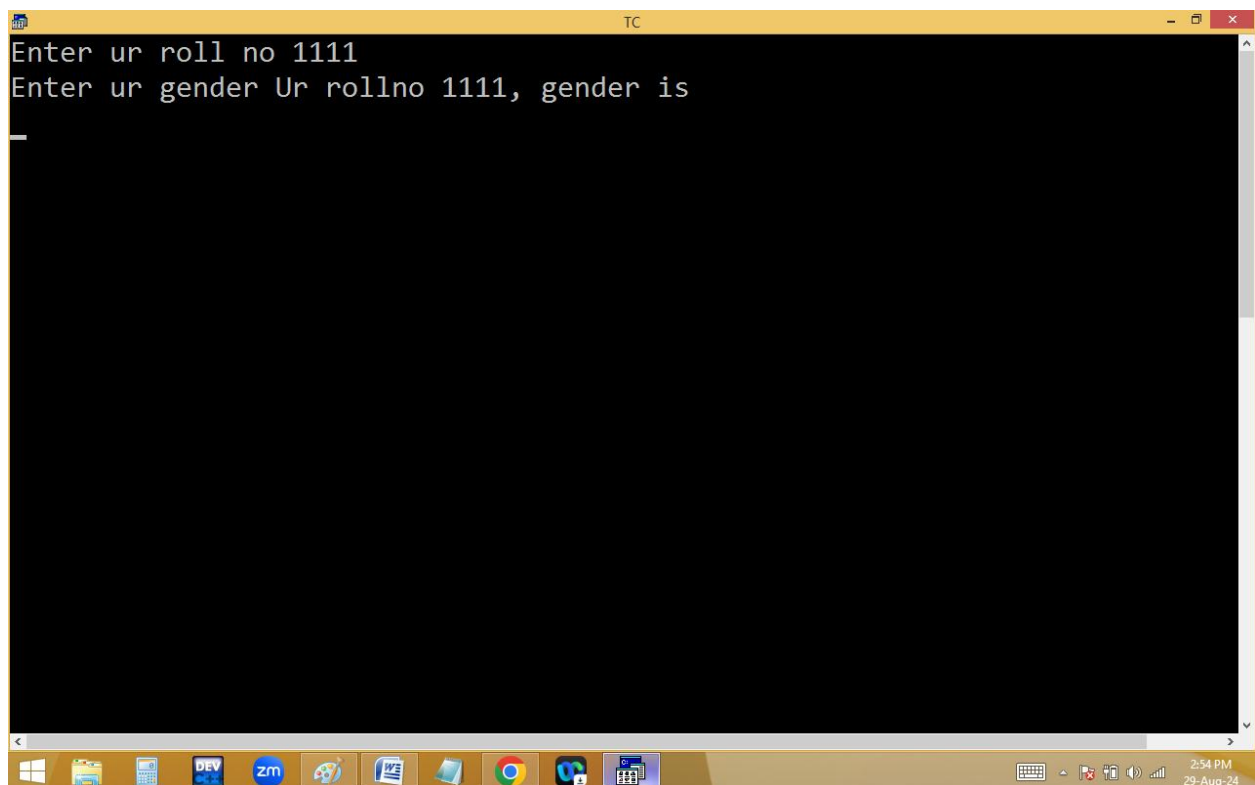
The screenshot shows the Turbo C++ (TC) IDE with a black background, displaying the output of the program. The text shown is:

```
Enter ur roll no 1234
Enter ur name Kishore Naidu
Ur rollno 1234, name is Kishore Naidu_
```

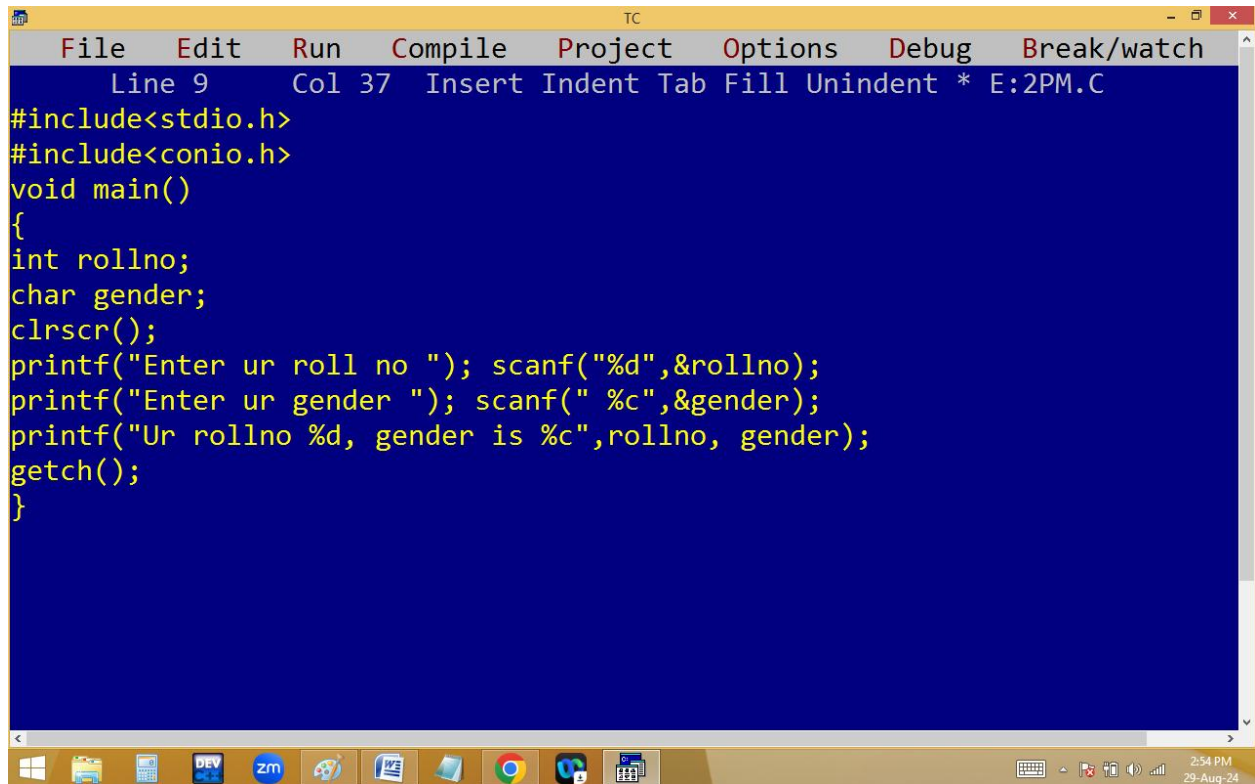
The Windows taskbar at the bottom shows the time as 2:53 PM on 29-Aug-24.



```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 10 Col 51 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int rollno;
char gender;
clrscr();
printf("Enter ur roll no "); scanf("%d",&rollno);
printf("Enter ur gender "); scanf("%c",&gender);
printf("Ur rollno %d, gender is %c",rollno, gender);
getch();
}
```



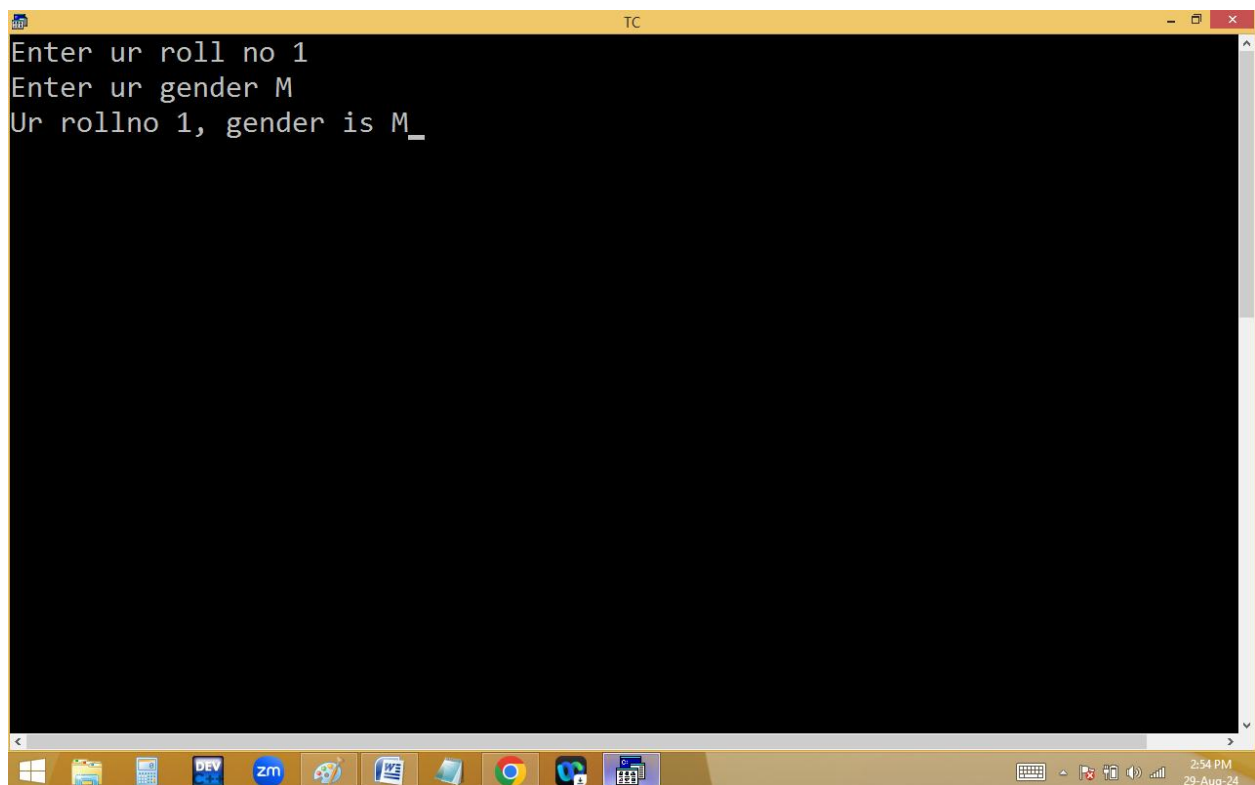
```
TC
Enter ur roll no 1111
Enter ur gender Ur rollno 1111, gender is
_
```



The screenshot shows the Turbo C++ (TC) IDE with a menu bar (File, Edit, Run, Compile, Project, Options, Debug, Break/watch) and a status bar (Line 9, Col 37, Insert, Indent, Tab, Fill, Unindent, * E:2PM.C). The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int rollno;
char gender;
clrscr();
printf("Enter ur roll no "); scanf("%d",&rollno);
printf("Enter ur gender "); scanf(" %c",&gender);
printf("Ur rollno %d, gender is %c",rollno, gender);
getch();
}
```

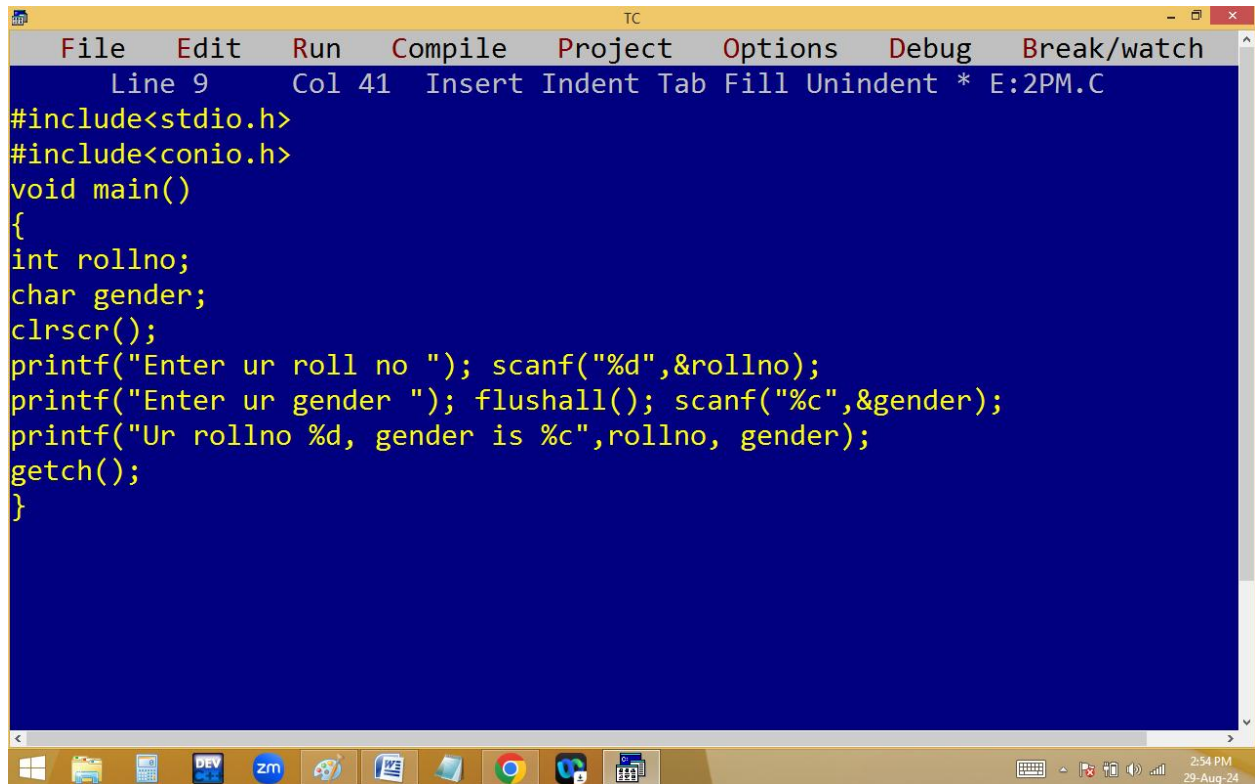
The Windows taskbar at the bottom shows the Start button and several application icons, including DEV, zm, and a folder icon. The system clock indicates 2:54 PM on 29-Aug-24.



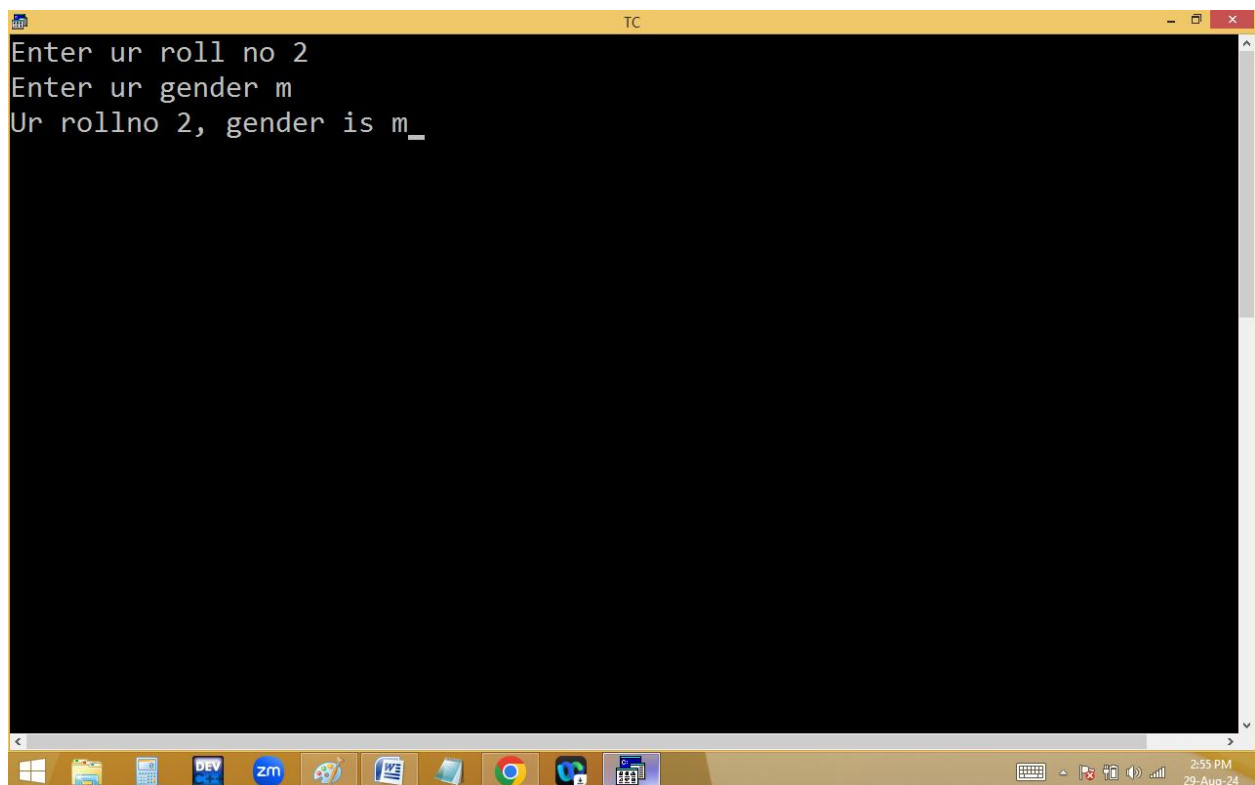
The screenshot shows the Turbo C++ (TC) IDE with the same menu bar and status bar as the first image. The output window displays the results of the program execution:

```
Enter ur roll no 1
Enter ur gender M
Ur rollno 1, gender is M_
```

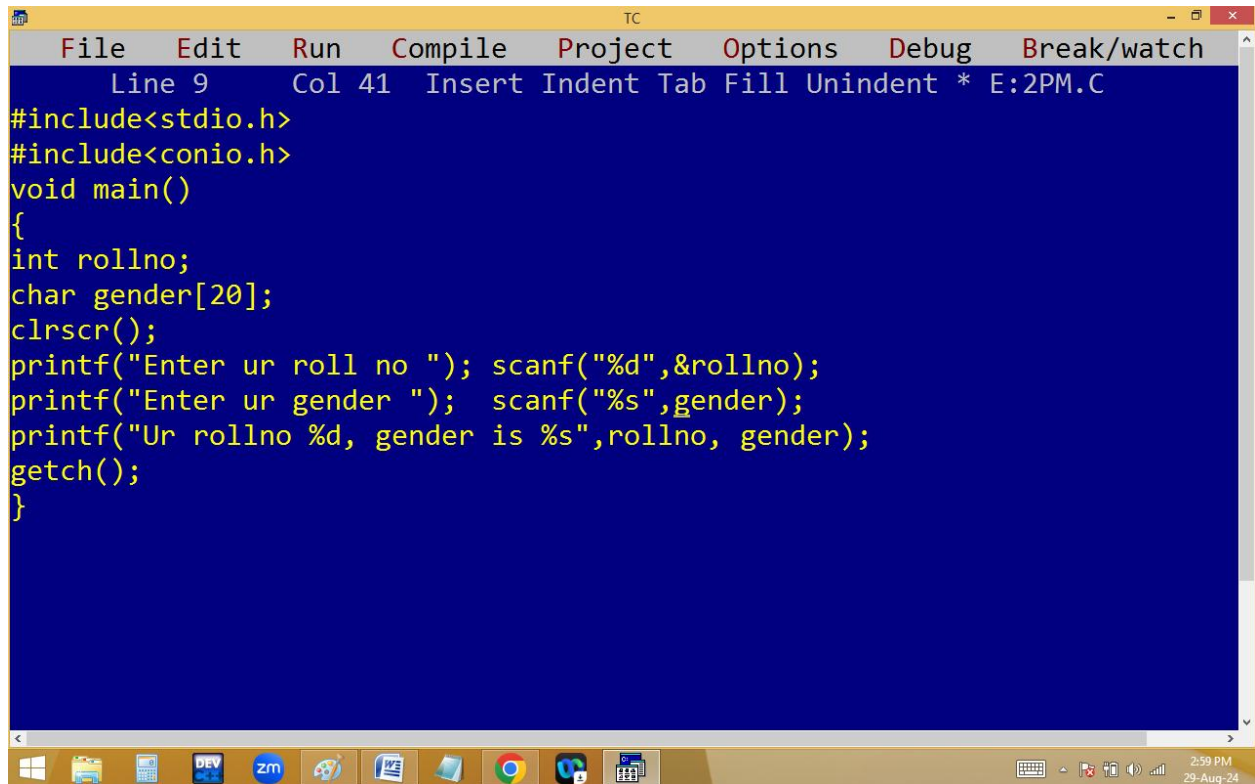
The Windows taskbar at the bottom is identical to the first image, showing the Start button, application icons, and the system clock at 2:54 PM on 29-Aug-24.



```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 9 Col 41 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int rollno;
char gender;
clrscr();
printf("Enter ur roll no "); scanf("%d",&rollno);
printf("Enter ur gender "); fflush(); scanf("%c",&gender);
printf("Ur rollno %d, gender is %c",rollno, gender);
getch();
}
```



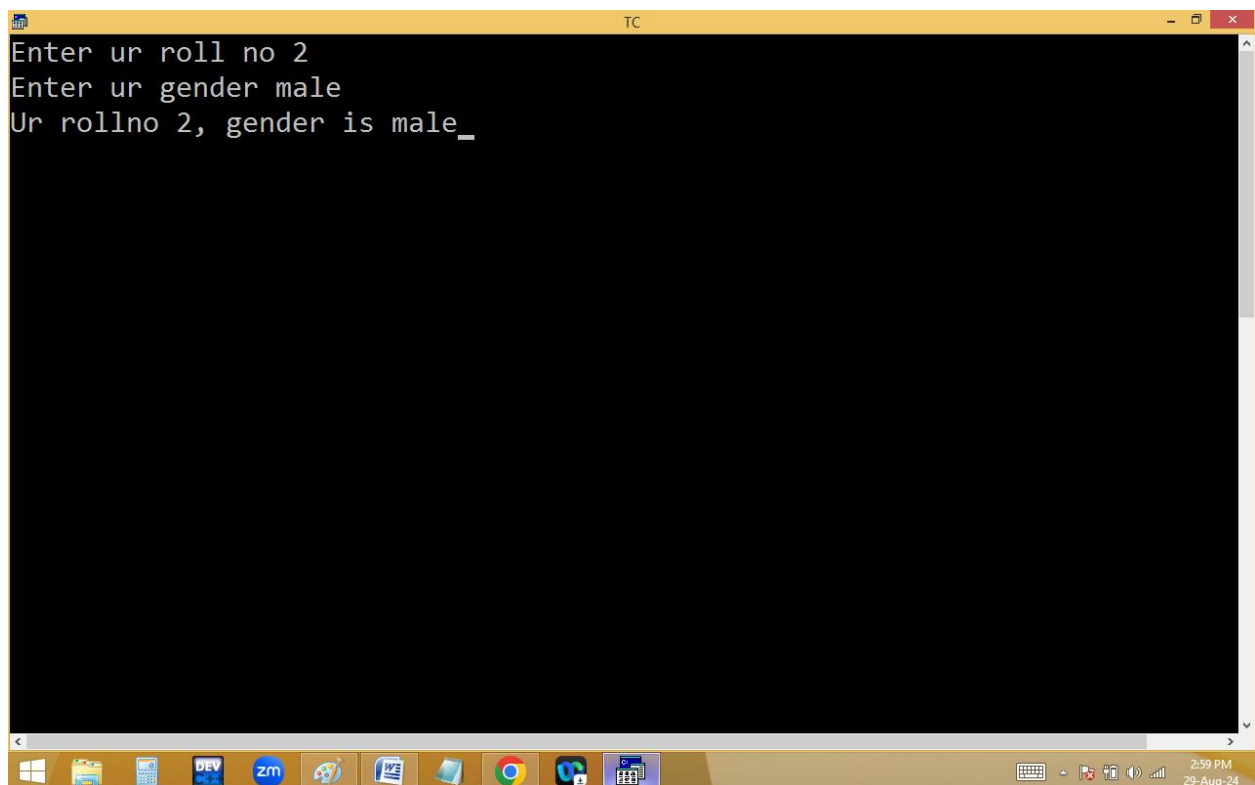
```
TC
Enter ur roll no 2
Enter ur gender m
Ur rollno 2, gender is m_
```



The screenshot shows the Turbo C++ (TC) IDE with a blue background. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates 'Line 9 Col 41 Insert Indent Tab Fill Unindent * E:2PM.C'. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int rollno;
char gender[20];
clrscr();
printf("Enter ur roll no "); scanf("%d",&rollno);
printf("Enter ur gender "); scanf("%s",gender);
printf("Ur rollno %d, gender is %s",rollno, gender);
getch();
}
```

The Windows taskbar at the bottom shows various icons including Windows, File Explorer, DEV, zm, and others. The system clock in the bottom right corner displays '2:59 PM 29-Aug-24'.

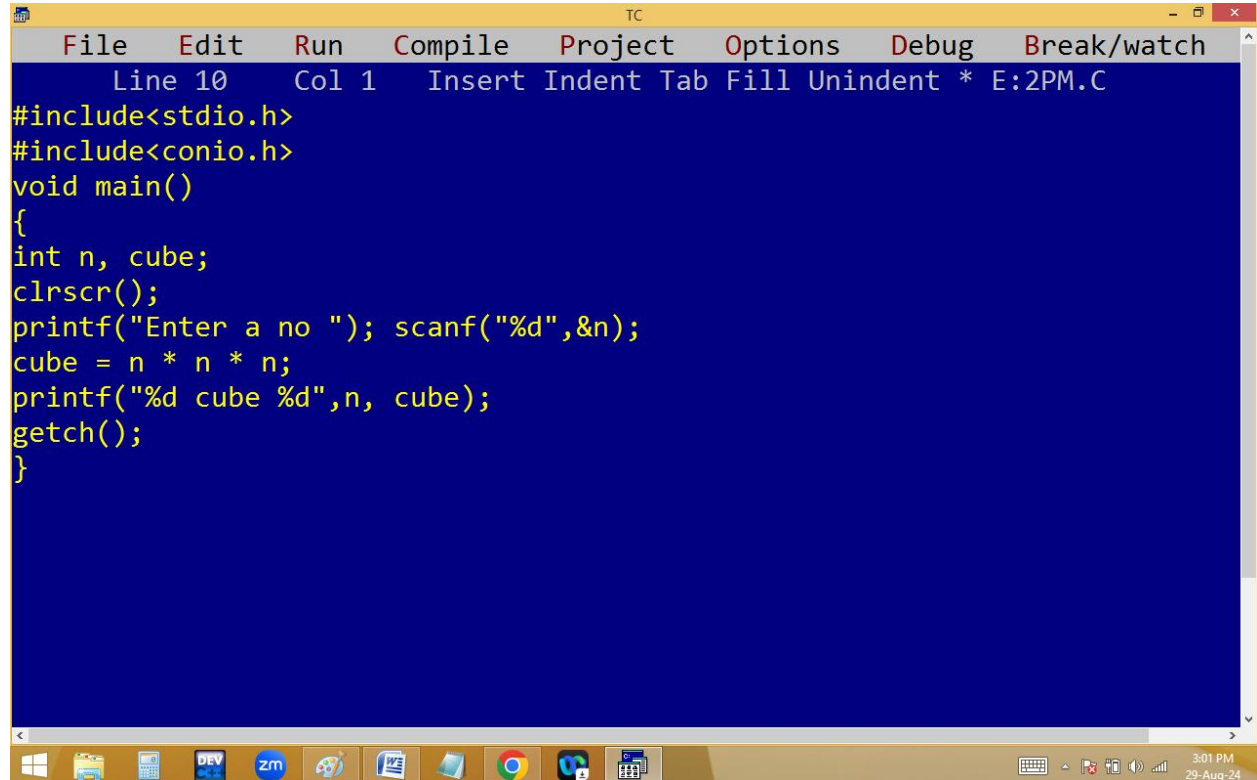


The screenshot shows the same Turbo C++ IDE window after the program has been executed. The output is displayed in the main window area:

```
Enter ur roll no 2
Enter ur gender male
Ur rollno 2, gender is male_
```

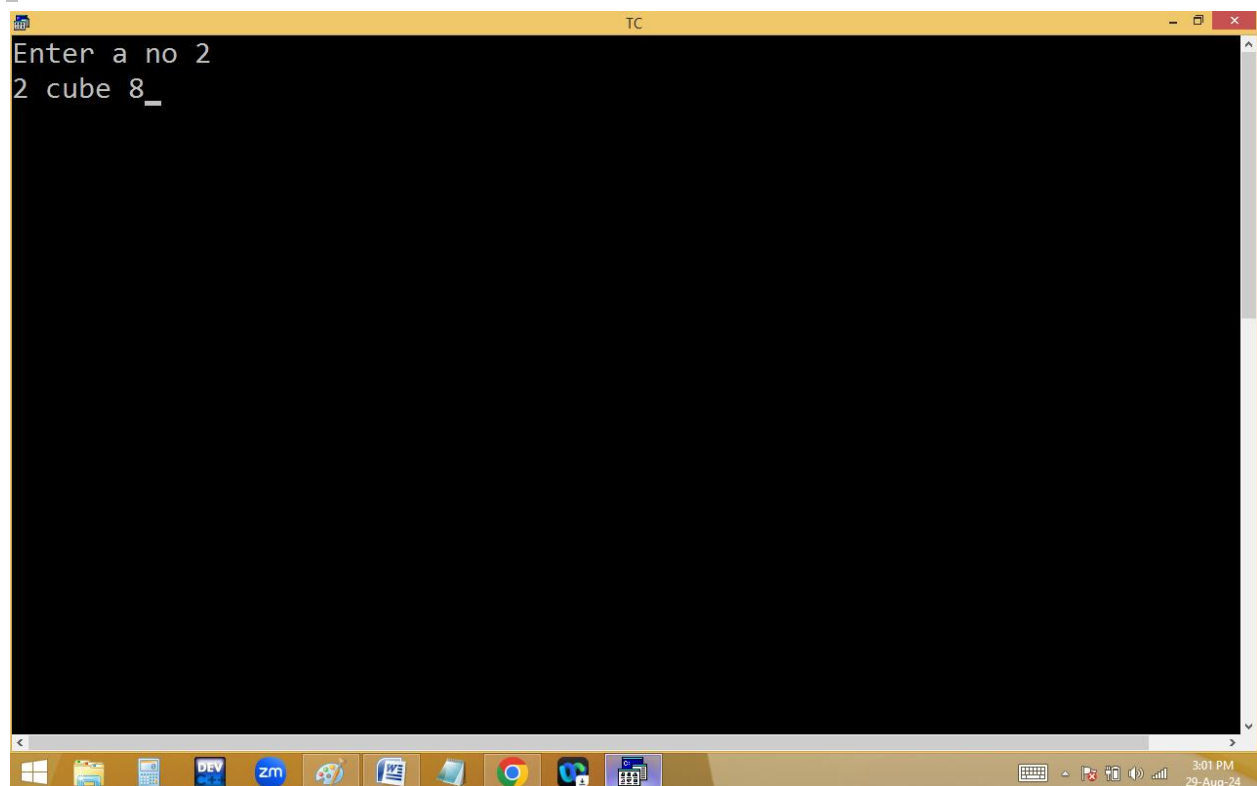
The Windows taskbar at the bottom remains the same, with the system clock still showing '2:59 PM 29-Aug-24'.

Finding cube value



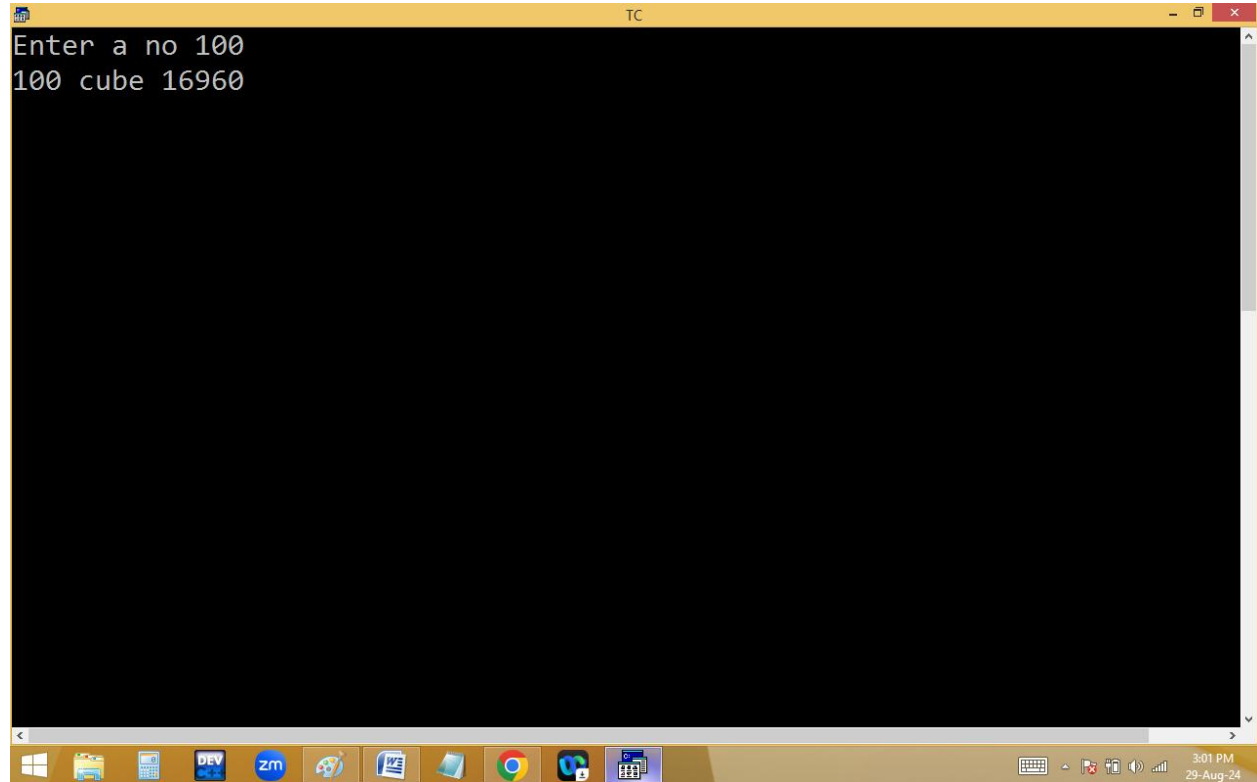
The screenshot shows the Turbo C++ (TC) IDE with a blue background. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the bottom indicates 'Line 10 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C'. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n, cube;
clrscr();
printf("Enter a no "); scanf("%d",&n);
cube = n * n * n;
printf("%d cube %d",n, cube);
getch();
}
```



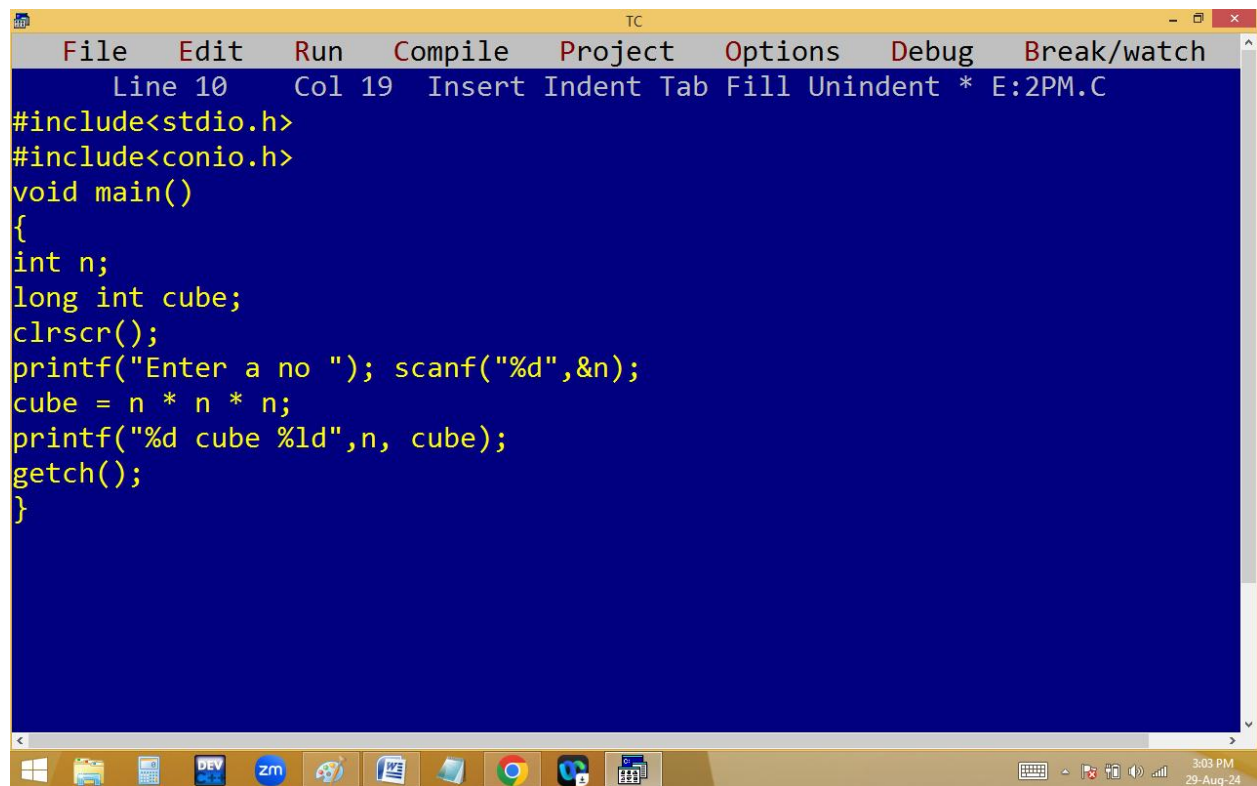
The screenshot shows the Turbo C++ (TC) IDE with a black background. The program has executed, and the output is displayed as follows:

```
Enter a no 2
2 cube 8_
```



A screenshot of a Turbo C++ (TC) console window. The window has a yellow title bar with the text "TC". The main area is black with white text. The text displayed is "Enter a no 100" on the first line and "100 cube 16960" on the second line. The Windows taskbar is visible at the bottom, showing various application icons and the system clock indicating 3:01 PM on 29-Aug-24.

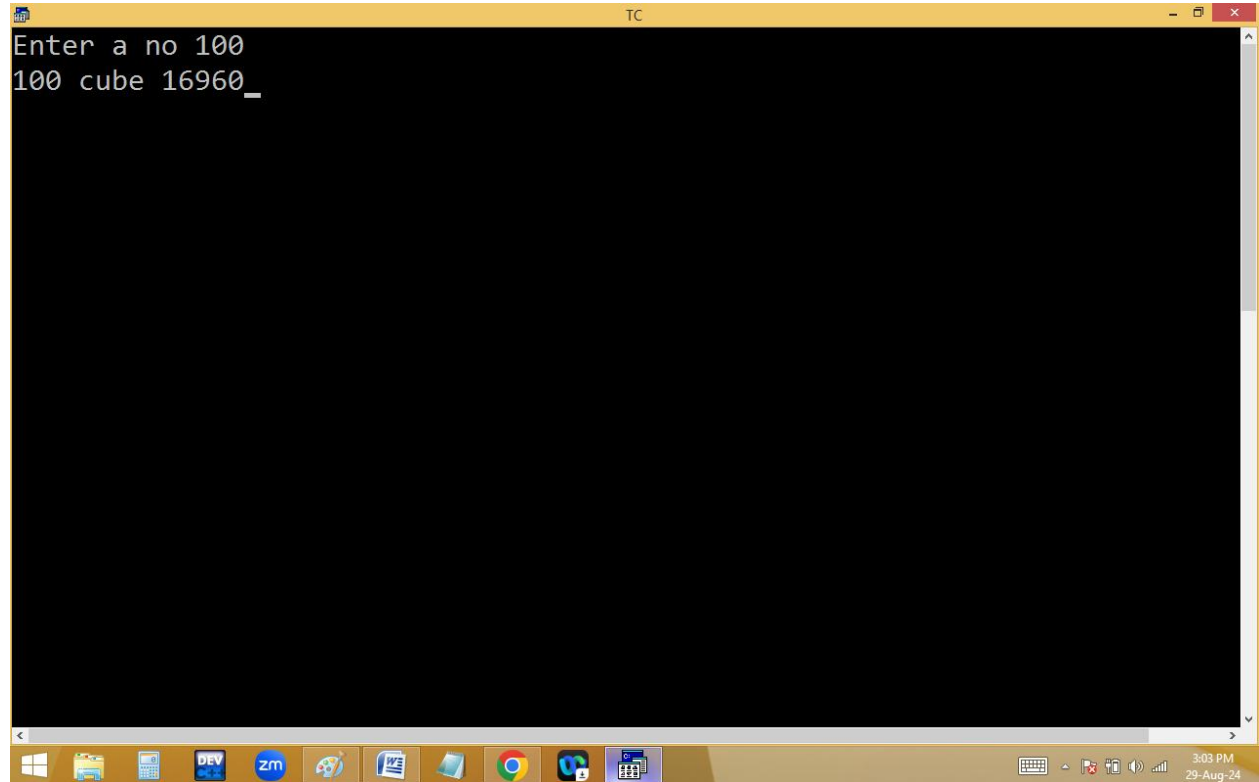
```
Enter a no 100
100 cube 16960
```

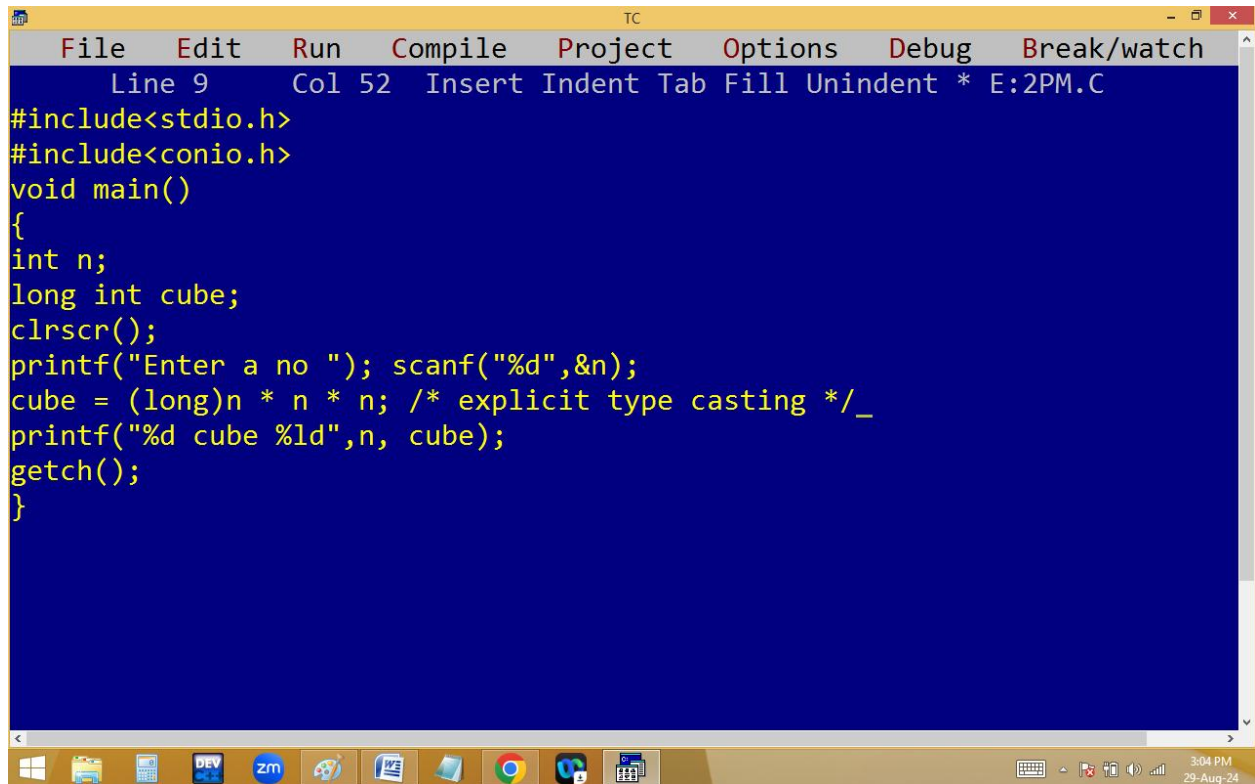


A screenshot of a Turbo C++ (TC) editor window. The window has a yellow title bar with the text "TC". Below the title bar is a menu bar with the following items: File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. Below the menu bar is a status bar that reads "Line 10 Col 19 Insert Indent Tab Fill Unindent * E:2PM.C". The main area has a blue background with yellow text showing the source code for a program that calculates the cube of a number. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n;
long int cube;
clrscr();
printf("Enter a no "); scanf("%d",&n);
cube = n * n * n;
printf("%d cube %ld",n, cube);
getch();
}
```

The Windows taskbar is visible at the bottom, showing various application icons and the system clock indicating 3:03 PM on 29-Aug-24.

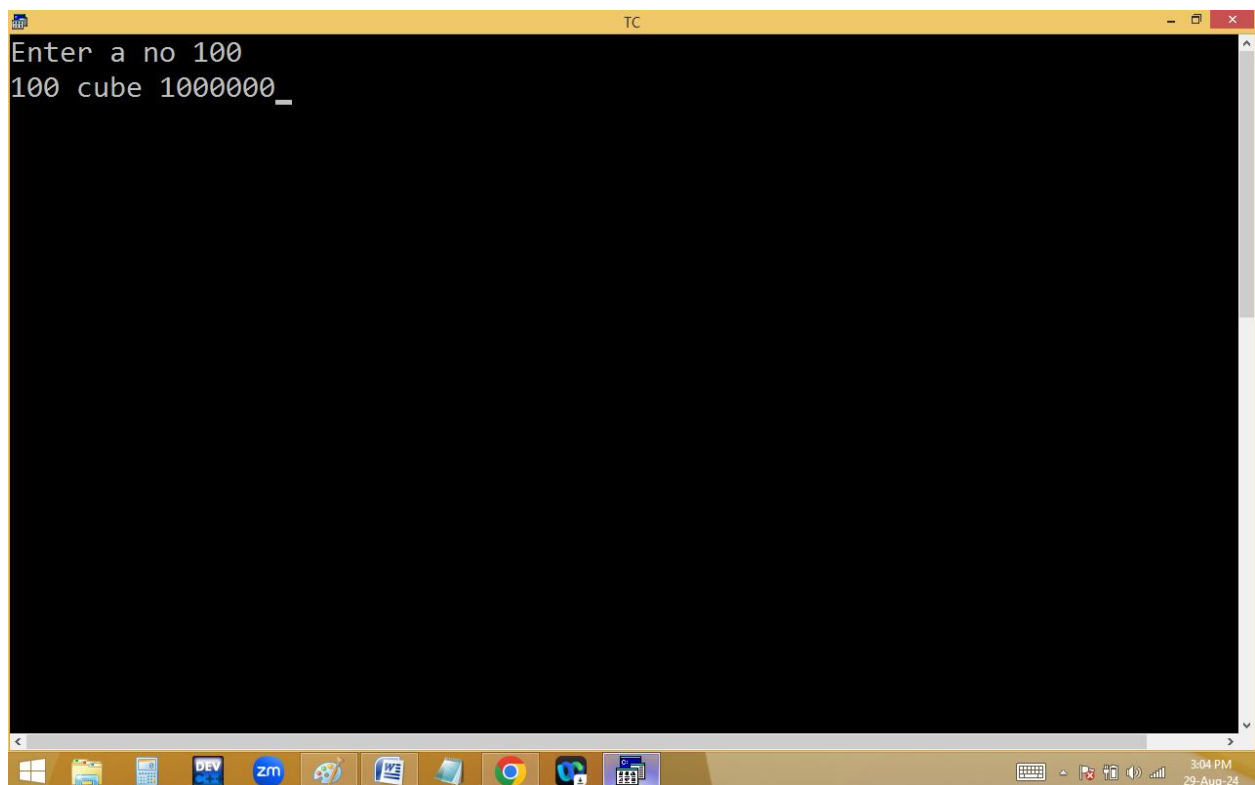




The screenshot shows the Turbo C++ (TC) IDE with a yellow title bar and menu bar. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates 'Line 9', 'Col 52', and 'Insert Indent Tab Fill Unindent * E:2PM.C'. The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n;
long int cube;
clrscr();
printf("Enter a no "); scanf("%d",&n);
cube = (long)n * n * n; /* explicit type casting */
printf("%d cube %ld",n, cube);
getch();
}
```

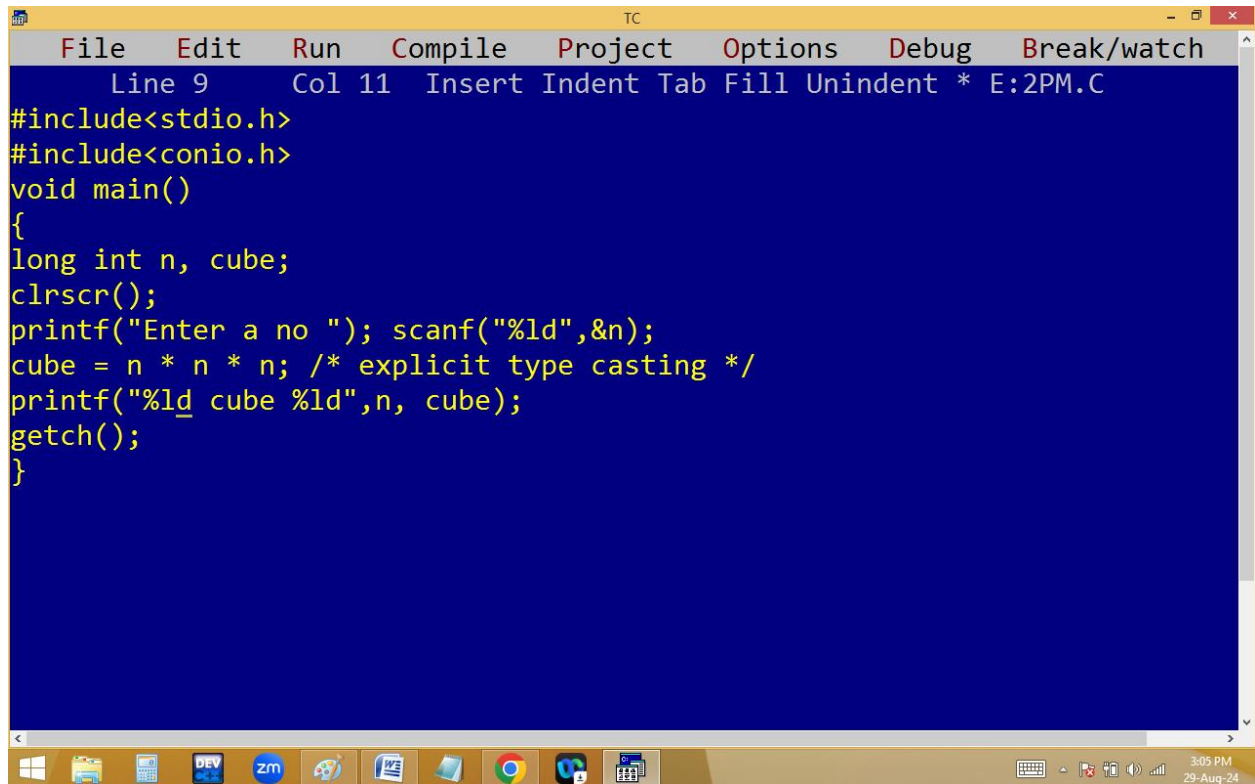
The Windows taskbar at the bottom shows various icons including Windows, File Explorer, DEV, zm, and others. The system clock in the bottom right corner displays '3:04 PM' and '29-Aug-24'.



The screenshot shows the Turbo C++ (TC) IDE with a yellow title bar and menu bar. The main editing area has a black background and displays the program's output:

```
Enter a no 100
100 cube 1000000_
```

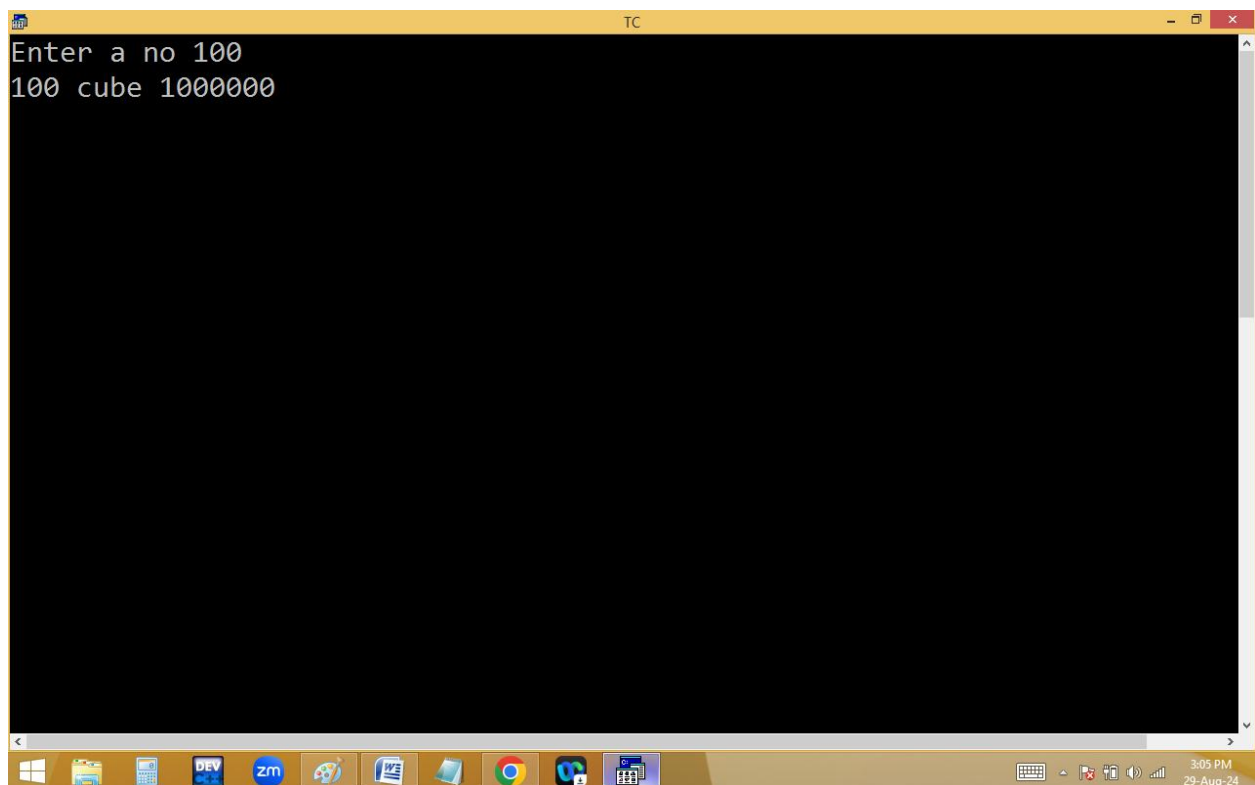
The Windows taskbar at the bottom is identical to the first screenshot, showing the same icons and system clock ('3:04 PM', '29-Aug-24').



The screenshot shows the Turbo C++ (TC) IDE with a yellow title bar and a menu bar containing File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates 'Line 9 Col 11 Insert Indent Tab Fill Unindent * E:2PM.C'. The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
long int n, cube;
clrscr();
printf("Enter a no "); scanf("%ld",&n);
cube = n * n * n; /* explicit type casting */
printf("%ld cube %ld",n, cube);
getch();
}
```

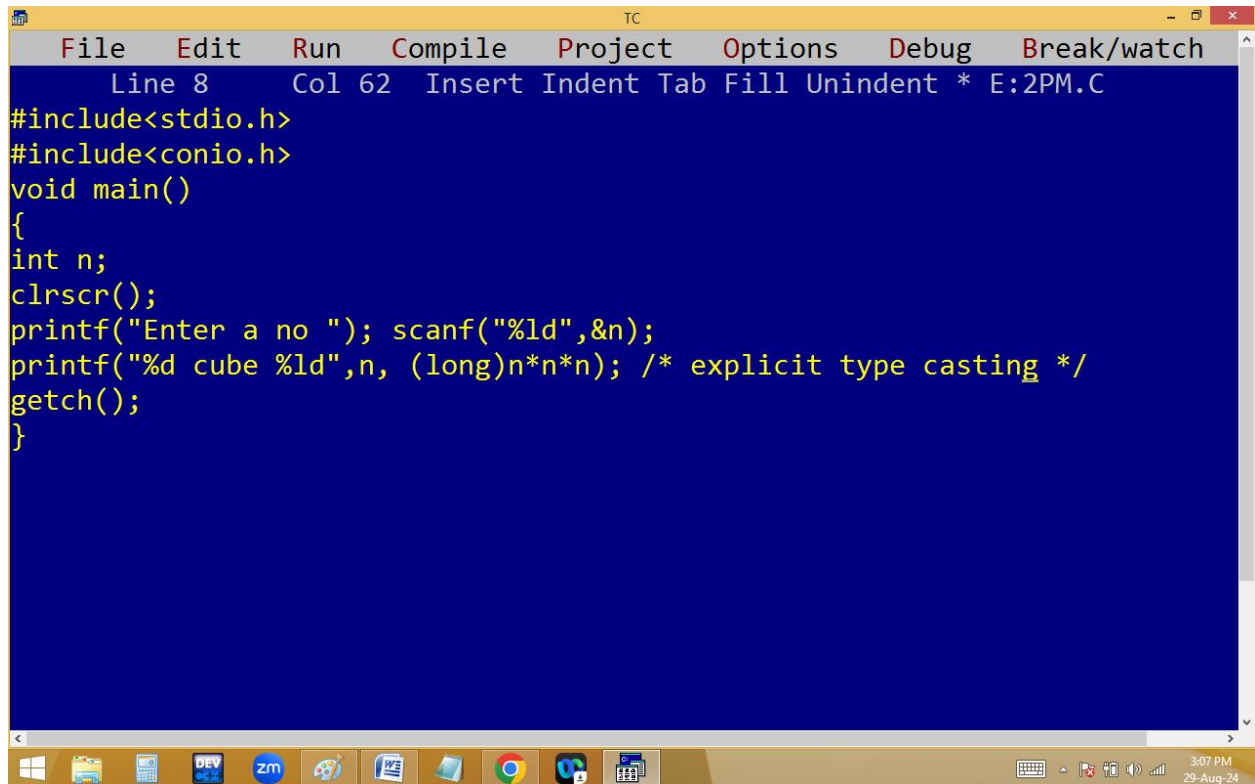
The Windows taskbar at the bottom shows various icons including Windows, File Explorer, DEV, zm, and others, along with system icons for keyboard, network, and volume, and a clock showing 3:05 PM on 29-Aug-24.



The screenshot shows the Turbo C++ (TC) IDE with a yellow title bar. The main editing area has a black background and displays the output of the program's execution:

```
Enter a no 100
100 cube 1000000
```

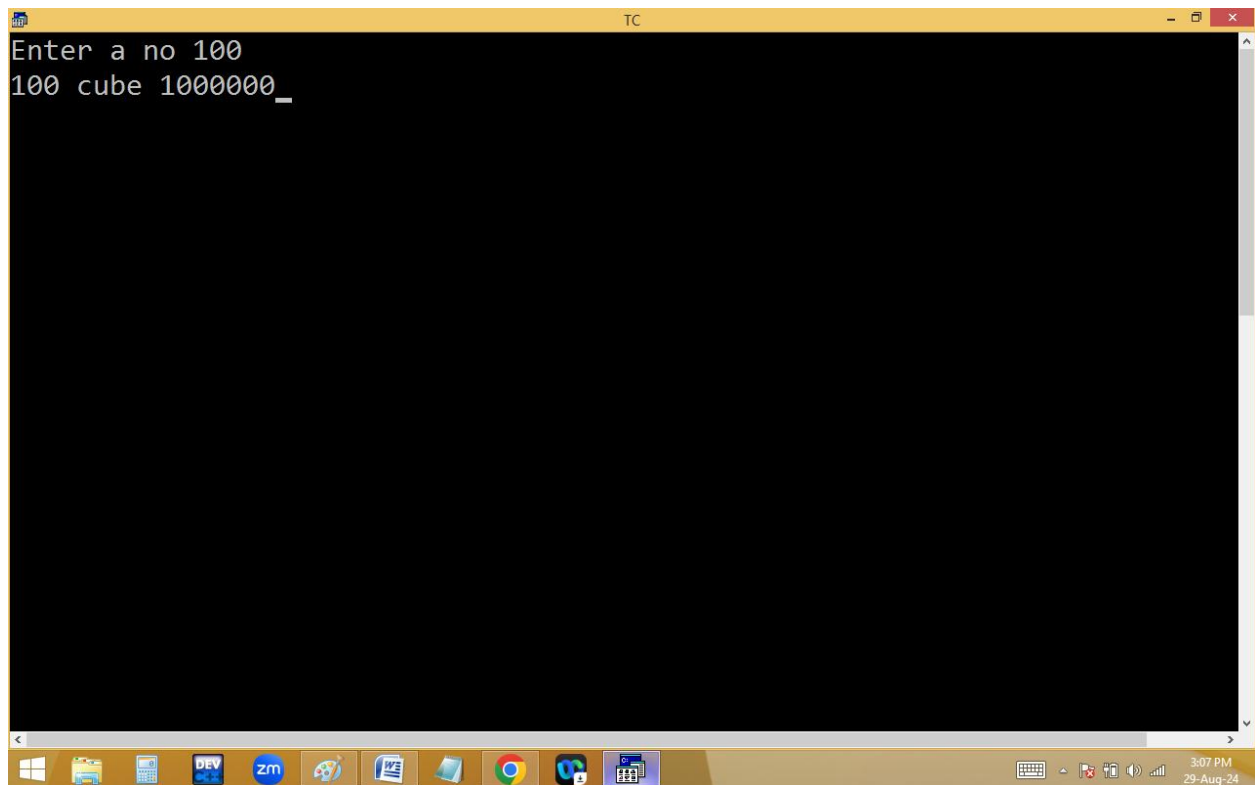
The Windows taskbar at the bottom is identical to the first screenshot, showing the same set of icons and system information (3:05 PM on 29-Aug-24).



The screenshot shows the Turbo C++ (TC) IDE with a yellow title bar and menu bar. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates 'Line 8 Col 62 Insert Indent Tab Fill Unindent * E:2PM.C'. The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n;
clrscr();
printf("Enter a no "); scanf("%ld",&n);
printf("%d cube %ld",n, (long)n*n*n); /* explicit type casting */
getch();
}
```

The Windows taskbar at the bottom shows various icons including Windows, File Explorer, DEV, zm, and others. The system clock in the bottom right corner displays '3:07 PM 29-Aug-24'.

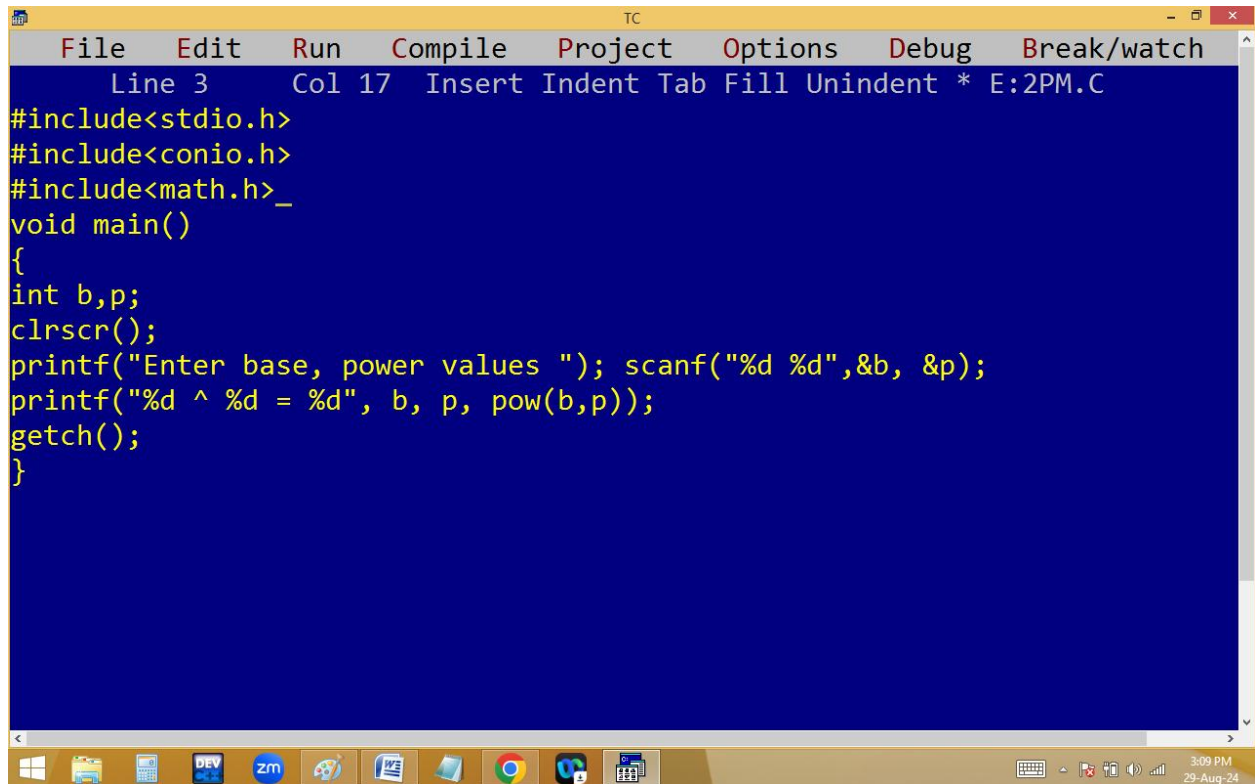


The screenshot shows the Turbo C++ (TC) IDE with a yellow title bar. The main editing area has a black background and displays the program's output:

```
Enter a no 100
100 cube 1000000_
```

The Windows taskbar at the bottom is identical to the first screenshot, showing the same icons and system clock ('3:07 PM 29-Aug-24').

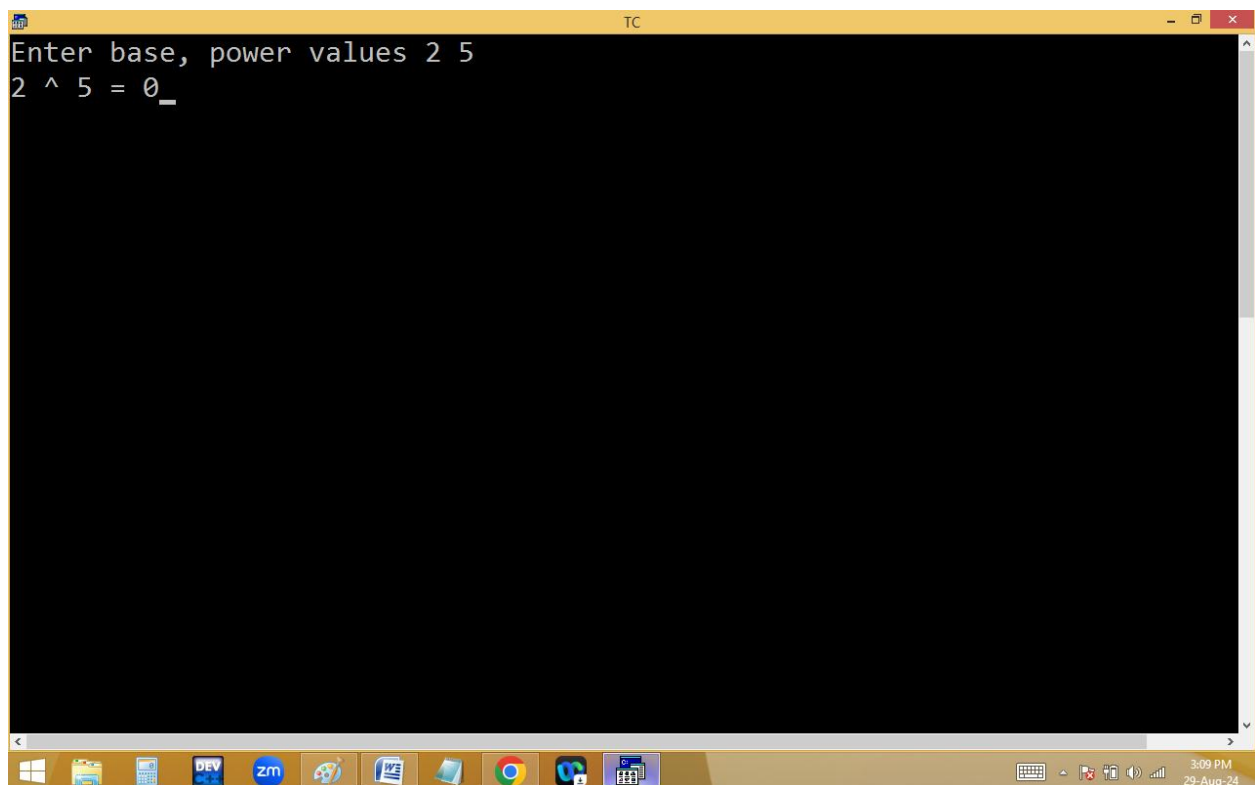
Finding power value:



The screenshot shows the Turbo C++ (TC) IDE with a blue background. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates 'Line 3 Col 17 Insert Indent Tab Fill Unindent * E:2PM.C'. The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
#include<math.h>_
void main()
{
int b,p;
clrscr();
printf("Enter base, power values "); scanf("%d %d",&b, &p);
printf("%d ^ %d = %d", b, p, pow(b,p));
getch();
}
```

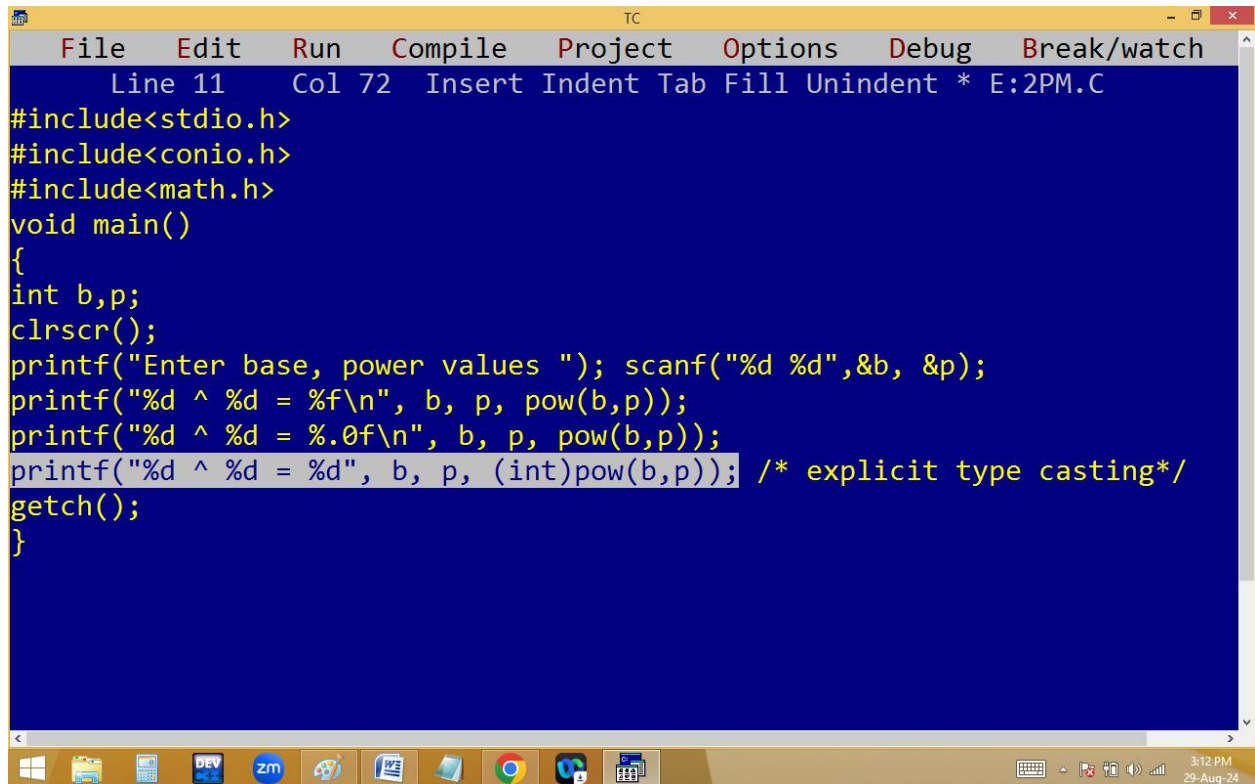
The Windows taskbar at the bottom shows various icons including Windows, File Explorer, DEV, zm, and others, along with the system clock showing 3:09 PM on 29-Aug-24.



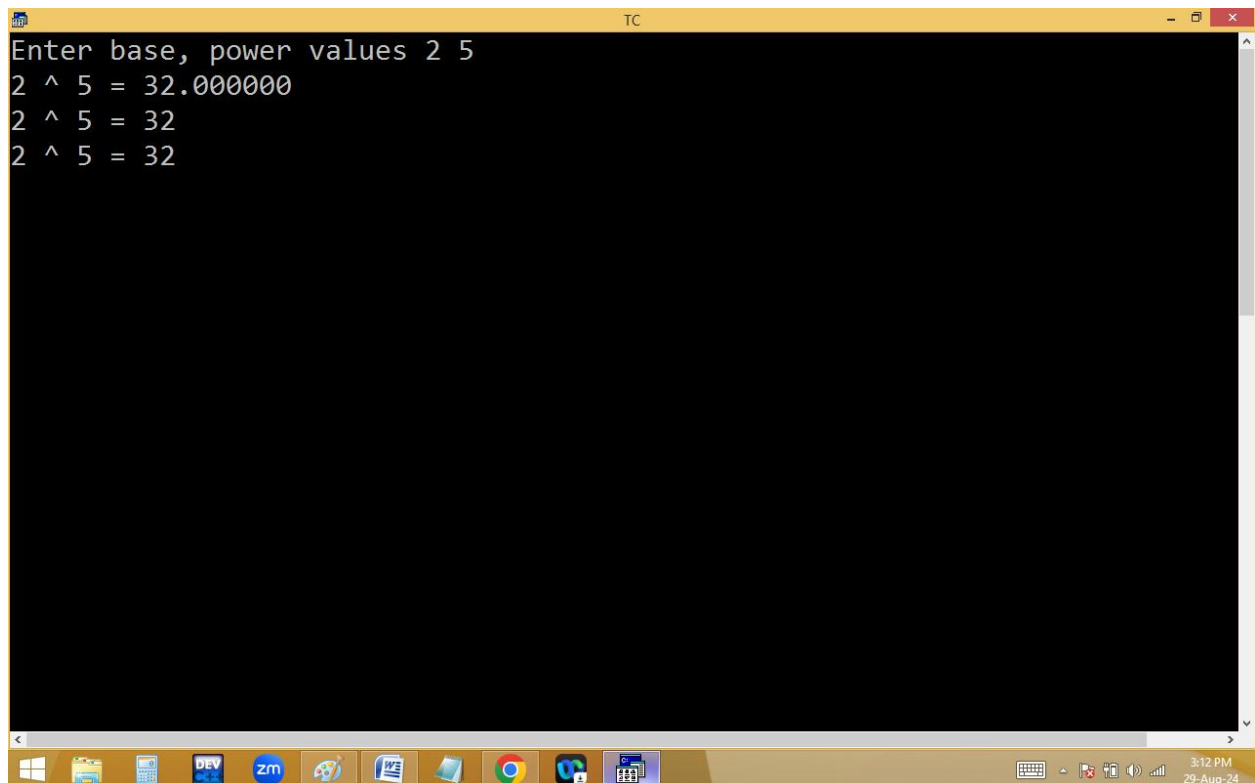
The screenshot shows the same Turbo C++ IDE window after execution. The output window displays the following text:

```
Enter base, power values 2 5
2 ^ 5 = 0_
```

The rest of the IDE interface, including the menu bar and taskbar, remains the same as in the first screenshot.

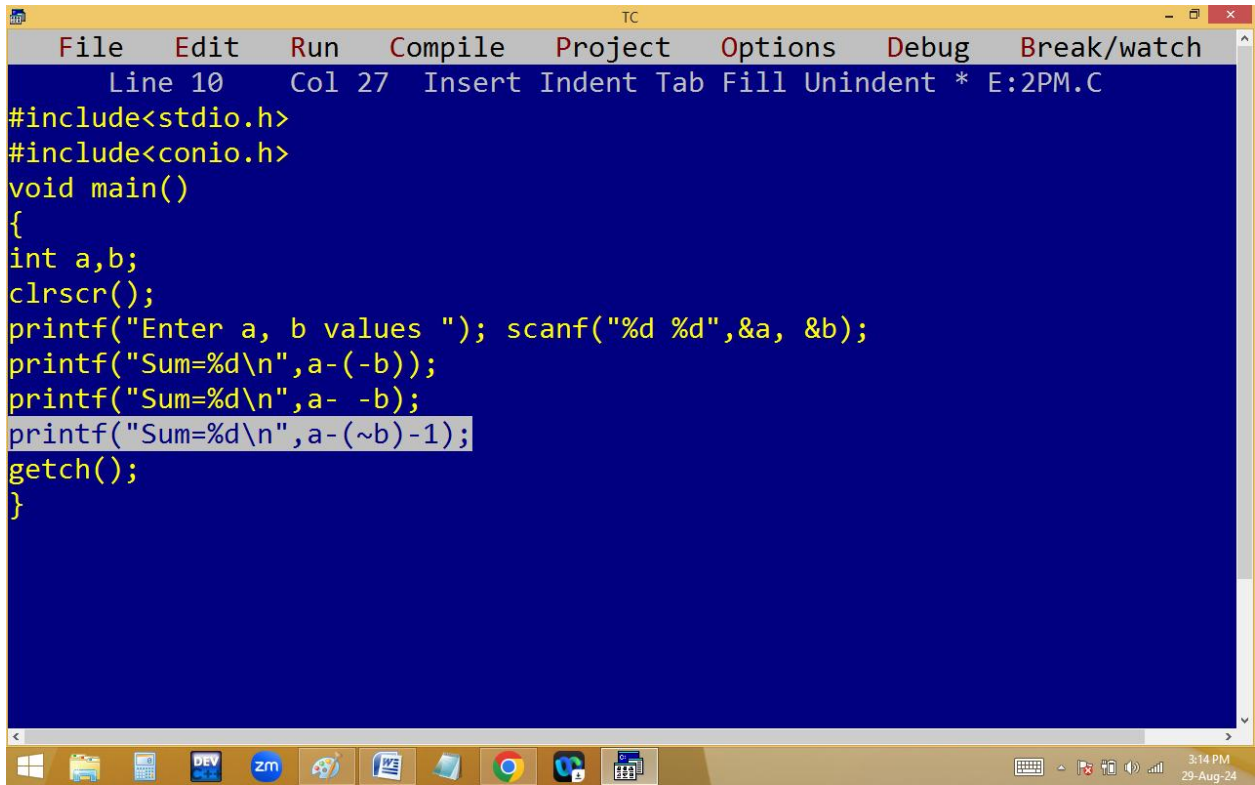


```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 11 Col 72 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
#include<math.h>
void main()
{
int b,p;
clrscr();
printf("Enter base, power values "); scanf("%d %d",&b, &p);
printf("%d ^ %d = %f\n", b, p, pow(b,p));
printf("%d ^ %d = %.0f\n", b, p, pow(b,p));
printf("%d ^ %d = %d", b, p, (int)pow(b,p)); /* explicit type casting*/
getch();
}
```



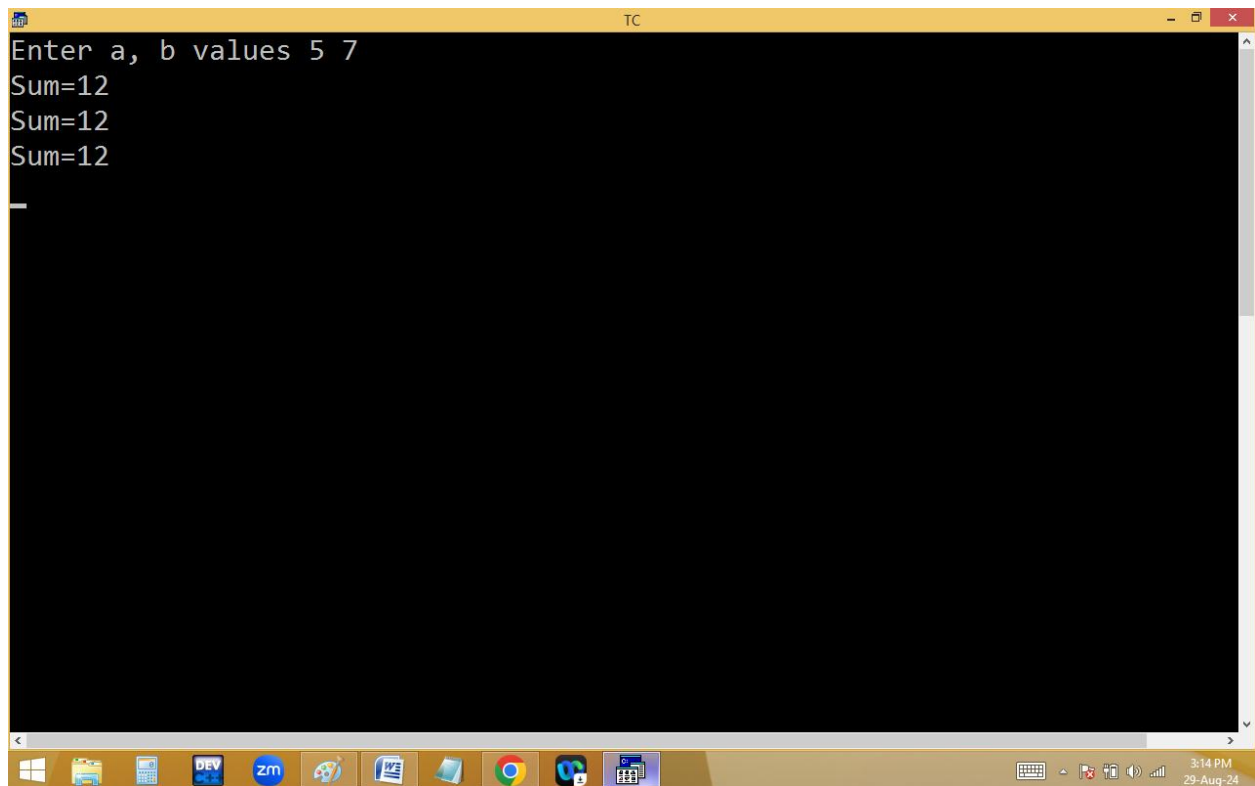
```
TC
Enter base, power values 2 5
2 ^ 5 = 32.000000
2 ^ 5 = 32
2 ^ 5 = 32
```

Adding of two numbers without using + operator:



```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 10 Col 27 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a,b;
clrscr();
printf("Enter a, b values "); scanf("%d %d",&a, &b);
printf("Sum=%d\n",a-(-b));
printf("Sum=%d\n",a- -b);
printf("Sum=%d\n",a-(~b)-1);
getch();
}
```

The screenshot shows a Turbo C++ IDE window titled 'TC'. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates 'Line 10 Col 27 Insert Indent Tab Fill Unindent * E:2PM.C'. The main editing area has a dark blue background and contains the following C code: `#include<stdio.h>`, `#include<conio.h>`, `void main()`, `{`, `int a,b;`, `clrscr();`, `printf("Enter a, b values "); scanf("%d %d",&a, &b);`, `printf("Sum=%d\n",a-(-b));`, `printf("Sum=%d\n",a- -b);`, `printf("Sum=%d\n",a-(~b)-1);`, `getch();`, and `}`. The Windows taskbar at the bottom shows various icons including the Start button, File Explorer, Calculator, DEV C++, Zoho Mail, and others. The system clock in the bottom right corner shows '3:14 PM' and '29-Aug-24'.



TC

```
Enter a, b values 5 7
Sum=12
Sum=12
Sum=12
```

The screenshot shows a Turbo C++ (TC) window with a black background and white text. The window title bar says 'TC'. The program prompts the user to enter values for 'a' and 'b', which are 5 and 7 respectively. The program then calculates the sum of these two values, which is 12, and displays it three times. The Windows taskbar is visible at the bottom, showing various application icons and the system clock indicating 3:14 PM on 29-Aug-24.

$$a-(\sim b)-1$$

$$5-(\sim 7)-1$$

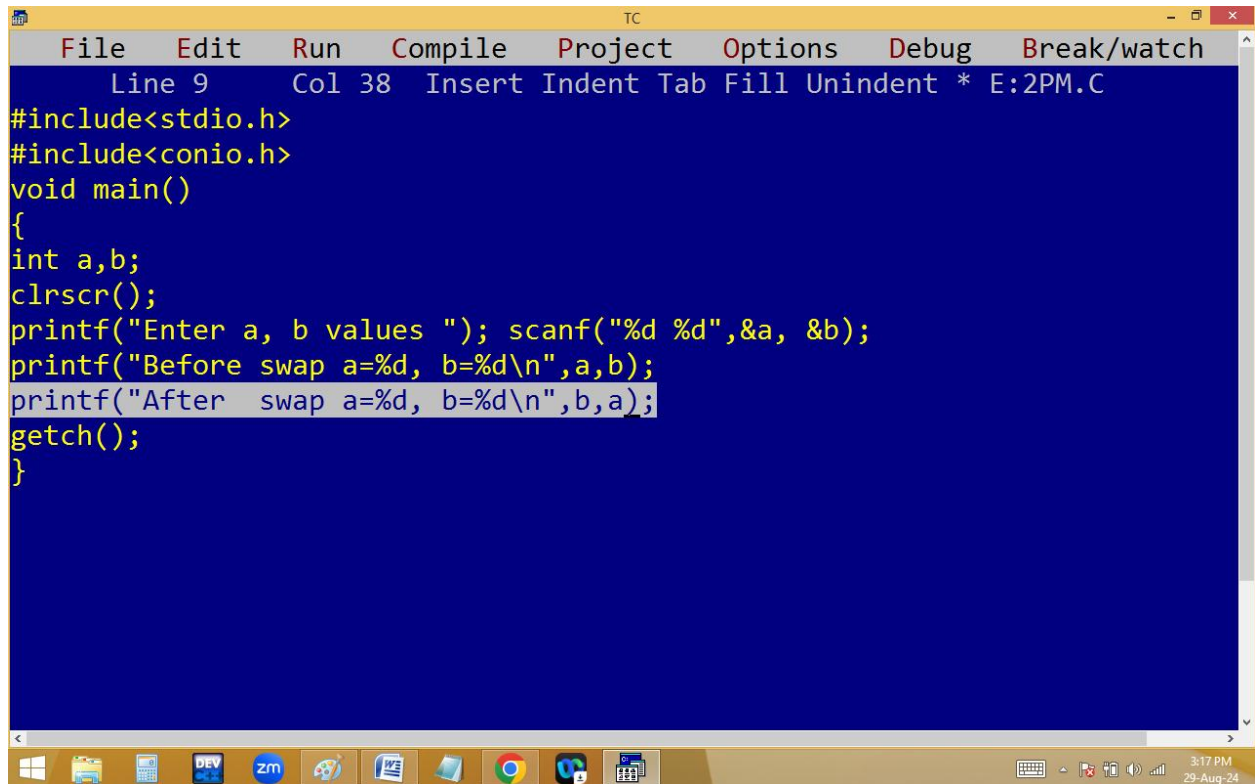
$$5-(-8)-1$$

$$5+8-1$$

$$13-1=12$$

Swap of two numbers [interchange]:

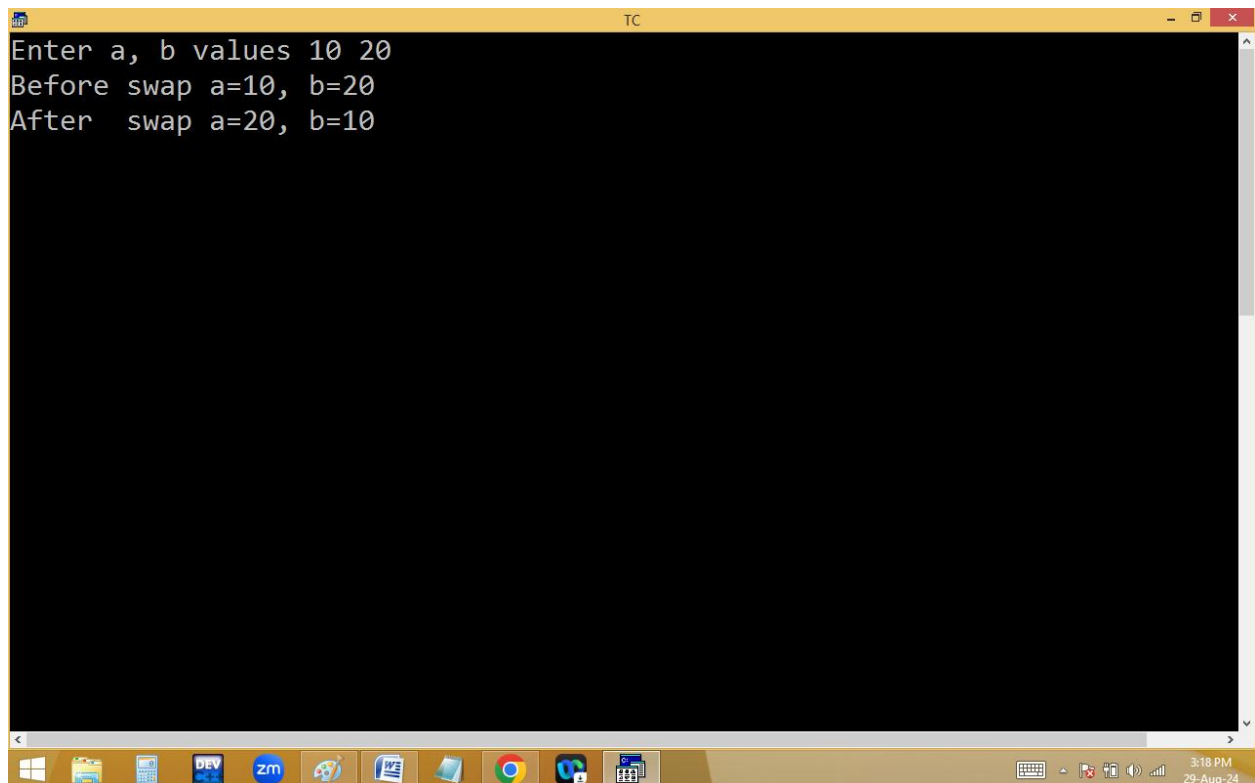
Method1: without using operators:



The screenshot shows the Turbo C++ (TC) IDE with a yellow title bar and menu bar. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates 'Line 9 Col 38 Insert Indent Tab Fill Unindent * E:2PM.C'. The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a,b;
clrscr();
printf("Enter a, b values "); scanf("%d %d",&a, &b);
printf("Before swap a=%d, b=%d\n",a,b);
printf("After swap a=%d, b=%d\n",b,a);
getch();
}
```

The taskbar at the bottom shows various application icons including Windows Explorer, DEV, zm, and others. The system clock in the bottom right corner shows 3:17 PM on 29-Aug-24.

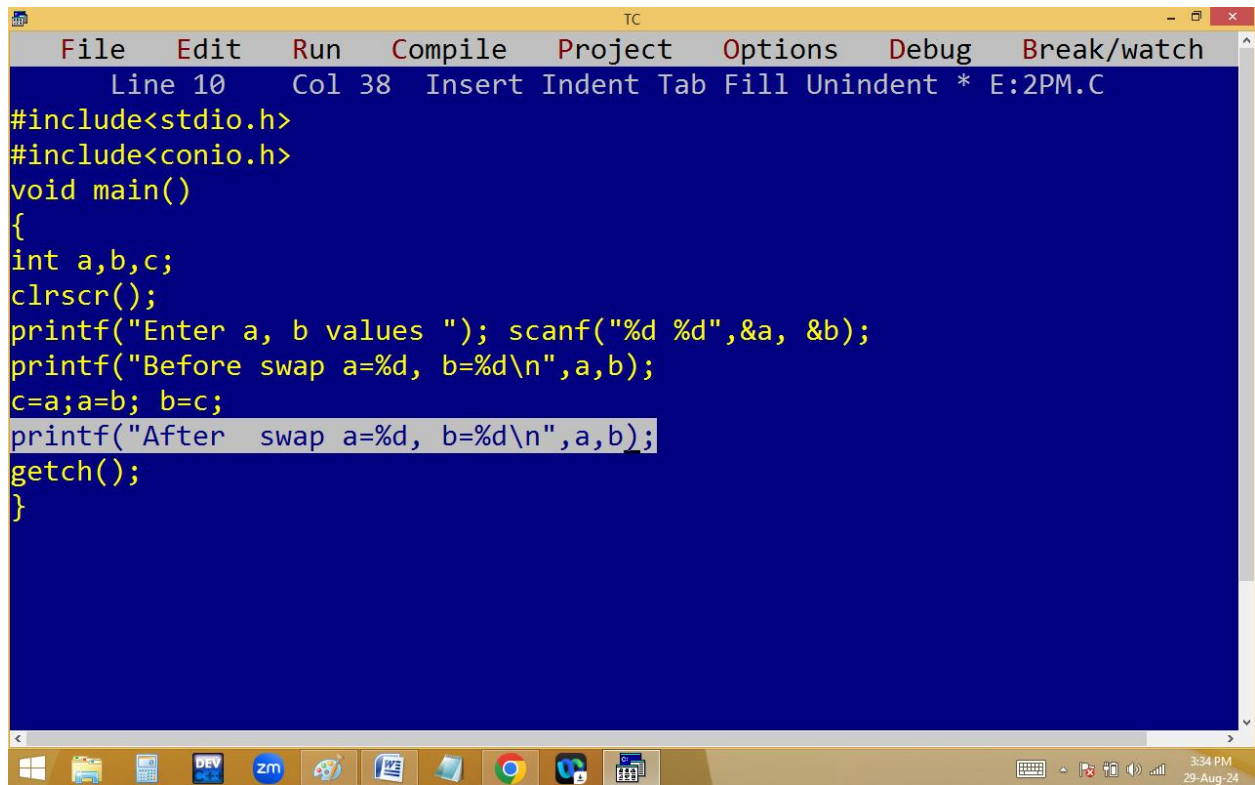


The screenshot shows the Turbo C++ (TC) IDE with a yellow title bar. The main editing area has a black background and displays the output of the program:

```
Enter a, b values 10 20
Before swap a=10, b=20
After swap a=20, b=10
```

The taskbar at the bottom is identical to the first screenshot, showing the same application icons and system clock (3:18 PM on 29-Aug-24).

Using 3rd variable:

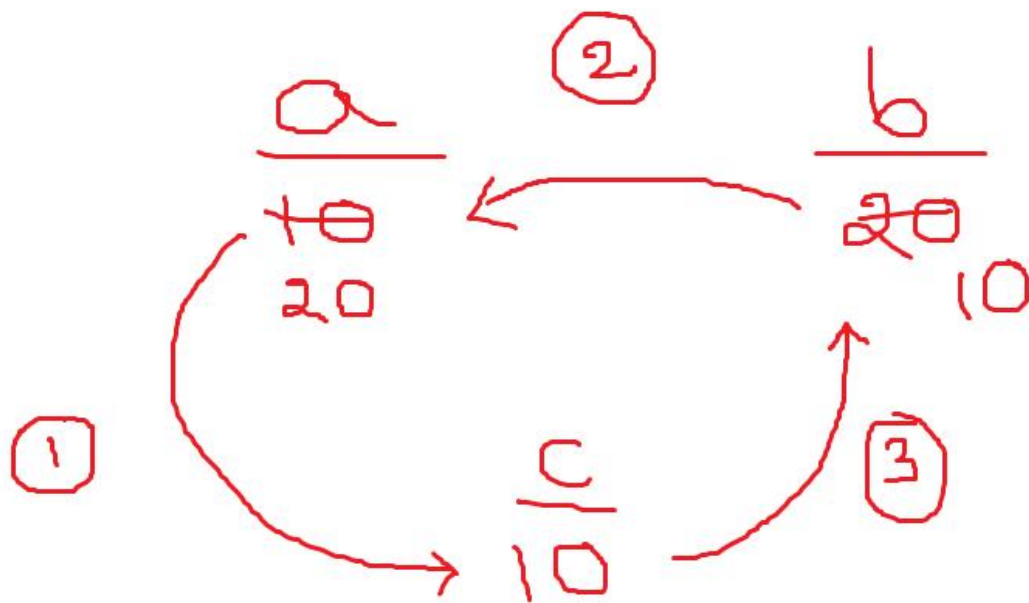


The screenshot shows the Turbo C++ (TC) IDE interface. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates 'Line 10 Col 38 Insert Indent Tab Fill Unindent * E:2PM.C'. The main editing area has a dark blue background with yellow text. The code is as follows:

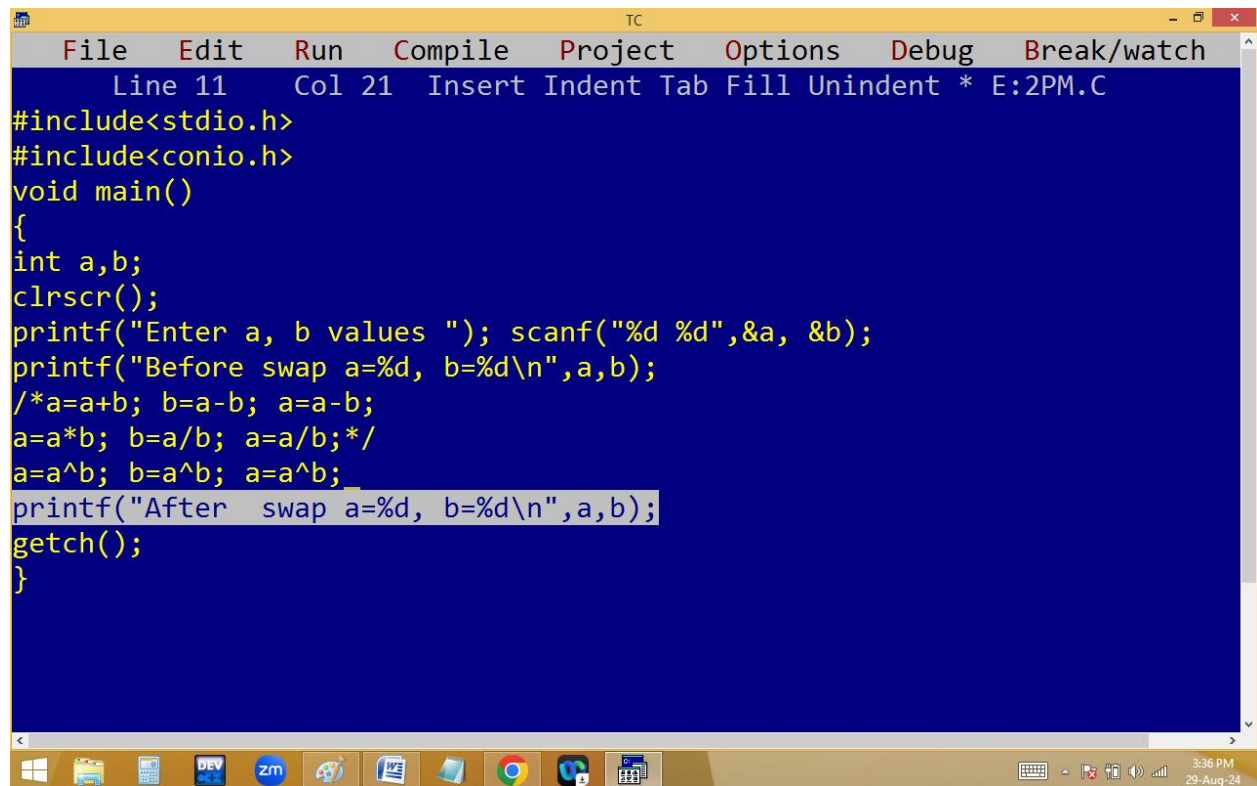
```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a,b,c;
    clrscr();
    printf("Enter a, b values "); scanf("%d %d",&a, &b);
    printf("Before swap a=%d, b=%d\n",a,b);
    c=a;a=b; b=c;
    printf("After swap a=%d, b=%d\n",a,b);
    getch();
}
```

The Windows taskbar is visible at the bottom, showing icons for File Explorer, Calculator, DEV C++, Zoho Mail, and other applications. The system clock in the bottom right corner shows 3:34 PM on 29-Aug-24.


```
TC
Enter a, b values 10 20
Before swap a=10, b=20
After swap a=20, b=10
```



Without using 3rd variable:



The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", "Debug", and "Break/watch". The status bar at the top indicates "Line 11", "Col 21", and "Insert Indent Tab Fill Unindent * E:2PM.C". The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a,b;
clrscr();
printf("Enter a, b values "); scanf("%d %d",&a, &b);
printf("Before swap a=%d, b=%d\n",a,b);
/*a=a+b; b=a-b; a=a-b;
a=a*b; b=a/b; a=a/b;*/
a=a^b; b=a^b; a=a^b;
printf("After swap a=%d, b=%d\n",a,b);
getch();
}
```

The taskbar at the bottom shows various icons including Windows, File Explorer, DEV, zm, a network icon, a folder icon, Google Chrome, a mail icon, and a task manager icon. The system clock in the bottom right corner shows "3:36 PM" and "29-Aug-24".

```
TC
Enter a, b values 3 7
Before swap a=3, b=7
After swap a=7, b=3
```

a=10 ==> 30 ==> 20

b=20 ==> 10

a=a+b==>10+20=30

b=a-b==>30-20=10

a=a-b==>30-10=20

a=10==>200 ==> 20

b=20==>10

a=a*b==>10*20=200

b=a/b==>200/20=10

a=a/b==>200/10=20

$$10 = 1010$$

$$20 = 10100$$

$$\begin{array}{r} 2 \overline{) 10} \\ 2 \overline{) 5-0} \\ 2 \overline{) 2-1} \\ 1-0 \end{array}$$

$$\begin{array}{r} 2 \overline{) 20} \\ 2 \overline{) 10-0} \\ 2 \overline{) 5-0} \\ 2 \overline{) 2-1} \\ 1-0 \end{array}$$

$$a = a \oplus b$$

$$a = 10 = 01010$$

$$b = 20 = \underline{10100}$$

$$11110 = 30$$

$$\begin{array}{cccc} 2^4 & 2^3 & 2^2 & 2^1 \\ | & | & | & | \\ 16 & 8 & 4 & 2 \end{array}$$

$$16 + 8 + 4 + 2 = 30$$

$$a = 30, b = 20$$

$$b = a \oplus b$$

$$a = 30 = 11110$$

$$b = 20 = \underline{10100}$$

$$01010 = 10$$

$$a = 30, b = 10$$

$$a = a \oplus b$$

$$a = 30 = 11110$$

$$b = 10 = \underline{01010}$$

$$10100 = 20$$

$$a = 20, b = 10 \checkmark$$

Home work:

1. Read a baby age in no of days and find the baby age in years, months, weeks and days.

total days

$$365 \overline{) 500} (1 - y$$

$$365$$

$$30 \overline{) 135} (4 - m$$

$$120$$

$$7 \overline{) 15} (2 - w$$

$$14$$

$$1 - d$$

2. Read a customer id, name, quantity purchased and rate of item. Find the amount, 35% discount and total.

